44. Serial Peripheral Interface (RSPIc)

In this section, “PCLK” is used to refer to PCLKA.

44.1 Overview

This MCU includes three channels of Serial Peripheral Interface (RSPI).
The RSPI channels are capable of high-speed, full-duplex or simplex (transmit-only) synchronous serial communications with multiple processors and peripheral devices.
Table 44.1 lists the specifications of the RSPI, and Figure 44.1 shows a block diagram of the RSPI.
In this section, x indicates A, B or C, and i indicates 0 to 3. Also, m as used with the RSPI command registers (SPCMDm) indicates 0 to 7.

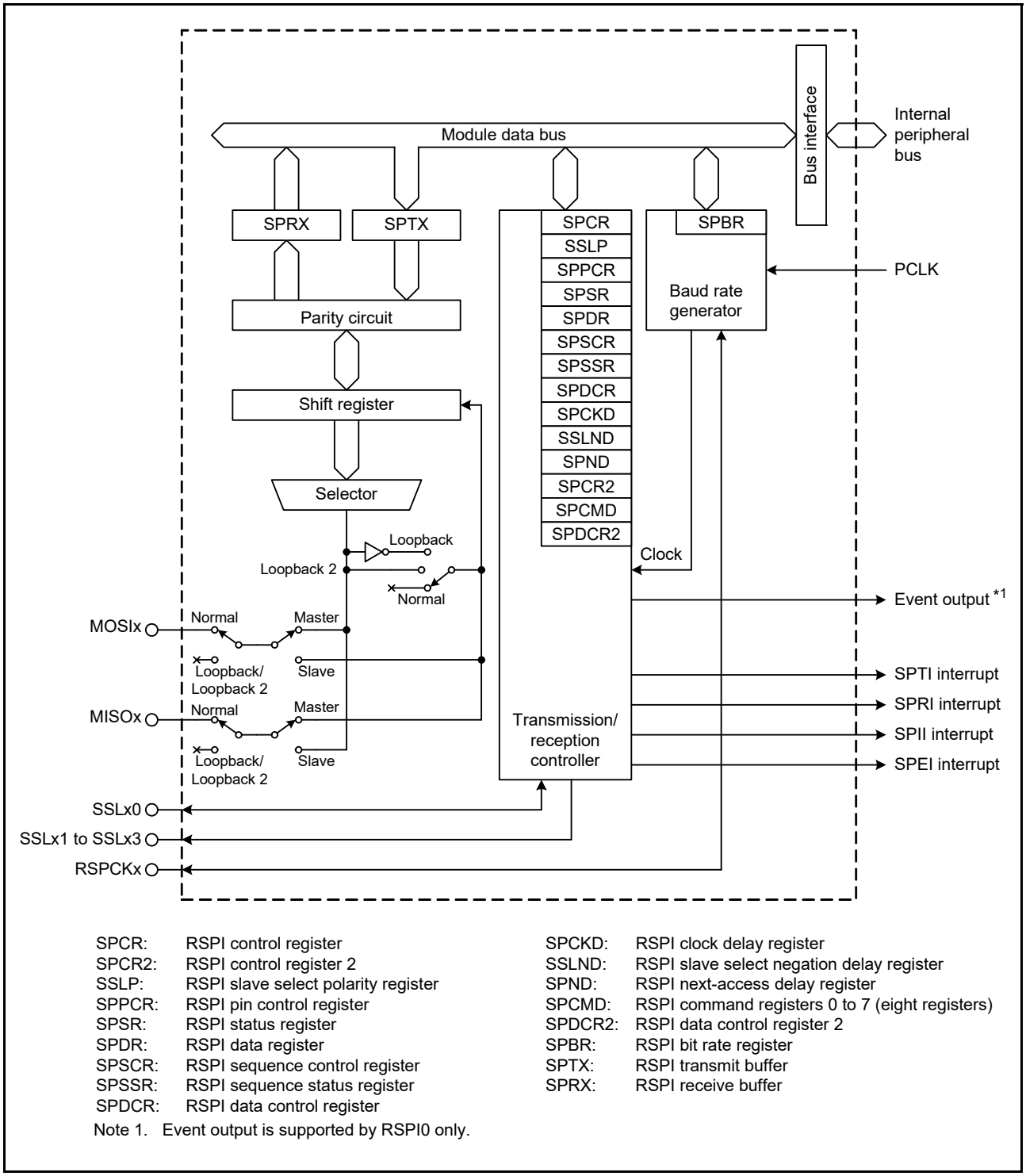
Table 44.1 RSPI Specifications (1/2)

Item	Description
Number of channels	Three channels
RSPI transfer functions	• Use of MOSI (master out/slave in), MISO (master in/slave out), SSL (slave select), and RSPCK (RSPI clock) signals allows serial communications through SPI operation (4-wire method) or clock synchronous operation (3-wire method). • Communication mode: Full-duplex or simplex (transmit-only) can be selected. • Switching of the polarity of RSPCK • Switching of the phase of RSPCK
Data format	• MSB first/LSB first selectable • Transfer bit length is selectable as 8, 9, 10, 11, 12, 13, 14, 15, 16, 20, 24, or 32 bits. • 128-bit transmit/receive buffers • Up to four frames can be transferred in one round of transmission/reception (each frame consisting of up to 32 bits). • Byte swapping of transmit and receive data is selectable.
Bit rate	• In master mode, the on-chip baud rate generator generates RSPCK by frequency-dividing PCLK (the division ratio ranges from divided by 2 to divided by 4096). • In slave mode, the minimum PCLK clock divided by 4 can be input as RSPCK (the maximum frequency of RSPCK is that of PCLK divided by 4). Width at high level: 2 cycles of PCLK; width at low level: 2 cycles of PCLK
Buffer configuration	• Double buffer configuration for the transmit/receive buffers • 128 bits for the transmit/receive buffers
Error detection	• Mode fault error detection • Overrun error detection*1 • Parity error detection • Underrun error detection
SSL control function	• Four SSL pins (SSLx0 to SSLx3) for each channel • In single-master mode, SSLx0 to SSLx3 pins are output. • In multi-master mode: SSLx0 pin for input, and SSLx1 to SSLx3 pins for either output or unused. • In slave mode: SSLx0 pin for input, and SSLx1 to SSLx3 pins for unused. • Controllable delay from SSL output assertion to RSPCK operation (RSPCK delay) Range: 1 to 8 RSPCK cycles (set in RSPCK-cycle units) • Controllable delay from RSPCK stop to SSL output negation (SSL negation delay) Range: 1 to 8 RSPCK cycles (set in RSPCK-cycle units) • Controllable wait for next-access SSL output assertion (next-access delay) Range: 1 to 8 RSPCK cycles (set in RSPCK-cycle units) • Function for changing SSL polarity
Control in master transfer	• A transfer of up to eight commands can be executed sequentially in looped execution. • For each command, the following can be set: SSL signal value, bit rate, RSPCK polarity/phase, transfer data length, MSB/LSB first, burst, RSPCK delay, SSL negation delay, and next-access delay • A transfer can be initiated by writing to the transmit buffer. • MOSI signal value specifiable in SSL negation • RSPCK auto-stop function

Table 44.1 RSPI Specifications (2/2)

Item	Description
Interrupt sources	• Interrupt sources Receive buffer full interrupt Transmit buffer empty interrupt Error interrupt (mode fault, overrun, underrun, or parity error) Idle interrupt
Event link function (output)	• The following events can be output to the event link controller. (RSPI0) Receive buffer full event Transmit buffer empty event Error event (mode fault, overrun, underrun, or parity error) Idle event Transmit end event
Others	• Function for initializing the RSPI • Loopback mode
Low power consumption function	Module stop state can be set.

Note 1. In master reception and when the RSPCK auto-stop function is enabled, an overrun error does not occur because the transfer clock is stopped at the timing of overrun error detection.



SPCR: RSPI control register
SPCR2: RSPI control register 2
SSLP: RSPI slave select polarity register
SPPCR: RSPI pin control register
SPSR: RSPI status register
SPDR: RSPI data register
SPSCR: RSPI sequence control register
SPSSR: RSPI sequence status register
SPDCR: RSPI data control register
SPCKD: RSPI clock delay register
SSLND: RSPI slave select negation delay register
SPND: RSPI next-access delay register
SPCMD: RSPI command registers 0 to 7 (eight registers)
SPDCR2: RSPI data control register 2
SPBR: RSPI bit rate register
SPTX: RSPI transmit buffer
SPRX: RSPI receive buffer

Note 1. Event output is supported by RSPI0 only.

Figure 44.1 RSPI Block Diagram

Table 44.2 lists the I/O pins used in the RSPI.
The RSPI automatically switches the I/O direction of the SSLx0 pin. SSLx0 is set as an output when the RSPI is a single master and as an input when the RSPI is a multi-master or a slave. Pins RSPCKx, MOSIx, and MISOx are automatically set as inputs or outputs according to the setting of master or slave and the level input on the SSLx0 pin.
Refer to section 44.3.2, Controlling RSPI Pins for details.

Table 44.2 RSPI Pin Configuration

Channel	Pin Name	I/O	Function
RSPI0	RSPCKA	I/O	Clock I/O
	MOSIA	I/O	Master transmit data I/O
	MISOA	I/O	Slave transmit data I/O
	SSLA0	I/O	Slave selection I/O
	SSLA1	Output	Slave selection output
	SSLA2	Output	Slave selection output
	SSLA3	Output	Slave selection output
RSPI1	RSPCKB	I/O	Clock I/O
	MOSIB	I/O	Master transmit data I/O
	MISOB	I/O	Slave transmit data I/O
	SSLB0	I/O	Slave selection I/O
	SSLB1	Output	Slave selection output
	SSLB2	Output	Slave selection output
	SSLB3	Output	Slave selection output
RSPI2	RSPCKC	I/O	Clock I/O
	MOSIC	I/O	Master transmit data I/O
	MISOC	I/O	Slave transmit data I/O
	SSLC0	I/O	Slave selection I/O
	SSLC1	Output	Slave selection output
	SSLC2	Output	Slave selection output
	SSLC3	Output	Slave selection output

44.2 Register Descriptions

44.2.1 RSPI Control Register (SPCR)

Address(es): RSPI0.SPCR 000D 0100h, RSPI1.SPCR 000D 0140h, RSPI2.SPCR 000D 0300h

	b7	b6	b5	b4	b3	b2	b1	b0
	SPRIE	SPE	SPTIE	SPEIE	MSTR	MODFEN	TXMD	SPMS
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	SPMS	RSPI Mode Select*1	0: SPI operation (4-wire method) 1: Clock synchronous operation (3-wire method)	R/W
b1	TXMD	Communications Operating Mode Select*1	0: Full-duplex communications (enables the receiver) 1: Transmit-only simplex communications (disables the receiver)	R/W
b2	MODFEN	Mode Fault Error Detection Enable *1	0: Disables the detection of mode fault error 1: Enables the detection of mode fault error	R/W
b3	MSTR	RSPI Master/Slave Mode Select*1	0: Slave mode 1: Master mode	R/W
b4	SPEIE	Error Interrupt Enable	0: Disables the generation of error interrupt requests 1: Enables the generation of error interrupt requests	R/W
b5	SPTIE	Transmit Buffer Empty Interrupt Enable	0: Disables the generation of transmit buffer empty interrupt requests 1: Enables the generation of transmit buffer empty interrupt requests	R/W
b6	SPE	RSPI Function Enable	0: Disables the RSPI function 1: Enables the RSPI function	R/W
b7	SPRIE	Receive Buffer Full Interrupt Enable	0: Disables the generation of receive buffer full interrupt requests 1: Enables the generation of receive buffer full interrupt requests	R/W

Note 1. Do not change the values of the MSTR, MODFEN, TXMD, and SPMS bits while the SPE bit is 1.

SPMS Bit (RSPI Mode Select)

The SPMS bit selects SPI operation (4-wire method) or clock synchronous operation (3-wire method).
The SSLx0 to SSLx3 pins are not used in clock synchronous operation. The RSPCKx, MOSIx, and MISOx pins handle communications. If clock synchronous operation is to proceed in master mode (SPCR.MSTR = 1), the SPCMDm.CPHA bit can be set to either 0 or 1. Set the CPHA bit to 1 if clock synchronous operation is to proceed in slave mode (SPCR.MSTR = 0). Do not set the CPHA bit to 0 when clock synchronous operation is to proceed in slave mode (SPCR.MSTR = 0).

TXMD Bit (Communications Operating Mode Select)

The TXMD bit selects full-duplex communications or transmit-only simplex communications.
When performing communications with the TXMD bit set to 1, the RSPI performs only transmit operations and not receive operations (refer to section 44.3.6, Communications Operating Mode).
When the TXMD bit is set to 1, receive buffer full interrupt requests cannot be used.

MODFEN Bit (Mode Fault Error Detection Enable)

The MODFEN bit enables or disables the detection of mode fault error (refer to section 44.3.9, Error Detection). In addition, the RSPI determines the I/O direction of the SSLx0 to SSLx3 pins based on combinations of the MODFEN and MSTR bits (refer to section 44.3.2, Controlling RSPI Pins).

MSTR Bit (RSPI Master/Slave Mode Select)

The MSTR bit selects master/slave mode of the RSPI. According to MSTR bit settings, the RSPI determines the direction of pins RSPCKx, MOSIx, MISOx, and SSLx0 to SSLx3.

SPEIE Bit (Error Interrupt Enable)

The SPEIE bit enables or disables the generation of error interrupt requests when the RSPI detects a mode fault error or underrun error and sets the SPSR.MODF flag to 1, when the RSPI detects an overrun error and sets the SPSR.OVRF flag to 1, or when the RSPI detects a parity error and sets the SPSR.PERF flag to 1 (refer to section 44.3.9, Error Detection).

SPTIE Bit (Transmit Buffer Empty Interrupt Enable)

The SPTIE bit enables or disables the generation of transmit buffer empty interrupt requests when the RSPI detects when the transmit buffer is empty.

A transmit buffer empty interrupt request when transmission starts is generated by setting the SPE and SPTIE bits to 1 at the same time or by setting the SPE bit to 1 after setting the SPTIE bit to 1.

Note that a transmit buffer interrupt is generated when the SPTIE bit is 1 even if the RSPI function is disabled (the SPTIE bit is changed to 0).

SPE Bit (RSPI Function Enable)

The SPE bit enables or disables the RSPI function.

When the SPSR.MODF flag is 1, the SPE bit cannot be set to 1. For details, refer to section 44.3.9, Error Detection.

Setting the SPE bit to 0 disables the RSPI function, and initializes a part of the module function. For details, refer to section 44.3.10, Initializing RSPI. Furthermore, a transmit buffer empty interrupt request is generated by the state of the SPE bit changing from 0 to 1 or from 1 to 0.

SPRIE Bit (Receive Buffer Full Interrupt Enable)

If the RSPI has detected a receive buffer full write after completion of a serial transfer, the SPRIE bit enables or disables the generation of a receive buffer full interrupt request.

44.2.2 RSPI Slave Select Polarity Register (SSLP)

Address(es): RSPI0.SSLP 000D 0101h, RSPI1.SSLP 000D 0141h, RSPI2.SSLP 000D 0301h

	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	SSL3P	SSL2P	SSL1P	SSL0P
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	SSL0P	SSL0 Signal Polarity Setting	0: SSL0 signal is active low 1: SSL0 signal is active high	R/W
b1	SSL1P	SSL1 Signal Polarity Setting	0: SSL1 signal is active low 1: SSL1 signal is active high	R/W
b2	SSL2P	SSL2 Signal Polarity Setting	0: SSL2 signal is active low 1: SSL2 signal is active high	R/W
b3	SSL3P	SSL3 Signal Polarity Setting	0: SSL3 signal is active low 1: SSL3 signal is active high	R/W
b7 to b4	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

Note: Do not change the SSLP register while the SPCR.SPE bit is 1.

44.2.3 RSPI Pin Control Register (SPPCR)

Address(es): RSPI0.SPPCR 000D 0102h, RSPI1.SPPCR 000D 0142h, RSPI2.SPPCR 000D 0302h

	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	MOIFE	MOIFV	—	—	SPLP2	SPLP
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	SPLP	RSPI Loopback	0: Normal mode 1: Loopback mode (data is inverted for transmission)	R/W
b1	SPLP2	RSPI Loopback 2	0: Normal mode 1: Loopback mode (data is not inverted for transmission)	R/W
b3, b2	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b4	MOIFV	MOSI Idle Fixed Value	0: The level output on the MOSIx pin during MOSI idling corresponds to low 1: The level output on the MOSIx pin during MOSI idling corresponds to high	R/W
b5	MOIFE	MOSI Idle Value Fixing Enable	0: MOSI output value equals final data from previous transfer 1: MOSI output value equals the value set in the MOIFV bit	R/W
b7, b6	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

Note: Do not change the SPPCR register while the SPCR.SPE bit is 1.

SPLP Bit (RSPI Loopback)

The SPLP bit selects the mode of the RSPI pins.

When the SPLP bit is set to 1, the RSPI shuts off the path between the MISOx pin and the shift register if the SPCR.MSTR bit is 1, and between the MOSIx pin and the shift register if the SPCR.MSTR bit is 0, and connects (inverts) the input path and output path for the shift register (loopback mode).

SPLP2 Bit (RSPI Loopback 2)

The SPLP2 bit selects the mode of the RSPI pins.

When the SPLP2 bit is set to 1, the RSPI shuts off the path between the MISOx pin and the shift register if the SPCR.MSTR bit is 1, and between the MOSIx pin and the shift register if the SPCR.MSTR bit is 0, and connects the input path and output path for the shift register (loopback mode).

MOIFV Bit (MOSI Idle Fixed Value)

If the MOIFE bit is 1 in master mode, the MOIFV bit determines the MOSIx pin output value during the SSL negation period (including the SSL retention period during a burst transfer).

MOIFE Bit (MOSI Idle Value Fixing Enable)

The MOIFE bit fixes the MOSIx output value when the RSPI in master mode is in an SSL negation period (including the SSL retention period during a burst transfer). When the MOIFE bit is 0, the RSPI outputs the last data from the previous serial transfer during the SSL negation period to the MOSIx pin. When the MOIFE bit is 1, the RSPI outputs the fixed value set in the MOIFV bit to the MOSIx pin.

44.2.4 RSPI Status Register (SPSR)

Address(es): RSPI0.SPSR 000D 0103h, RSPI1.SPSR 000D 0143h, RSPI2.SPSR 000D 0303h

	b7	b6	b5	b4	b3	b2	b1	b0
	SPRF	—	SPTEF	UDRF	PERF	MODF	IDLNF	OVRF
Value after reset:	0	0	1	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	OVRF	Overrun Error Flag	0: No overrun error occurs 1: An overrun error occurs	R/(W) *1
b1	IDLNF	Idle Flag	0: RSPI is in the idle state 1: RSPI is in the transfer state	R
b2	MODF	Mode Fault Error Flag	0: Neither a mode fault error nor an underrun error occurs 1: A mode fault error or an underrun error occurs	R/(W) *1
b3	PERF	Parity Error Flag	0: No parity error occurs 1: A parity error occurs	R/(W) *1
b4	UDRF	Underrun Error Flag	This flag is used with the MODF flag to check the state in terms of mode fault errors and underrun errors. b4 b2 0 0: Neither a mode fault error nor an underrun error occurs 0 1: A mode fault error occurs 1 1: An underrun error occurs	R/(W) *1, *2
b5	SPTEF	Transmit Buffer Empty Flag	0: Transmit buffer has valid data 1: Transmit buffer has no valid data	R*3
b6	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b7	SPRF	Receive Buffer Full Flag	0: Receive buffer has no valid data 1: Receive buffer has valid data	R*3

Note 1. Only 0 can be written to clear the flag after reading 1.
Note 2. Clear the MODF and UDRF flags at the same time.
Note 3. The write value should be 1.

OVRF Flag (Overrun Error Flag)

The OVRF flag indicates the occurrence of an overrun error. In master mode (when the SPCR.MSTR bit is 1) and when the RSPCK clock auto-stop function is enabled (the SPCR2.SCKASE bit is 1), an overrun error does not occur; accordingly this flag does not become 1. For details, refer to section 44.3.9.1, Overrun Error.

[Setting condition]

- When the next data reception ends while the SPCR.TXMD bit is 0 and the receive buffer is full.

[Clearing condition]

- When the SPSR register is read while the OVRF flag is 1, and then 0 is written to the OVRF flag.

IDLNF Flag (Idle Flag)

The IDLNF flag indicates the transfer status of the RSPI.

[Setting condition]

Master mode

- When both of the conditions in master mode under the [Clearing condition] below are not satisfied.

Slave mode

- When the SPCR.SPE bit is set to 1 (enables the RSPI function).

[Clearing condition]

Master mode

- When the SPCR.SPE bit is set to 0 (disables the RSPI function)
- When all of the following conditions are satisfied.
  1. The transmit buffer is empty (the SPTEF flag is 1)
  2. The SPSSR.SPCP[2:0] bits are 000b
  3. The transmission of the last bit has been completed and the time specified by the SSLND.SLNDL[2:0] and SPND.SPNDL[2:0] bits has elapsed

Slave mode

- When the SPCR.SPE bit is set to 0 (disables the RSPI function).

MODF Flag (Mode Fault Error Flag)

Indicates the occurrence of a mode fault error or an underrun error. The UDRF flag indicates whether the error is a mode fault error or an underrun error.

[Setting condition]

Multi-master mode

- When the input level of the SSLxi pin changes to the active level while the SPCR.MSTR bit is 1 (master mode) and the SPCR.MODFEN bit is 1 (mode fault error detection is enabled), the RSPI detects a mode fault error.

Slave mode

- When the SSLxi pin is negated before the RSPCK cycle necessary for data transfer ends while the SPCR.MSTR bit is 0 (slave mode) and the SPCR.MODFEN bit is 1 (mode fault error detection is enabled), the RSPI detects a mode fault error.
- When the serial transfer starts while the SPCR.SPE bit is 1 (enables the RSPI function) and the transmit data are not ready for output, the RSPI detects an underrun error.

The active level of the SSLxi signal is determined by the SSLP.SSLiP bit (SSLi signal polarity setting bit).

[Clearing condition]

- When the SPSR register is read while the MODF flag is 1, and then 0 is written to the MODF flag.

PERF Flag (Parity Error Flag)

Indicates the occurrence of a parity error.

[Setting condition]

- When a data reception ends while the SPCR.TXMD bit is 0 and the SPCR2.SPPE bit is 1, the RSPI detects a parity error.

[Clearing condition]

- When the SPSR register is read while the PERF flag is 1, and then 0 is written to the PERF flag.

UDRF Flag (Underrun Error Flag)

Indicates the occurrence of an underrun error. When this flag becomes 1, the MODF flag becomes 1 too. When the MODF flag is 1 and this flag is 0, the error is a mode fault error.

[Setting condition]

- When the serial transfer starts while the SPCR.MSTR bit is 0 (slave mode), the SPCR.SPE bit is 1 (enables the RSPI function), and the transmit data are not ready for output, the RSPI detects an underrun error

[Clearing condition]

- When 0 is written to the UDRF flag after reading the SPSR register while the UDRF flag is 1

SPTEF Flag (Transmit Buffer Empty Flag)

Indicates whether the transmit buffer (SPTX) in the RSPI data register has valid data.

[Setting condition]

- When the SPCR.SPE bit is set to 0 (disables the RSPI function).
- When the number of frames of transmit data specified by the SPDCR.SPFC[1:0] bits have been transferred from the transmit buffer to the shift register.

[Clearing condition]

- When the number of frames of transmit data specified by the SPDCR.SPFC[1:0] bits is written to the SPDR register.

The SPDR register can be set only when the SPTEF flag is 1. The data in the transmit buffer is not updated when the SPDR register is set while the SPTEF flag is 0.

SPRF Flag (Receive Buffer Full Flag)

Indicates whether the receive buffer (SPRX) in the RSPI data register has valid data.

[Setting condition]

- When the number of frames of receive data specified by the SPDCR.SPFC[1:0] bits is transferred from shift register to the receive buffer (SPRX) while the SPCR.TXMD bit is 0 (full duplex) and the SPRF flag is 0.
  Note that the SPRF flag does not become 1 when the OVRF flag is 1.

[Clearing condition]

- When all of the received data are read from the SPDR register.

44.2.5 RSPI Data Register (SPDR)

- When accessing in longword size

Address(es): RSPI0.SPDR 000D 0104h, RSPI1.SPDR 000D 0144h, RSPI2.SPDR 000D 0304h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

- When accessing in word size

Address(es): RSPI0.SPDR.H 000D 0104h, RSPI1.SPDR.H 000D 0144h, RSPI2.SPDR.H 000D 0304h

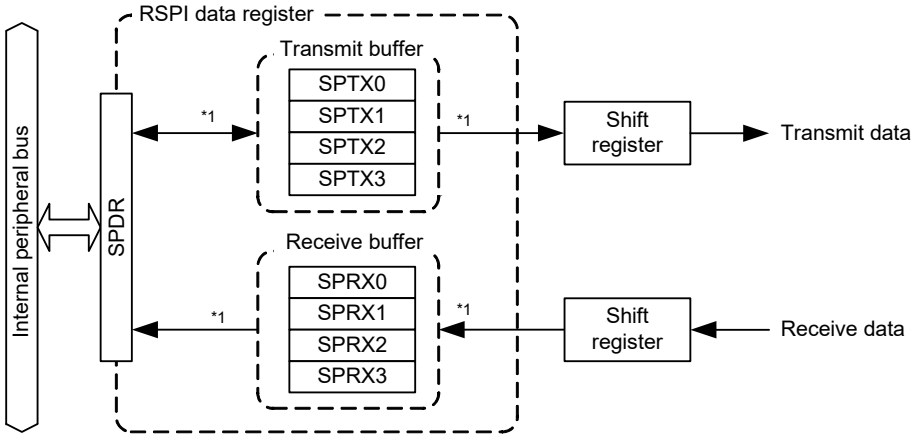
	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	H															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

- When accessing in byte size

Address(es): RSPI0.SPDR.HH 000D 0104h, RSPI1.SPDR.HH 000D 0144h, RSPI2.SPDR.HH 000D 0304h

	b7	b6	b5	b4	b3	b2	b1	b0
	HH							
Value after reset:	0	0	0	0	0	0	0	0

The SPDR register is the interface with the buffers that hold data for transmission and reception by the RSPI.
When accessing in longwords (the SPLW bit is 1 and the SPBYT bit is 0), access the SPDR register in 32-bit units.
When accessing in words (the SPLW bit is 0 and the SPBYT bit is 0), access the SPDR.H register in 16-bit units.
When accessing in bytes (the SPBYT bit is 1), access the SPDR.HH register in 8-bit units.
The transmit buffer (SPTX) and receive buffer (SPRX) are independent but are both mapped to the SPDR register.
Figure 44.2 shows the Configuration of the SPDR Register.



Note 1. The destination buffer and stage for access is automatically switched by hardware.

Figure 44.2 Configuration of the SPDR Register

The transmit and receive buffers each have four stages. The number of stages to be used is selectable by the number of frames specification bits in the RSPI data control register (SPDCR.SPFC[1:0]). The eight stages of the buffer are all mapped to the single address of the SPDR register.

Data written to the SPDR register are written to a transmit-buffer stage (SPTXn) (n = 0 to 3) and then transmitted from the buffer. The receive buffer holds received data on completion of reception. The receive buffer is not updated if an overrun is generated.

Furthermore, if the data length is other than 32 bits, bits not referred to in SPTXn (n = 0 to 3) are stored in the corresponding bits in SPRXn. For example, if the data length is 9 bits, received data are stored in the SPRXn[8:0] bits and the SPTXn[31:9] bits are stored in the SPRXn[31:9] bits.

(1) Bus Interface

The SPDR register is the interface with 32-bit wide transmit and receive buffers, each of which has four stages, for a total of 32 bytes. In other words, the 32 bytes are mapped to the 4-byte address space for the SPDR register. Furthermore, the unit of access for the SPDR register is selected by the SPDCR.SPLW bit and the SPDCR.SPBYT bit.
Data for transmission should be flush with the LSB end of the register. Received data are stored flush with the LSB end.
Operations involved in writing to and reading from the SPDR register are described below.

(a) Writing

Data written to the SPDR register are written to a transmit buffer (SPTXn). This is not influenced by the value of the SPDCR.SPRDTD bit unlike when reading from the SPDR register.
The transmit buffer includes a transmit buffer write pointer which is automatically updated to indicate the next stage each time data are written to the SPDR register.
Figure 44.3 shows the configuration of the bus interface with the transmit buffer in the case of writing to the SPDR register.

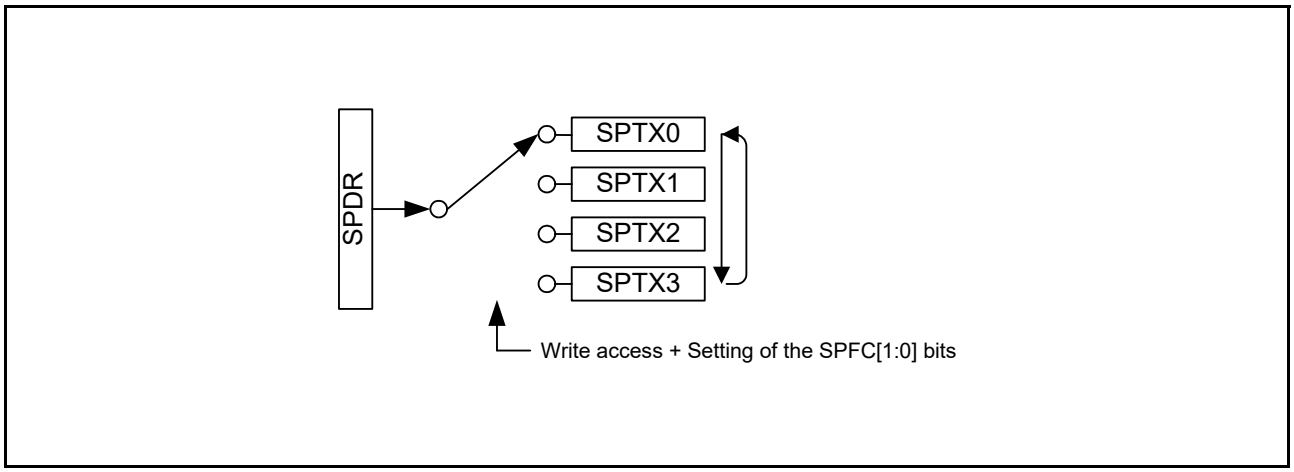


Figure 44.3 Configuration of the SPDR Register (Writing)

The sequence for switching the transmit buffer write pointer differs with the setting of the number of frames specification bits in the RSPI data control register (SPDCR.SPFC[1:0]).

- Settings of the SPFC[1:0] bits and sequence of switching the pointer among SPTX0 to SPTX3.
  When the SPFC[1:0] bits are 00b: SPTX0 → SPTX0 → SPTX0 → …
  When the SPFC[1:0] bits are 01b: SPTX0 → SPTX1 → SPTX0 → SPTX1 → …
  When the SPFC[1:0] bits are 10b: SPTX0 → SPTX1 → SPTX2 → SPTX0 → SPTX1 → …
  When the SPFC[1:0] bits are 11b: SPTX0 → SPTX1 → SPTX2 → SPTX3 → SPTX0 → SPTX1 → …

When 1 is written to the RSPI function enable bit in the RSPI control register (SPCR.SPE) while the bit's current value is 0, SPTX0 will be the destination the next time writing proceeds.
When writing to the transmit buffer (SPTXn) after generation of the transmit buffer empty interrupt (after the SPSR.SPTEF flag becomes 1), write the number of frames set by the number of frames specification bits (SPFC[1:0]) in the RSPI data control register (SPDCR). Even if the number of frames is written to the transmit buffer (SPTXn), the value of the buffer is not updated after completion of the writing and before generation of the next transmit buffer empty interrupt (while the SPSR.SPTEF flag is 0).

(b) Reading

The SPDR register can be read to read the value of a receive buffer (SPRXn) or a transmit buffer (SPTXn). The setting of the RSPI receive/transmit data select bit in the RSPI data control register (SPDCR.SPRDTD) selects whether reading is of the receive or transmit buffer.

The sequence of reading the SPDR register is controlled by independent pointers, receive buffer read pointer and transmit buffer read pointer.

Figure 44.4 shows the configuration of the bus interface with the receive and transmit buffers in the case of reading from the SPDR register.

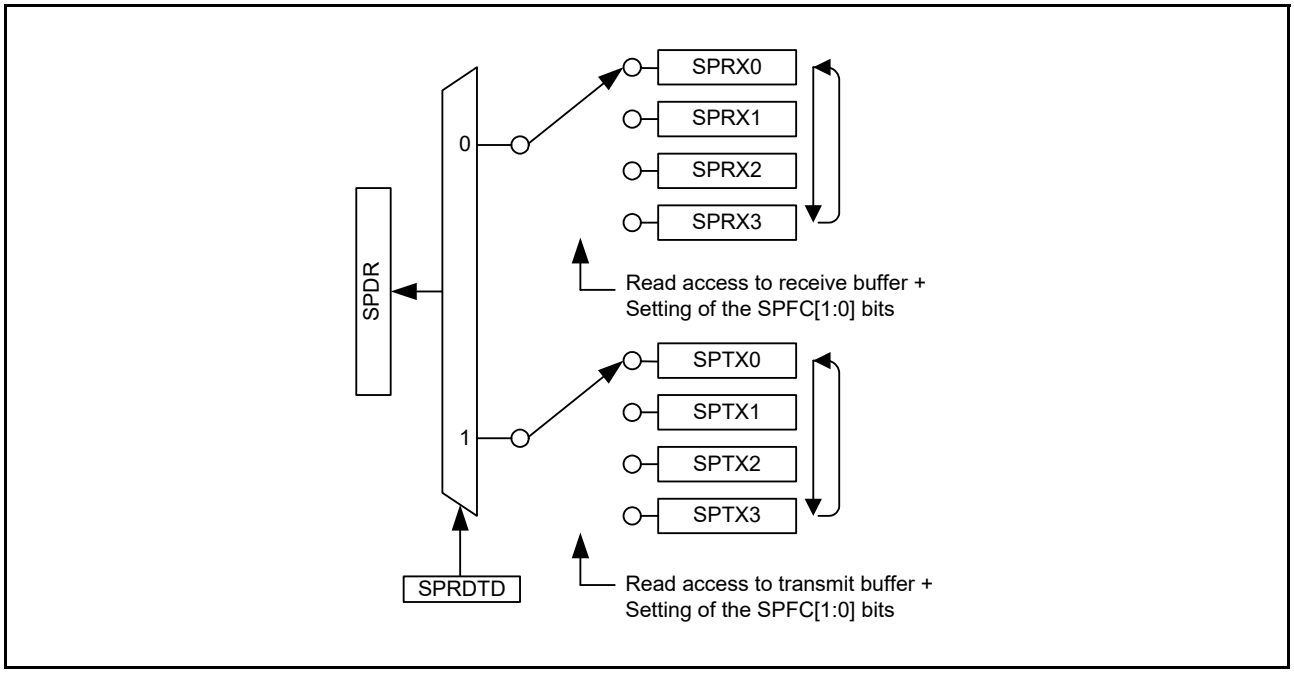


Figure 44.4 Configuration of the SPDR Register (Reading)

Reading the receive buffer switches the receive buffer read pointer to the next buffer automatically.

The sequence of switching the receive buffer read pointer is the same as that for the transmit buffer write pointer. However, when 1 is written to the RSPI function enable bit in the RSPI control register (SPCR.SPE) while the bit's current value is 0, SPRX0 will be indicated by the buffer read pointer the next time reading proceeds.

The transmit buffer read pointer is updated when writing to the SPDR register, and not updated when reading from the transmit buffer. When reading from the transmit buffer, the value most recently written to the SPDR register is read. However, after generation of the transmit buffer empty interrupt, the values read from the transmit buffer are all 0 in the interval after completion of writing the number of frames of data specified in the number of frames specification bits (SPDCR.SPFC[1:0]) and before generation of the next buffer empty interrupt (while the SPSR.SPTEF flag is 0).

44.2.6 RSPI Sequence Control Register (SPSCR)

Address(es): RSPI0.SPSCR 000D 0108h, RSPI1.SPSCR 000D 0148h, RSPI2.SPSCR 000D 0308h

	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	SPSLN[2:0]		
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b2 to b0	SPSLN[2:0]	RSPI Sequence Length Specification	b2 b0 Sequence Length — Referenced SPCMD0 to SPCMD7 registers (No.) 0 0 0: 1 — 0→0→… 0 0 1: 2 — 0→1→0→… 0 1 0: 3 — 0→1→2→0→… 0 1 1: 4 — 0→1→2→3→0→… 1 0 0: 5 — 0→1→2→3→4→0→… 1 0 1: 6 — 0→1→2→3→4→5→0→… 1 1 0: 7 — 0→1→2→3→4→5→6→0→… 1 1 1: 8 — 0→1→2→3→4→5→6→7→0→… The order in which the SPCMD0 to SPCMD7 registers are to be referenced is changed according to the sequence length that is set in these bits. The relationship among the setting of these bits, sequence length, and the SPCMD0 to SPCMD7 registers referenced by the RSPI is shown above. However, the RSPI in slave mode references the SPCMD0 register.	R/W
b7 to b3	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

The SPSCR register sets the sequence length when the RSPI operates in master mode. When changing the SPSCR.SPSLN[2:0] bits while both the SPCR.MSTR and SPCR.SPE bits are 1, the bits should be changed while the SPSR.IDLNF flag is 0.

SPSLN[2:0] Bits (RSPI Sequence Length Specification)

The SPSLN[2:0] bits specify a sequence length when the RSPI in master mode performs sequential operations. The RSPI in master mode changes the SPCMD0 to SPCMD7 registers to be referenced and the order in which they are referenced according to the sequence length that is set in the SPSLN[2:0] bits.
In slave mode, the SPCMD0 register is referred.

44.2.7 RSPI Sequence Status Register (SPSSR)

Address(es): RSPI0.SPSSR 000D 0109h, RSPI1.SPSSR 000D 0149h, RSPI2.SPSSR 000D 0309h

	b7	b6	b5	b4	b3	b2	b1	b0
	—	SPECM[2:0]			—	SPCP[2:0]		
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b2 to b0	SPCP[2:0]	RSPI Command Pointer	b2 b0 0 0 0: SPCMD0 0 0 1: SPCMD1 0 1 0: SPCMD2 0 1 1: SPCMD3 1 0 0: SPCMD4 1 0 1: SPCMD5 1 1 0: SPCMD6 1 1 1: SPCMD7	R
b3	—	Reserved	This bit is read as 0.	R
b6 to b4	SPECM[2:0]	RSPI Error Command	b6 b4 0 0 0: SPCMD0 0 0 1: SPCMD1 0 1 0: SPCMD2 0 1 1: SPCMD3 1 0 0: SPCMD4 1 0 1: SPCMD5 1 1 0: SPCMD6 1 1 1: SPCMD7	R
b7	—	Reserved	This bit is read as 0.	R

The SPSSR register indicates the sequence control status when the RSPI operates in master mode.
Any writing to the SPSSR register is ignored.

SPCP[2:0] Bits (RSPI Command Pointer)

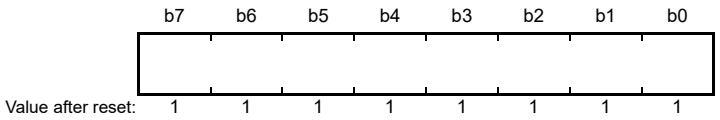
The SPCP[2:0] bits indicate the SPCMDm register that is currently pointed to by the pointer during sequence control by the RSPI.
For the RSPI's sequence control, refer to section 44.3.11.1, Master Mode Operation.

SPECM[2:0] Bits (RSPI Error Command)

The SPECM[2:0] bits indicate the SPCMDm register that is specified by the SPCP[2:0] bits when an error is detected during sequence control by the RSPI. The RSPI updates the SPECM[2:0] bits only when an error is detected. If both the SPSR.OVRF and SPSR.MODF flags are 0 and there is no error, the values of the SPECM[2:0] bits have no meaning.
For the RSPI's error detection function, refer to section 44.3.9, Error Detection. For the RSPI's sequence control, refer to section 44.3.11.1, Master Mode Operation.

44.2.8 RSPI Bit Rate Register (SPBR)

Address(es): RSPI0.SPBR 000D 010Ah, RSPI1.SPBR 000D 014Ah, RSPI2.SPBR 000D 030Ah



The SPBR register sets the bit rate in master mode. Do not change the SPBR register while both the SPCR.MSTR and SPCR.SPE bits are 1.

When the RSPI is used in slave mode, the bit rate depends on the bit rate of the input clock (bit rate satisfying the electrical characteristics should be used) regardless of the settings of the SPBR register and the SPCMDm.BRDV[1:0] bits (bit rate division setting bits).
The bit rate is determined by combinations of the SPBR register setting and the SPCMDm.BRDV[1:0] bit setting. The equation for calculating the bit rate is given below. In the equation, n denotes the SPBR register setting (0, 1, 2, …, 255), and N denotes the BRDV[1:0] bit setting (0, 1, 2, 3).

$$\text{Bit rate} = \frac{f(PCLK)}{2 \times (n+1) \times 2^{N}}$$

Table 44.3 lists examples of the relationship among the SPBR register settings, the BRDV[1:0] bit settings, and bit rates.
Use the bit rate that meets electrical characteristics based on the AC specifications of the target device.

Table 44.3 Relationship among SPBR Settings, BRDV[1:0] Settings, and Bit Rates

SPBR (n)	BRDV[1:0] Bits (N)	Division Ratio	Bit Rate							
			PCLK = 32 MHz	PCLK = 36 MHz	PCLK = 40 MHz	PCLK = 50 MHz	PCLK = 60 MHz	PCLK = 80 MHz	PCLK = 100 MHz	PCLK = 120 MHz
0	0	2	16.0 Mbps	18.0 Mbps	20.0 Mbps	25.0 Mbps	30.0 Mbps	40.0 Mbps	—	—
1	0	4	8.00 Mbps	9.00 Mbps	10.0 Mbps	12.5 Mbps	15.0 Mbps	20.0 Mbps	25.0 Mbps	30.0 Mbps
2	0	6	5.33 Mbps	6.00 Mbps	6.67 Mbps	8.33 Mbps	10.0 Mbps	13.3 Mbps	16.7 Mbps	20.0 Mbps
3	0	8	4.00 Mbps	4.50 Mbps	5.00 Mbps	6.25 Mbps	7.50 Mbps	10.0 Mbps	12.5 Mbps	15.0 Mbps
4	0	10	3.20 Mbps	3.60 Mbps	4.00 Mbps	5.00 Mbps	6.00 Mbps	8.00 Mbps	10.0 Mbps	12.0 Mbps
5	0	12	2.67 Mbps	3.00 Mbps	3.33 Mbps	4.16 Mbps	5.00 Mbps	6.67 Mbps	8.33 Mbps	10.0 Mbps
5	1	24	1.33 Mbps	1.50 Mbps	1.67 Mbps	2.08 Mbps	2.50 Mbps	3.33 Mbps	4.17 Mbps	5.00 Mbps
5	2	48	667 kbps	750 kbps	833 kbps	1.04 Mbps	1.25 Mbps	1.67 Mbps	2.08 Mbps	2.50 Mbps
5	3	96	333 kbps	375 kbps	417 kbps	521 kbps	625 kbps	833 kbps	1.04 Mbps	1.25 Mbps
255	3	4096	7.81 kbps	8.80 kbps	9.78 kbps	12.2 kbps	14.6 kbps	19.5 kbps	24.4 kbps	29.3 kbps

44.2.9 RSPI Data Control Register (SPDCR)

Address(es): RSPI0.SPDCR 000D 010Bh, RSPI1.SPDCR 000D 014Bh, RSPI2.SPDCR 000D 030Bh

	b7	b6	b5	b4	b3	b2	b1	b0
	—	SPBYT	SPLW	SPRDTD	—	—	SPFC[1:0]	
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b1, b0	SPFC[1:0]	Number of Frames Specification	b1 b0 0 0: 1 frame 0 1: 2 frames 1 0: 3 frames 1 1: 4 frames	R/W
b3, b2	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b4	SPRDTD	RSPI Receive/Transmit Data Select	0: The SPDR values are read from the receive buffer 1: The SPDR values are read from the transmit buffer (but only if the transmit buffer is empty)	R/W
b5	SPLW	RSPI Longword Access/ Word Access Specification*1	0: The SPDR register is accessed in words 1: The SPDR register is accessed in longwords	R/W
b6	SPBYT	RSPI Byte Access Specification	0: The SPDR register is accessed in words or longwords (the SPLW bit is enabled) 1: The SPDR register is accessed in bytes (the SPLW bit is disabled)	R/W
b7	—	Reserved	This bit is read as 0. The write value should be 0.	R/W

Note 1. Set the SPBYT bit to 0, when accessing the SPDR register in words or longwords.

Up to four frames can be transmitted or received in one round of transmission or reception activation. The amount of data in each transfer is controlled by the combination of the SPCMDm.SPB[3:0] bits, the SPSCR.SPSLN[2:0] bits, and the SPDCR.SPFC[1:0] bits.
When changing the SPDCR.SPFC[1:0] bits while the SPCR.SPE bit is 1, the bits should be changed while the SPSR.IDLNF flag is 0.

SPFC[1:0] Bits (Number of Frames Specification)

The SPFC[1:0] bits specify the number of frames that can be stored in the SPDR register (per transfer activation). Up to four frames can be transmitted or received in one round of transmission or reception, and the amount of data is determined by the combination of the SPSCR.SPSLN[2:0] bits, and the SPDCR.SPFC[1:0] bits. Furthermore, the setting of the SPFC[1:0] bits adjusts the number of frames for generation of receive buffer full interrupt, and start of transmission or generation of transmit buffer empty interrupts.
When the number of frames of transmit data specified by SPFC[1:0] bits is written to the SPDR register, the SPSR.SPTEF flag becomes 0 and transmission starts. Then, when the specified number of frames of transmit data has been transferred to the shift register, the SPTEF flag becomes 1 and the RSPI transmit buffer empty interrupt is generated.
When the number of frames specified by the SPFC[1:0] bits are received, the SPSR.SPRF flag becomes 1 and the receive buffer full interrupt is generated.
Table 44.4 lists the frame configurations that can be stored in the SPDR register and examples of combinations of settings for transmission and reception. Do not select the combinations of settings other than those shown in the examples.

Table 44.4 Settable Combinations of SPSLN[2:0] Bits and SPFC[1:0] Bits

Setting	SPSLN[2:0]	SPFC[1:0]	Number of Frames in a Single Sequence	Number of Frames at which Transmit Buffer or Receive Buffer Status Becomes “Has Valid Data”
1-1	000b	00b	1	1
1-2	000b	01b	2	2
1-3	000b	10b	3	3
1-4	000b	11b	4	4
2-1	001b	01b	2	2
2-2	001b	11b	4	4
3	010b	10b	3	3
4	011b	11b	4	4
5	100b	00b	5	1
6	101b	00b	6	1
7	110b	00b	7	1
8	111b	00b	8	1

SPRDTD Bit (RSPI Receive/Transmit Data Select)

The SPRDTD bit selects whether the SPDR register reads values from the receive buffer or from the transmit buffer.
If reading is from the transmit buffer, the value written to the SPDR register immediately beforehand is read.
When reading the transmit buffer, do so before writing of the number of frames set in the SPFC[1:0] bits is finished and after generation of the transmit buffer empty interrupt (While the SPSR.SPTEF flag is 1).
For details, refer to section 44.2.5, RSPI Data Register (SPDR).

SPLW Bit (RSPI Longword Access/Word Access Specification)

The SPLW bit specifies the access width for the SPDR register. This bit setting is enabled when the SPBYT bit is 0.
Access to the SPDR register in words when the SPLW bit is 0 and in longwords when the SPLW bit is 1.
Also, when the SPLW bit is 0, set the SPCMDm.SPB[3:0] bits (RSPI data length setting bits) to 8 to 16 bits. Do not select 20, 24, or 32 bits.

SPBYT Bit (RSPI Byte Access Specification)

The SPBYT bit specifies the access width for the SPDR register. Access to the SPDR register according to the SPLW bit setting when the SPBYT bit is 0. Access to the SPDR register in bytes when the SPBYT bit is 1.
When the SPBYT bit is 1, set the SPCMDm.SPB[3:0] bits (RSPI data length setting bits) to 8 bits. Do not select 9 to 16, 20,24, or 32 bits.

44.2.10 RSPI Clock Delay Register (SPCKD)

Address(es): RSPI0.SPCKD 000D 010Ch, RSPI1.SPCKD 000D 014Ch, RSPI2.SPCKD 000D 030Ch

	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	SCKDL[2:0]		
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b2 to b0	SCKDL[2:0]	RSPCK Delay Setting	b2 b0 0 0 0: 1 RSPCK 0 0 1: 2 RSPCK 0 1 0: 3 RSPCK 0 1 1: 4 RSPCK 1 0 0: 5 RSPCK 1 0 1: 6 RSPCK 1 1 0: 7 RSPCK 1 1 1: 8 RSPCK	R/W
b7 to b3	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

The SPCKD register sets a period from the beginning of SSLxi signal assertion to RSPCK oscillation (RSPCK delay) when the SPCMDm.SCKDEN bit is 1. Do not change the SPCKD register while both the SPCR.MSTR and SPCR.SPE bits are 1.

SCKDL[2:0] Bits (RSPCK Delay Setting)

The SCKDL[2:0] bits set an RSPCK delay value when the SPCMDm.SCKDEN bit is 1.
When using the RSPI in slave mode, set the SCKDL[2:0] bits to 000b.

44.2.11 RSPI Slave Select Negation Delay Register (SSLND)

Address(es): RSPI0.SSLND 000D 010Dh, RSPI1.SSLND 000D 014Dh, RSPI2.SSLND 000D 030Dh

	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	SLNDL[2:0]		
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b2 to b0	SLNDL[2:0]	SSL Negation Delay Setting	b2 b0 0 0 0: 1 RSPCK 0 0 1: 2 RSPCK 0 1 0: 3 RSPCK 0 1 1: 4 RSPCK 1 0 0: 5 RSPCK 1 0 1: 6 RSPCK 1 1 0: 7 RSPCK 1 1 1: 8 RSPCK	R/W
b7 to b3	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

The SSLND register sets a period (SSL negation delay) from the transmission of a final RSPCK edge to the negation of the SSLxi signal during a serial transfer by the RSPI in master mode. Do not change the SSLND register while both the SPCR.MSTR and SPCR.SPE bits are 1.

SLNDL[2:0] Bits (SSL Negation Delay Setting)

The SLNDL[2:0] bits set an SSL negation delay value when the SPCMDm.SLNDEN bit is 1.
When using the RSPI in slave mode, set the SLNDL[2:0] bits to 000b.

44.2.12 RSPI Next-Access Delay Register (SPND)

Address(es): RSPI0.SPND 000D 010Eh, RSPI1.SPND 000D 014Eh, RSPI2.SPND 000D 030Eh

	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	SPNDL[2:0]		
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b2 to b0	SPNDL[2:0]	RSPI Next-Access Delay Setting	b2 b0 0 0 0: 1 RSPCK + 2 PCLK 0 0 1: 2 RSPCK + 2 PCLK 0 1 0: 3 RSPCK + 2 PCLK 0 1 1: 4 RSPCK + 2 PCLK 1 0 0: 5 RSPCK + 2 PCLK 1 0 1: 6 RSPCK + 2 PCLK 1 1 0: 7 RSPCK + 2 PCLK 1 1 1: 8 RSPCK + 2 PCLK	R/W
b7 to b3	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

SPND sets a non-active period (next-access delay) of the SSLxi signal after termination of a serial transfer when the SPCMDm.SPNDEN bit is 1. Do not change the SPND register while both the SPCR.MSTR and SPCR.SPE bits are 1.

SPNDL[2:0] Bits (RSPI Next-Access Delay Setting)

The SPNDL[2:0] bits set a next-access delay when the SPCMDm.SPNDEN bit is 1.
When using the RSPI in slave mode, set the SPNDL[2:0] bits to 000b.

44.2.13 RSPI Control Register 2 (SPCR2)

Address(es): RSPI0.SPCR2 000D 010Fh, RSPI1.SPCR2 000D 014Fh, RSPI2.SPCR2 000D 030Fh

	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	SCKASE	PTE	SPIIE	SPOE	SPPE
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	SPPE	Parity Enable*1	0: Does not add the parity bit to transmit data and does not check the parity bit of receive data 1: Adds the parity bit to transmit data and checks the parity bit of receive data	R/W
b1	SPOE	Parity Mode*1	0: Selects even parity for use in transmission and reception 1: Selects odd parity for use in transmission and reception	R/W
b2	SPIIE	Idle Interrupt Enable	0: Disables the generation of idle interrupt requests 1: Enables the generation of idle interrupt requests	R/W
b3	PTE	Parity Self-Diagnosis	0: Disables the self-diagnosis function of the parity circuit 1: Enables the self-diagnosis function of the parity circuit	R/W
b4	SCKASE	RSPCK Auto-Stop Function Enable*1	0: Disables the RSPCK auto-stop function 1: Enables the RSPCK auto-stop function	R/W
b7 to b5	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

Note 1. Do not change the SPPE, SPOE, and SCKASE bits while the SPCR.SPE bit is 1.

SPPE Bit (Parity Enable)

The SPPE bit enables or disables the parity function.

SPOE Bit (Parity Mode)

The SPOE bit specifies odd or even parity.

When even parity is set, parity bit addition is performed so that the total number of 1-bits in the transmit/receive character plus the parity bit is even. Similarly, when odd parity is set, parity bit addition is performed so that the total number of 1-bits in the transmit/receive character plus the parity bit is odd.

The SPOE bit is valid only when the SPPE bit is 1.

SPIIE Bit (Idle Interrupt Enable)

The SPIIE bit enables or disables the generation of idle interrupt requests when the RSPI being in the idle state is detected and the SPSR.IDLNF flag is set to 0.

PTE Bit (Parity Self-Diagnosis)

The PTE bit enables the self-diagnosis function of the parity circuit in order to check whether the parity function is operating correctly.

SCKASE Bit (RSPCK Auto-Stop Function Enable)

The SCKASE bit enables or disables the RSPCK auto-stop function. When this function is enabled, the RSPCK clock is stopped before an overrun error occurs when data is received in master mode. For details, refer to section 44.3.9.1, Overrun Error.

44.2.14 RSPI Command Register m (SPCMDm) (m = 0 to 7)

Address(es): RSPI0.SPCMD0 000D 0110h, RSPI0.SPCMD1 000D 0112h, RSPI0.SPCMD2 000D 0114h, RSPI0.SPCMD3 000D 0116h, RSPI0.SPCMD4 000D 0118h, RSPI0.SPCMD5 000D 011Ah, RSPI0.SPCMD6 000D 011Ch, RSPI0.SPCMD7 000D 011Eh, RSPI1.SPCMD0 000D 0150h, RSPI1.SPCMD1 000D 0152h, RSPI1.SPCMD2 000D 0154h, RSPI1.SPCMD3 000D 0156h, RSPI1.SPCMD4 000D 0158h, RSPI1.SPCMD5 000D 015Ah, RSPI1.SPCMD6 000D 015Ch, RSPI1.SPCMD7 000D 015Eh, RSPI2.SPCMD0 000D 0310h, RSPI2.SPCMD1 000D 0312h, RSPI2.SPCMD2 000D 0314h, RSPI2.SPCMD3 000D 0316h, RSPI2.SPCMD4 000D 0318h, RSPI2.SPCMD5 000D 031Ah, RSPI2.SPCMD6 000D 031Ch, RSPI2.SPCMD7 000D 031Eh

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	SCKDEN	SLNDEN	SPNDEN	LSBF	SPB[3:0]				SSLKP	SSLA[2:0]			BRDV[1:0]		CPOL	CPHA
Value after reset:	0	0	0	0	0	1	1	1	0	0	0	0	1	1	0	1

Bit	Symbol	Bit Name	Description	R/W
b0	CPHA	RSPCK Phase Setting	0: Data sampling on odd edge, data variation on even edge 1: Data variation on odd edge, data sampling on even edge	R/W
b1	CPOL	RSPCK Polarity Setting	0: RSPCK is low when idle 1: RSPCK is high when idle	R/W
b3, b2	BRDV[1:0]	Bit Rate Division Setting	b3 b2 0 0: These bits select the base bit rate 0 1: These bits select the base bit rate divided by 2 1 0: These bits select the base bit rate divided by 4 1 1: These bits select the base bit rate divided by 8	R/W
b6 to b4	SSLA[2:0]	SSL Signal Assertion Setting	b6 b4 0 0 0: SSL0 0 0 1: SSL1 0 1 0: SSL2 0 1 1: SSL3 1 x x: Setting prohibited	R/W
b7	SSLKP	SSL Signal Level Keeping	0: Negates all SSL signals upon completion of transfer 1: Keeps the SSL signal level from the end of transfer until the beginning of the next access (burst transfer)	R/W
b11 to b8	SPB[3:0]	RSPI Data Length Setting	b11 b8 0100 to 0111: 8 bits 1 0 0 0: 9 bits 1 0 0 1: 10 bits 1 0 1 0: 11 bits 1 0 1 1: 12 bits 1 1 0 0: 13 bits 1 1 0 1: 14 bits 1 1 1 0: 15 bits 1 1 1 1: 16 bits 0 0 0 0: 20 bits 0 0 0 1: 24 bits 0010, 0011: 32 bits	R/W
b12	LSBF	RSPI LSB First	0: MSB first 1: LSB first	R/W
b13	SPNDEN	RSPI Next-Access Delay Enable	0: A next-access delay of 1 RSPCK + 2 PCLK 1: A next-access delay is equal to the setting of the RSPI next-access delay register (SPND)	R/W
b14	SLNDEN	SSL Negation Delay Setting Enable	0: An SSL negation delay of 1 RSPCK 1: An SSL negation delay is equal to the setting of the RSPI slave select negation delay register (SSLND)	R/W
b15	SCKDEN	RSPCK Delay Setting Enable	0: An RSPCK delay of 1 RSPCK 1: An RSPCK delay is equal to the setting of the RSPI clock delay register (SPCKD)	R/W

x: Don’t care

The SPCMDm register is used to set a transfer format for the RSPI in master mode. Each channel has eight RSPI command registers (SPCMD0 to SPCMD7). Some of the bits in the SPCMD0 register is used to set a transfer mode for the RSPI in slave mode. The RSPI in master mode sequentially references the SPCMDm register according to the settings in the SPSCR.SPSLN[2:0] bits, and executes the serial transfer that is set in the referenced SPCMDm register. SPCMDm register should be set while the transmit buffer is empty (data for the next transfer is not set) and before setting of the data that is to be transmitted when that SPCMDm register is referenced.
An SPCMDm register that is referenced by the RSPI in master mode can be checked by means of the SPSSR.SPCP[2:0] bits. Do not change the SPCMDm register while the SPCR.MSTR bit is 0 and the SPCR.SPE bit is 1.

CPHA Bit (RSPCK Phase Setting)

The CPHA bit sets an RSPCK phase of the RSPI in master mode or slave mode. Data communications between RSPI modules require the same RSPCK phase setting between the modules.

CPOL Bit (RSPCK Polarity Setting)

The CPOL bit sets an RSPCK polarity of the RSPI in master mode or slave mode. Data communications between RSPI modules require the same RSPCK polarity setting between the modules.

BRDV[1:0] Bits (Bit Rate Division Setting)

The BRDV[1:0] bits are used to determine the bit rate. A bit rate is determined by combinations of the settings in the BRDV[1:0] bits and the SPBR register (refer to section 44.2.8, RSPI Bit Rate Register (SPBR)). The settings in the SPBR register determine the base bit rate. The settings in the BRDV[1:0] bits are used to select a bit rate which is obtained by dividing the base bit rate by 1, 2, 4, or 8. In the SPCMDm register, different BRDV[1:0] bit settings can be specified. This enables execution of serial transfers at a different bit rate for each command.

SSLA[2:0] Bits (SSL Signal Assertion Setting)

The SSLA[2:0] bits control the SSLxi signal assertion when the RSPI performs serial transfers in master mode.
Setting the SSLA[2:0] bits controls the assertion for the SSLxi signal. When an SSLxi signal is asserted, its polarity is determined by the set value in the corresponding SSLP. When the SSLA[2:0] bits are set to 000b in multi-master mode, serial transfers are performed with all the SSL signals in the negated state (as the SSLx0 pin acts as input).
When using the RSPI in slave mode, set the SSLA[2:0] bits to 000b.

SSLKP Bit (SSL Signal Level Keeping)

When the RSPI in master mode performs a serial transfer, the SSLKP bit specifies whether the SSLxi signal level for the current command is to be kept or negated between the SSL negation timing associated with the current command and the SSL assertion timing associated with the next command.
Setting the SSLKP bit to 1 enables a burst transfer. For details, refer to section 44.3.11.1, Master Mode Operation (4) Burst Transfer.
When using the RSPI in slave mode, the SSLKP bit should be set to 0.

SPB[3:0] Bits (RSPI Data Length Setting)

The SPB[3:0] bits set a transfer data length for the RSPI in master mode or slave mode. When the SPDCR.SPBYT is 1, set the SPB[3:0] bits to 0100b (8 bits). When the SPBYT bit is 0 and the SPDCR.SPLW bit is 0, set the SPB[3:0] bits to 0100b (8 bits) to 1111b (16 bits).

LSBF Bit (RSPI LSB First)

The LSBF bit sets the data format of the RSPI in master mode or slave mode to MSB first or LSB first.

SPNDEN Bit (RSPI Next-Access Delay Enable)

The SPNDEN bit sets the period from the time the RSPI in master mode terminates a serial transfer and sets the SSLxi signal inactive until the RSPI enables the SSLxi signal assertion for the next access (next-access delay). If the SPNDEN bit is 0, the RSPI sets the next-access delay to 1 RSPCK + 2 PCLK. If the SPNDEN bit is 1, the RSPI inserts a next-access delay in compliance with the SPND setting.

When using the RSPI in slave mode, the SPNDEN bit should be set to 0.

SLNDEN Bit (SSL Negation Delay Setting Enable)

The SLNDEN bit sets the period from the time the RSPI in master mode stops RSPCK oscillation until the RSPI sets the SSLxi signal inactive (SSL negation delay). If the SLNDEN bit is 0, the RSPI sets the SSL negation delay to 1 RSPCK. If the SLNDEN bit is 1, the RSPI negates the SSL signal at an SSL negation delay in compliance with the SSLND setting.

When using the RSPI in slave mode, the SLNDEN bit should be set to 0.

SCKDEN Bit (RSPCK Delay Setting Enable)

The SCKDEN bit sets the period from the point when the RSPI in master mode activates the SSLxi signal until the RSPCK starts oscillation (RSPI clock delay). If the SCKDEN bit is 0, the RSPI sets the RSPCK delay to 1 RSPCK. If the SCKDEN bit is 1, the RSPI starts the oscillation of RSPCK at an RSPCK delay in compliance with the SPCKD setting.

When using the RSPI in slave mode, the SCKDEN bit should be set to 0.

44.2.15 RSPI Data Control Register 2 (SPDCR2)

Address(es): RSPI0.SPDCR2 000D 0120h, RSPI1.SPDCR2 000D 0160h, RSPI2.SPDCR2 000D 0320h

	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	BYSW
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	BYSW	Byte Swap	0: Byte swapping of SPDR data disabled 1: Byte swapping of SPDR data enabled	R/W
b7 to b1	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

This register is used to enable or disable byte swapping of transmit and receive data.
The SPCR.SPE bit should be 0 when rewriting this register.

BYSW Bit (Byte Swap)

On transmit, this bit specifies that data bytes written in the SPDR register will be swapped before being transmitted. On receive, this bit specifies that received bytes will be swapped before the data is transferred to the SPDR register. This bit setting is enabled when the SPDCR.SPBYT bit is 0.
When using byte swap, set the SPCMD.SPB[3:0] bits to 1111b (16 bits), 0010b (32 bits), or 0011b (32 bits). Also, set the SPCR2.SPPE bit to 0 (parity bit not added). For details, refer to sections 44.3.4.3 Byte Swap Transmission and 44.3.4.4 Byte Swap Reception.

44.3 Operation

In this section, the serial transfer period means a period from the beginning of driving valid data to the fetching of the final valid data.

44.3.1 Overview of RSPI Operations

The RSPI is capable of synchronous serial transfers in slave mode (SPI operation), single-master mode (SPI operation), multi-master mode (SPI operation), slave mode (clock synchronous operation), and master mode (clock synchronous operation). A particular mode of the RSPI can be selected by using the MSTR, MODFEN, and SPMS bits in SPCR. Table 44.5 lists the relationship between RSPI modes and SPCR settings, and a description of each mode.

Table 44.5 Relationship between RSPI Modes and SPCR Settings and Description of Each Mode

	SPI Operation			Clock Synchronous Operation	
Mode	**Slave**	**Single-Master**	**Multi-Master**	**Slave**	**Master**
MSTR bit setting	0	1	1	0	1
MODFEN bit setting	0 or 1	0	1	0	0
SPMS bit setting	0	0	0	1	1
RSPCKx signal	Input	Output	Output/Hi-Z*1	Input	Output
MOSIx signal	Input	Output	Output/Hi-Z*1	Input	Output
MISOx signal	Output/Hi-Z*2	Input	Input	Output	Input
SSLx0 signal	Input	Output	Input	Hi-Z*3	Hi-Z*3
SSLx1 to SSLx3 signals	Hi-Z*3	Output	Output/Hi-Z*1	Hi-Z*3	Hi-Z*3
SSL polarity change function	Supported	Supported	Supported	—	—
Transfer rate	Up to PCLK/4	Up to PCLK/2	Up to PCLK/2	Up to PCLK/4	Up to PCLK/2
Clock source	RSPCK input	On-chip baud rate generator	On-chip baud rate generator	RSPCK input	On-chip baud rate generator
Clock polarity	Two				
Clock phase	Two	Two	Two	One (CPHA = 1)	Two
First transfer bit	MSB/LSB				
Transfer data length	8 to 16, 20, 24, 32 bits				
Burst transfer	Possible (CPHA = 1)	Possible (CPHA = 0,1)	Possible (CPHA = 0,1)	—	—
RSPCK delay control	Not supported	Supported	Supported	Not supported	Supported
SSL negation delay control	Not supported	Supported	Supported	Not supported	Supported
Next-access delay control	Not supported	Supported	Supported	Not supported	Supported
Transfer activation method	SSL input active or RSPCK oscillation	Transmit buffer is written to when a transmit buffer empty interrupt request is generated or the SPTEF flag is 1	Transmit buffer is written to when a transmit buffer empty interrupt request is generated or the SPTEF flag is 1	RSPCK oscillation	Transmit buffer is written to when a transmit buffer empty interrupt request is generated or the SPTEF flag is 1
Sequence control	Not supported	Supported	Supported	Not supported	Supported
Transmit buffer empty detection	Supported				
Receive buffer full detection	Supported*4				
Overrun error detection	Supported*4	Supported*4, *6	Supported*4, *6	Supported*4	Supported*4, *6
Underrun error detection	Supported	Not supported	Not supported	Supported	Not supported
Parity error detection	Supported*4, *5				
Mode fault error detection	Supported (MODFEN = 1)	Not supported	Supported	Not supported	Not supported

Note 1. When SSLx0 is asserted by another master device, the pin becomes Hi-Z.
Note 2. When SSLx0 is negated or the SPCR.SPE bit is 0, the pin becomes Hi-Z.
Note 3. This function is not supported in this mode.
Note 4. When the SPCR.TXMD bit is 1, the detections of receiver buffer full, overrun error, and parity error are not performed.
Note 5. When the SPCR2.SPPE bit is 0, the detection of parity error is not performed.

Note 6. When the SPCR2.SCKASE bit is 1, the detection of overrun error is not performed.

44.3.2 Controlling RSPI Pins

The RSPI in single-master mode (SPI operation) or multi-master mode (SPI operation) determines MOSI signal values during the SSL negation period (including the SSL retention period during a burst transfer) according to MOIFE and MOIFV bit settings in SPPCR, as listed in Table 44.6.

Table 44.6 MOSI Signal Value Determination during SSL Negation Period

MOIFE Bit	MOIFV Bit	MOSIx Signal Value during SSL Negation Period
0	0, 1	Final data from previous transfer
1	0	Low
1	1	High

44.3.3 RSPI System Configuration Examples

44.3.3.1 Single Master/Single Slave (with This MCU Acting as Master)

Figure 44.5 shows a single-master/single-slave RSPI system configuration example when this MCU is used as a master. In the single-master/single-slave configuration, the SSLx0 to SSLx3 output of this MCU (master) are not used. The SSL input of the SPI slave is fixed to the low level, and the SPI slave is maintained in a select state.*1
This MCU (master) drives the RSPCKx and MOSIx. The SPI slave drives the MISO.

Note 1. In the transfer format corresponding to the case where the SPCMDm.CPHA bit is 0, there are slave devices for which the SSL signal cannot be fixed to the active level. In situations where the SSL signal cannot be fixed, the SSLxi output of this MCU should be connected to the SSL input of the slave device.

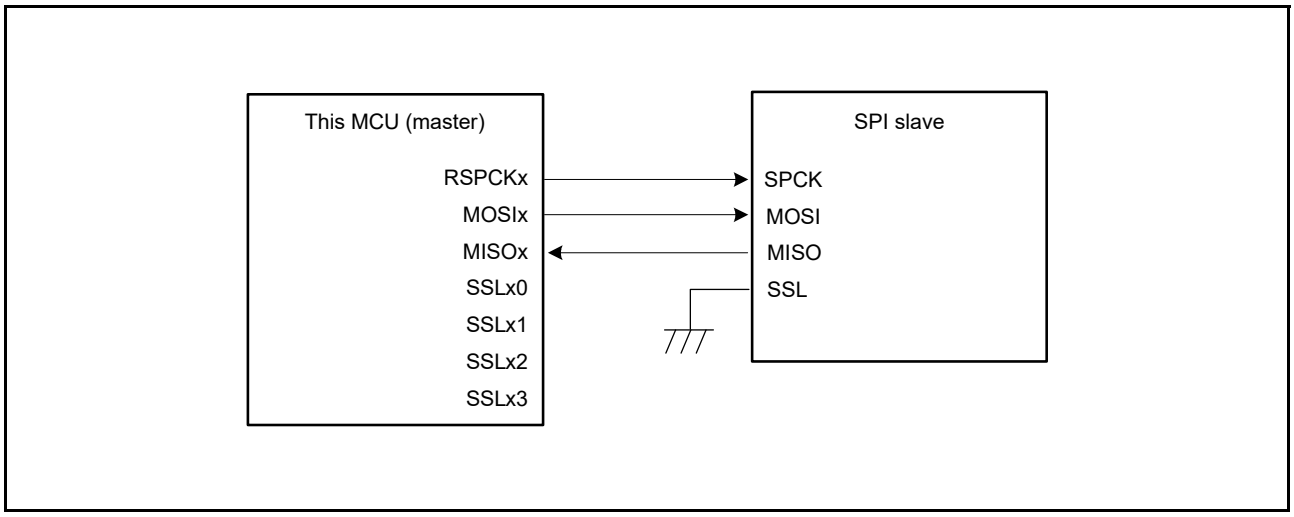


Figure 44.5 Single-Master/Single-Slave Configuration Example (This MCU = Master)

44.3.3.2 Single Master/Single Slave (with This MCU Acting as Slave)

Figure 44.6 shows a single-master/single-slave RSPI system configuration example when this MCU is used as a slave. When this MCU is to operate as a slave, the SSLx0 pin is used as SSL input. The SPI master drives the RSPCK and MOSI. This MCU (slave) drives the MISOx.*1

In the single-slave configuration in which the SPCMDm.CPHA bit is set to 1, the SSLx0 input of this MCU (slave) is fixed to the low level, this MCU (slave) is maintained in a select state, and in this manner it is possible to execute serial transfer (Figure 44.7).

Note 1. When SSLx0 is at the non-active level, the pin state is Hi-Z.

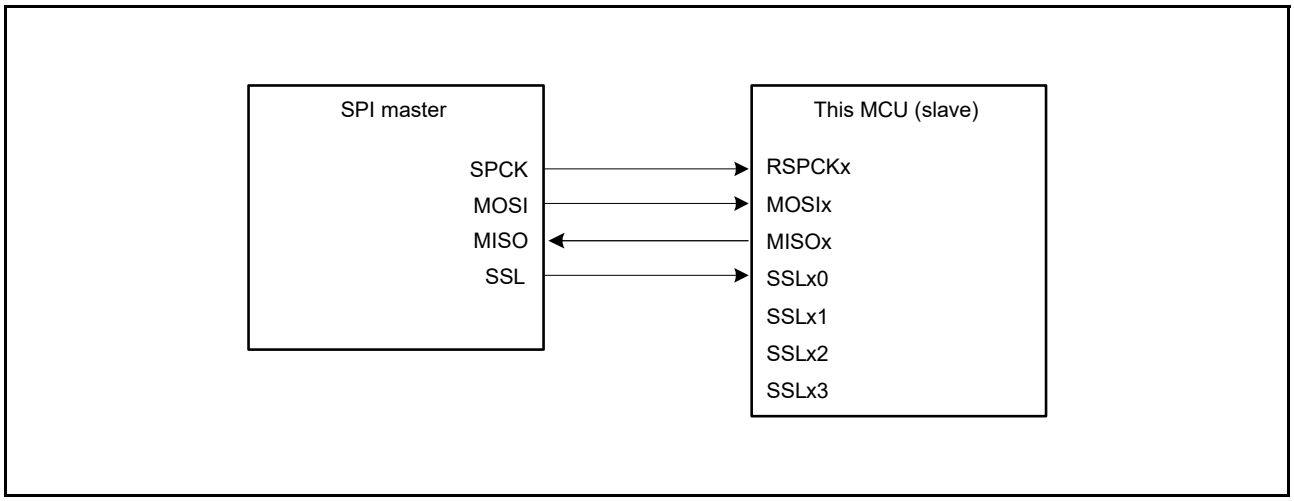


Figure 44.6 Single-Master/Single-Slave Configuration Example (This MCU = Slave, CPHA = 0)

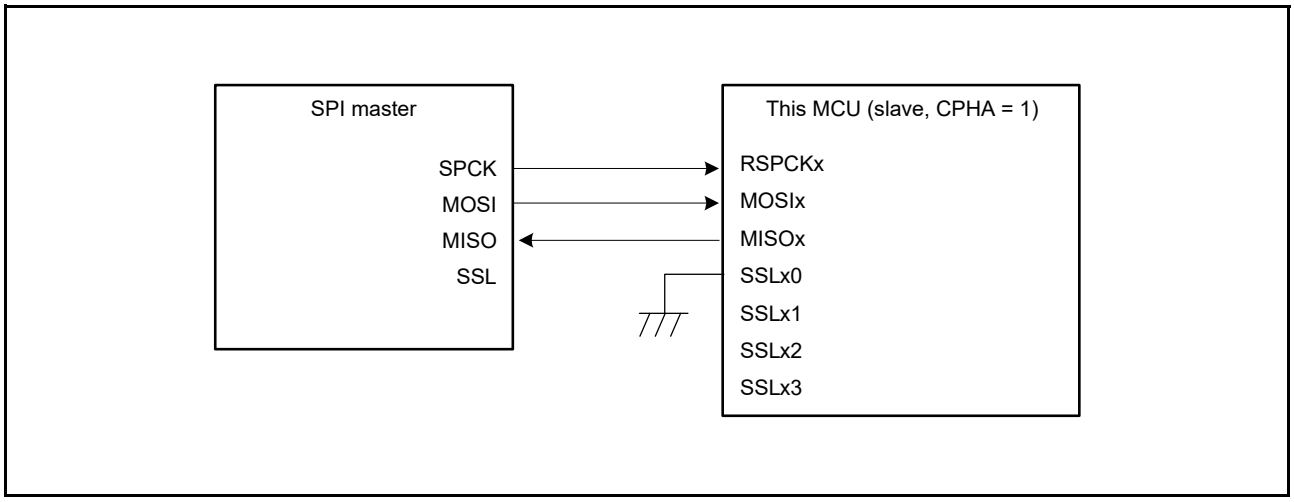


Figure 44.7 Single-Master/Single-Slave Configuration Example (This MCU = Slave, CPHA = 1)

44.3.3.3 Single Master/Multi-Slave (with This MCU Acting as Master)

Figure 44.8 shows a single-master/multi-slave RSPI system configuration example when this MCU is used as a master. In the example of Figure 44.8, the RSPI system is comprised of this MCU (master) and four slaves (SPI slave 0 to SPI slave 3).

The RSPCKx and MOSIx outputs of this MCU (master) are connected to the RSPCK and MOSI inputs of SPI slave 0 to SPI slave 3. The MISO outputs of SPI slave 0 to SPI slave 3 are all connected to the MISOx input of this MCU (master). SSLx0 to SSLx3 outputs of this MCU (master) are connected to the SSL inputs of SPI slave 0 to SPI slave 3, respectively.

This MCU (master) drives RSPCKx, MOSIx, and SSLx0 to SSLx3. Of the SPI slave 0 to SPI slave 3, the slave that receives low-level input into the SSL input drives MISO.

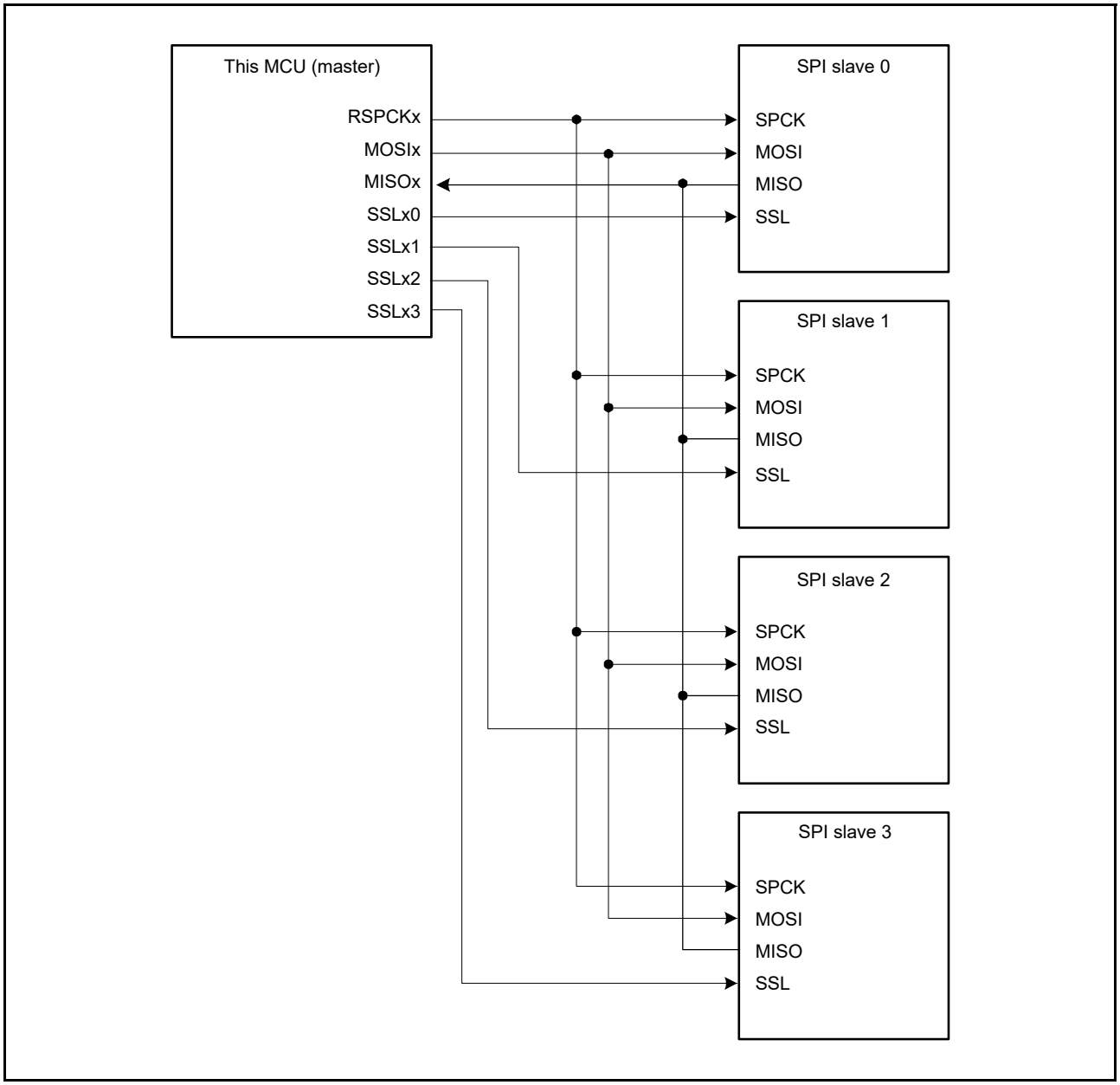


Figure 44.8 Single-Master/Multi-Slave Configuration Example (This MCU = Master)

44.3.3.4 Single Master/Multi-Slave (with This MCU Acting as Slave)

Figure 44.9 shows a single-master/multi-slave RSPI system configuration example when this MCU is used as a slave. In the example of Figure 44.9, the RSPI system is comprised of an SPI master and two MCUs (slave X and slave Y).
The SPCK and MOSI outputs of the SPI master are connected to the RSPCKx and MOSIx inputs of the MCUs (slave X and slave Y). The MISOx outputs of the MCUs (slave X and slave Y) are all connected to the MISO input of the SPI master. SSLX and SSLY outputs of the SPI master are connected to the SSLx0 inputs of the MCUs (slave X and slave Y), respectively.
The SPI master drives SPCK, MOSI, SSLX, and SSLY. Of the MCUs (slave X and slave Y), the slave that receives low-level input into the SSLx0 input drives MISOx.

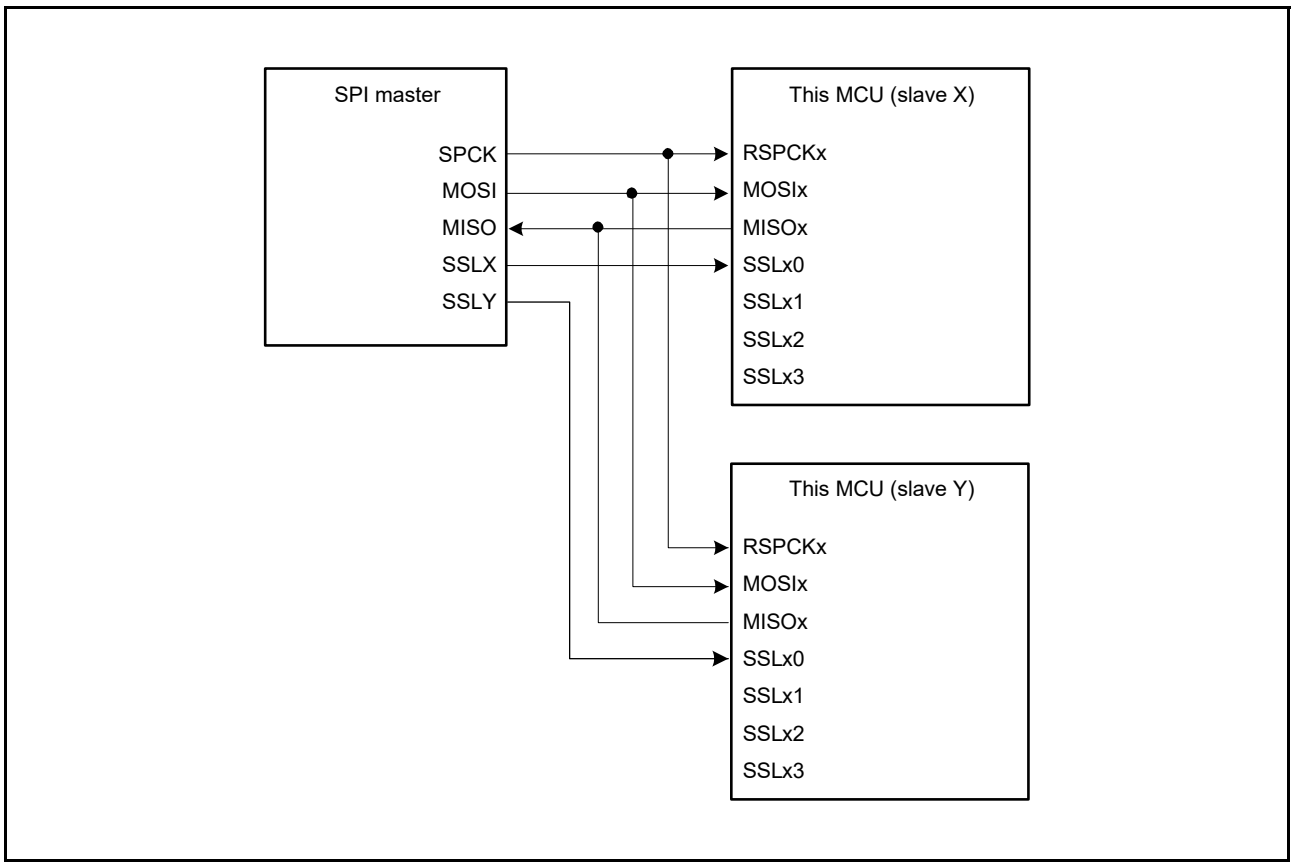


Figure 44.9 Single-Master/Multi-Slave Configuration Example (This MCU = Slave)

44.3.3.5 Multi-Master/Multi-Slave (with This MCU Acting as Master)

Figure 44.10 shows a multi-master/multi-slave RSPI system configuration example when this MCU is used as a master. In the example of Figure 44.10, the RSPI system is comprised of two MCUs (master X and master Y) and two SPI slaves (SPI slave 1 and SPI slave 2).

The RSPCKx and MOSIx outputs of the MCUs (master X and master Y) are connected to the RSPCK and MOSI inputs of SPI slaves 1 and 2. The MISO outputs of SPI slaves 1 and 2 are connected to the MISOx inputs of the MCUs (master X and master Y). Any generic port Y output from this MCU (master X) is connected to the SSLx0 input of this MCU (master Y). Any generic port X output of this MCU (master Y) is connected to the SSLx0 input of this MCU (master X). The SSLx1 and SSLx2 outputs of the MCUs (master X and master Y) are connected to the SSL inputs of the SPI slaves 1 and 2. In this configuration example, since the system can be comprised solely of SSLx0 input, and SSLx1 and SSLx2 outputs for slave connections, the SSLx3 output of this MCU is not required.

This MCU drives RSPCKx, MOSIx, SSLx1, and SSLx2 when the SSLx0 input level is high. When the SSLx0 input level is low, this MCU detects a mode fault error, sets RSPCKx, MOSIx, SSLx1, and SSLx2 to Hi-Z, and releases the RSPI bus right to the other master. Of the SPI slaves 1 and 2, the slave that receives low-level input into the SSL input drives MISO.

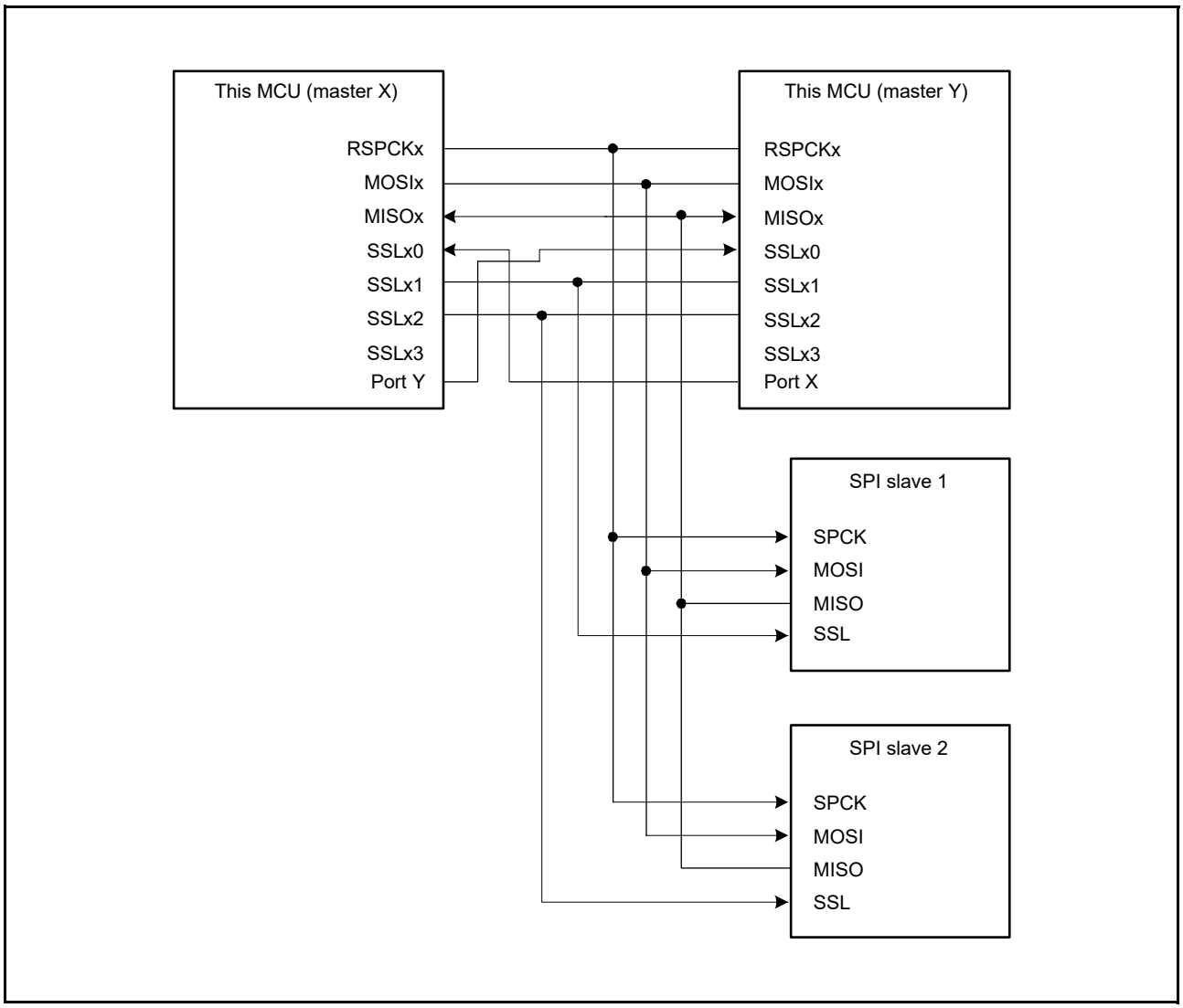


Figure 44.10 Multi-Master/Multi-Slave Configuration Example (This MCU = Master)

44.3.3.6 Master (Clock Synchronous Operation)/Slave (Clock Synchronous Operation) (with This MCU Acting as Master)

Figure 44.11 shows a master (clock synchronous operation)/slave (clock synchronous operation) RSPI system configuration example when this MCU is used as a master. In the master (clock synchronous operation)/slave (clock synchronous operation) configuration, SSLx0 to SSLx3 of this MCU (master) are not used.
This MCU (master) drives the RSPCKx and MOSIx. The SPI slave drives the MISO.

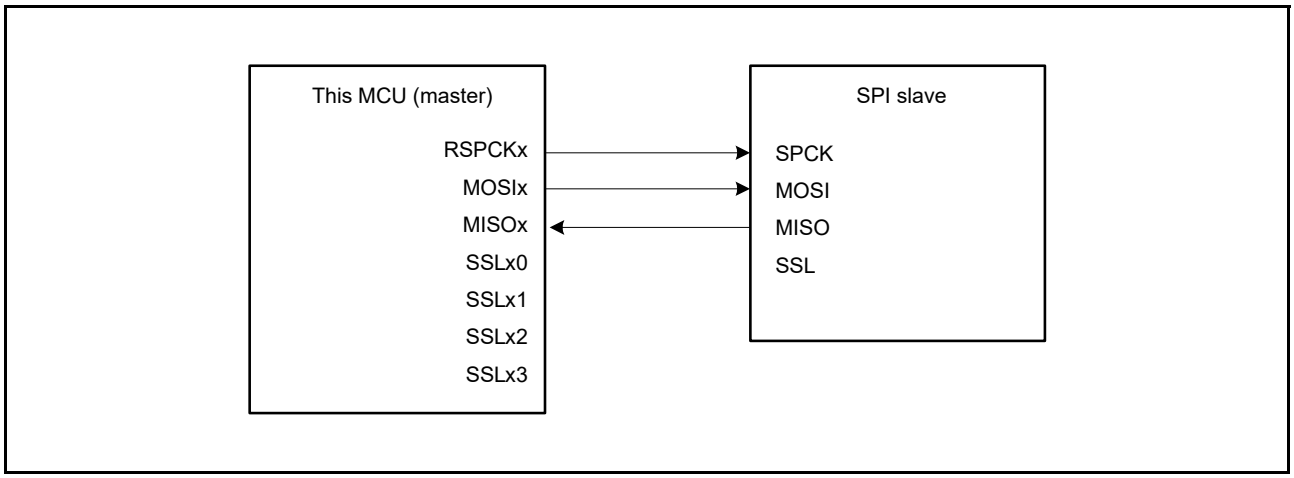


Figure 44.11 Master (Clock Synchronous Operation)/Slave (Clock Synchronous Operation) Configuration Example (This MCU = Master)

44.3.3.7 Master (Clock Synchronous Operation)/Slave (Clock Synchronous Operation) (with This MCU Acting as Slave)

Figure 44.12 shows a master (clock synchronous operation)/slave (clock synchronous operation) RSPI system configuration example when this MCU is used as a slave. When this MCU is to operate as a slave (clock synchronous operation), this MCU (slave) drives the MISOx and the SPI master drives the SPCK and MOSI. In addition, SSLx0 to SSLx3 of this MCU (slave) are not used.
Only in the single-slave configuration in which the SPCMDm.CPHA bit is set to 1, this MCU (slave) can execute serial transfer.

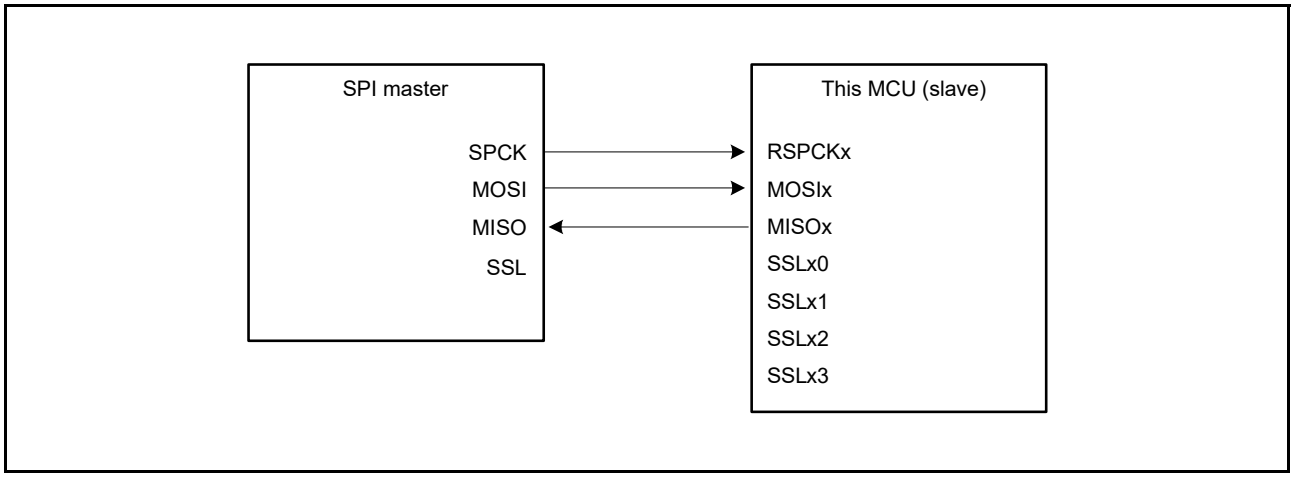


Figure 44.12 Master (Clock Synchronous Operation)/Slave (Clock Synchronous Operation) Configuration Example (This MCU = Slave, CPHA = 1)

44.3.4 Data Format

The RSPI's data format depends on the settings in RSPI command register m (SPCMDm) (m = 0 to 7) and the parity enable bit in RSPI control register 2 (SPCR2.SPPE) and RSPI data control register 2 (SPDCR2). Regardless of whether the MSB or LSB is first, the RSPI treats the range from the LSB bit in the RSPI data register (SPDR) to the selected data length as transfer data.
The format of one frame of data before or after transfer is shown below.

(a) With Parity Disabled

When parity is disabled, transmission or reception of data proceeds with the length in bits selected in the RSPI data length setting bits in RSPI command register m (SPCMDm.SPB[3:0]).

(b) With Parity Enabled

When parity is enabled, transmission or reception of data proceeds with the length in bits selected in the RSPI data length setting bits in RSPI command register m (SPCMDm.SPB[3:0]). In this case, however, the last bit is a parity bit.

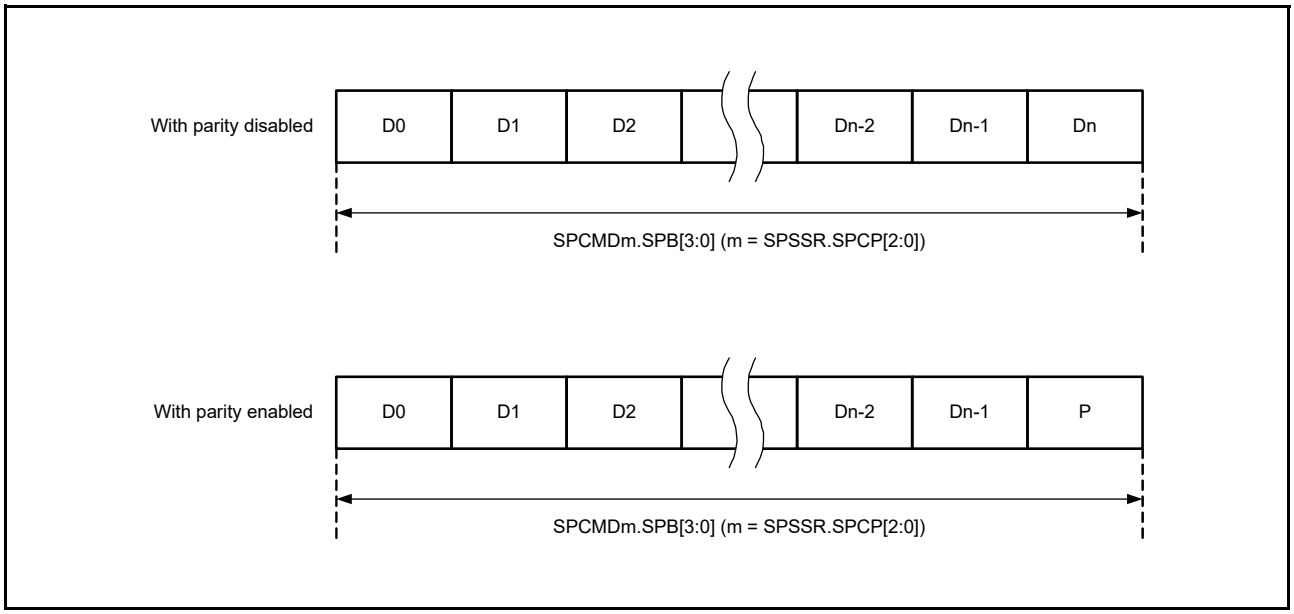


Figure 44.13 Outline of the Data Format (with Parity Disabled/Enabled)

44.3.4.1 When Parity is Disabled (SPCR2.SPPE = 0)

When parity is disabled, data for transmission are copied to the shift register with no prior processing. A description of the connection between the RSPI data register (SPDR) and the shift register in terms of the combination of MSB or LSB first and data length is given below.

(1) MSB First Transfer (32-Bit Data)

Figure 44.14 shows details of operations by the RSPI data register (SPDR) and the shift register in transfer with parity disabled, an RSPI data length of 32 bits, and MSB first selected.
In transmission, bits T31 to T00 from the current stage of the transmit buffer are copied to the shift register. Data for transmission are shifted out from the shift register in order from T31, through T30, and so on to T00.
In reception, received data are shifted in bit by bit through bit 0 of the shift register. When bits R31 to R00 have been collected after input of the required number of cycles of RSPCK, the value in the shift register is copied to the receive buffer.

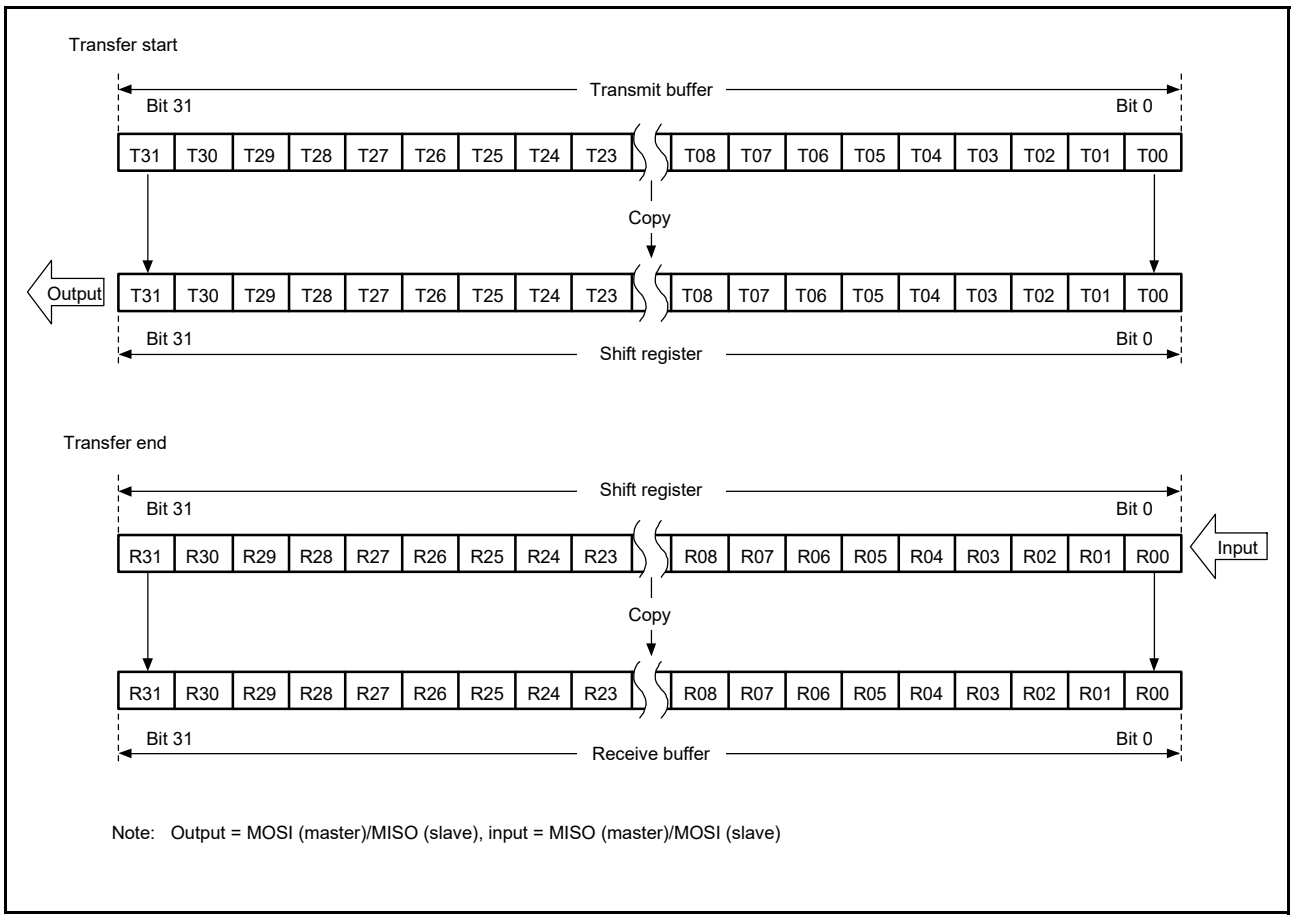


Figure 44.14 MSB First Transfer (32-Bit Data, Parity Disabled)

(2) MSB First Transfer (24-Bit Data)

Figure 44.15 shows details of operations by the RSPI data register (SPDR) and the shift register in transfer with parity disabled, 24 bits as the RSPI data length for an example that is not 32 bits, and MSB first selected.
In transmission, the lower 24 bits (T23 to T00) from the current stage of the transmit buffer are copied to the shift register. Data for transmission are shifted out from the shift register in order from T23, through T22, and so on to T00.
In reception, received data are shifted in bit by bit through bit 0 of the shift register. When bits R23 to R00 have been collected after input of the required number of cycles of RSPCK, the value in the shift register is copied to the receive buffer. At this time, the upper 8 bits of the transmit buffer are stored in the upper 8 bits of the receive buffer. Writing 0 to bits T31 to T24 at the time of transmission leads to 0 being set in the upper 8 bits of the receive buffer.

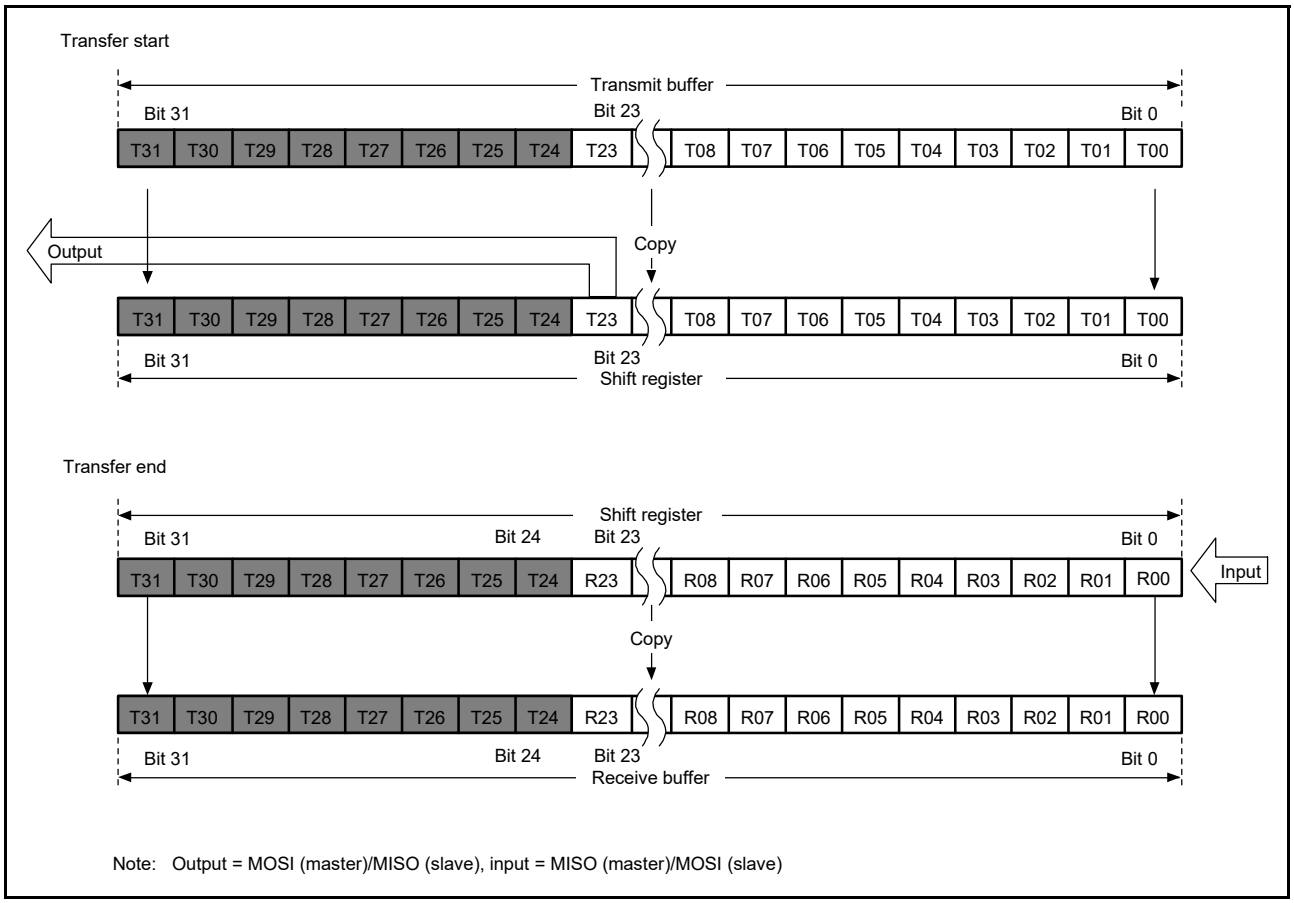


Figure 44.15 MSB First Transfer (24-Bit Data, Parity Disabled)

(3) LSB First Transfer (32-Bit Data)

Figure 44.16 shows details of operations by the RSPI data register (SPDR) and the shift register in transfer with parity disabled, an RSPI data length of 32 bits, and LSB first selected.

In transmission, bits T31 to T00 from the current stage of the transmit buffer are reordered bit by bit to obtain the order T00 to T31 for copying to the shift register. Data for transmission are shifted out from the shift register in order from T00, through T01, and so on to T31.

In reception, received data are shifted in bit by bit through bit 0 of the shift register. When bits R00 to R31 have been collected after input of the required number of cycles of RSPCK, the value in the shift register is copied to the receive buffer.

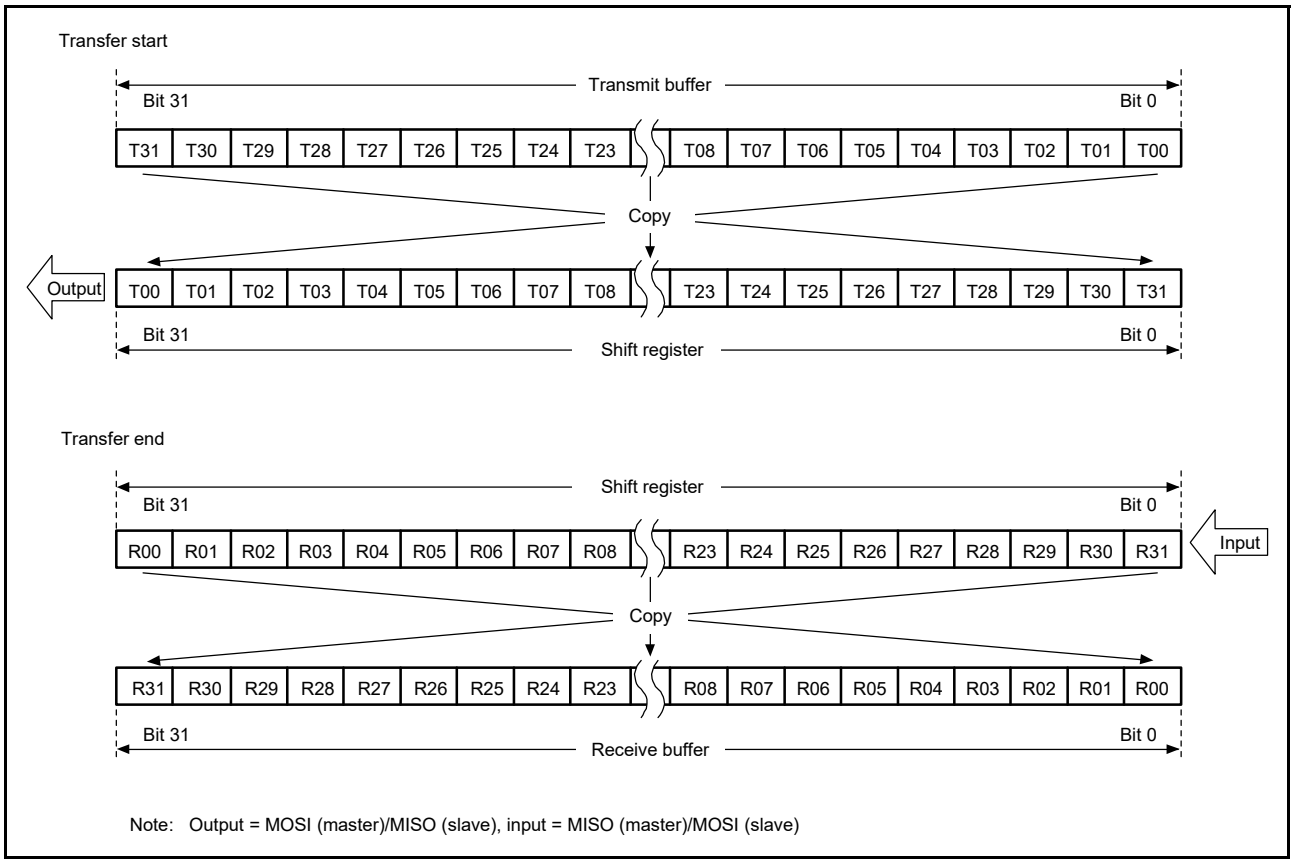


Figure 44.16 LSB First Transfer (32-Bit Data, Parity Disabled)

(4) LSB First Transfer (24-Bit Data)

Figure 44.17 shows details of operations by the RSPI data register (SPDR) and the shift register in transfer with parity disabled, 24 bits as the RSPI data length for an example that is not 32 bits, and LSB first selected.

In transmission, the lower 24 bits (T23 to T00) from the current stage of the transmit buffer are reordered bit by bit to obtain the order T00 to T23 for copying to the shift register. Data for transmission are shifted out from the shift register in order from T00, through T01, and so on to T23.

In reception, received data are shifted in bit by bit through bit 8 of the shift register. When bits R00 to R23 have been collected after input of the required number of cycles of RSPCK, the value in the shift register is copied to the receive buffer. At this time, the upper 8 bits of the transmit buffer are stored in the upper 8 bits of the receive buffer. Writing 0 to bits T31 to T24 at the time of transmission leads to 0 being set in the upper 8 bits of the receive buffer.

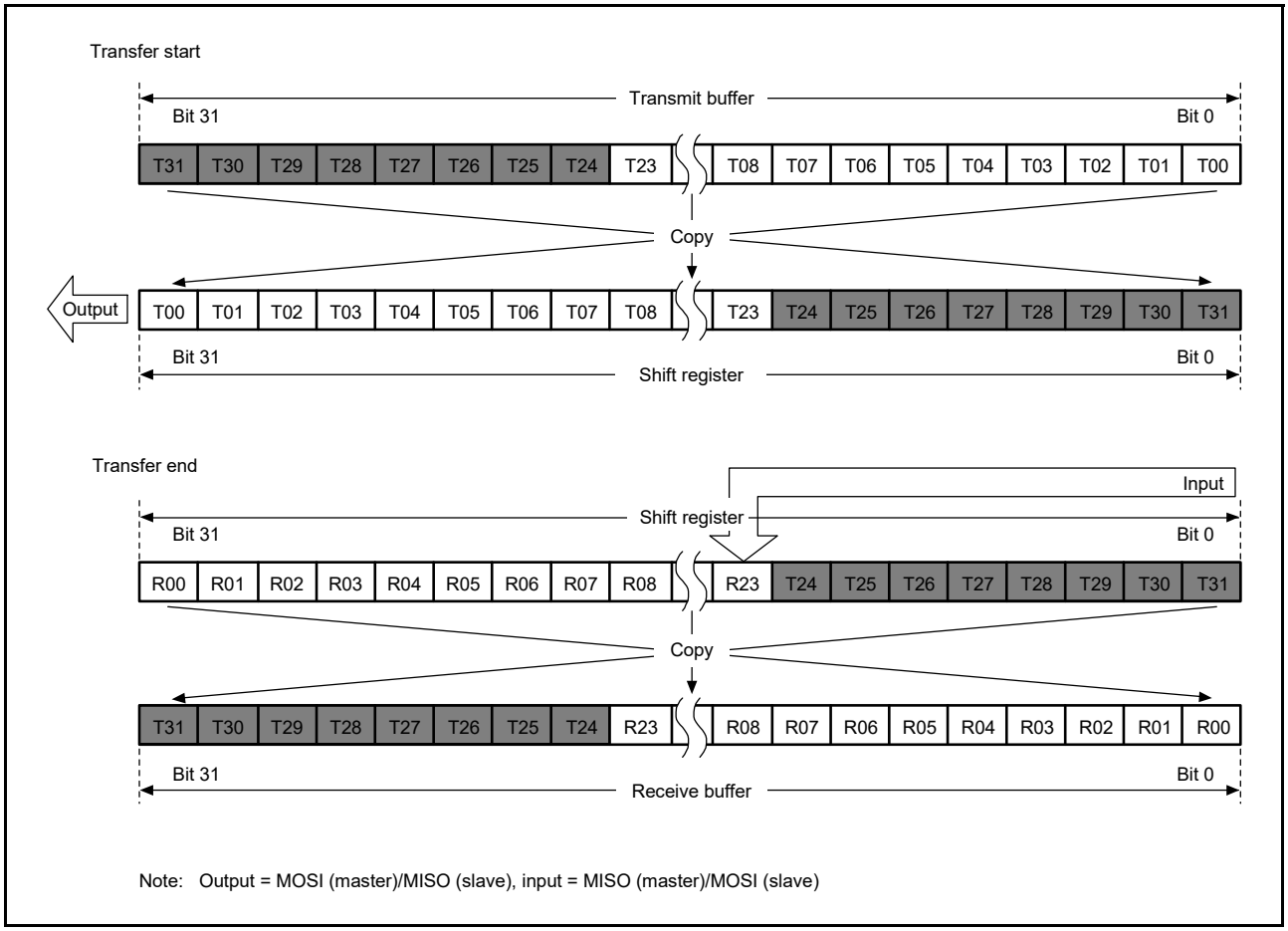


Figure 44.17 LSB First Transfer (24-Bit Data, Parity Disabled)

44.3.4.2 When Parity is Enabled (SPCR2.SPPE = 1)

When parity is enabled, the lowest-order bit of the data for transmission becomes a parity bit. Hardware calculates the value of the parity bit.

(1) MSB First Transfer (32-Bit Data)

Figure 44.18 shows details of operations by the RSPI data register (SPDR) and the shift register in transfer with parity enabled, an RSPI data length of 32 bits, and MSB first selected.

In transmission, the value of the parity bit (P) is calculated from bits T31 to T01. This replaces the final bit, T00, and the whole is copied to the shift register. Data are transmitted in the order T31, T30, …, T01, and P.

In reception, received data are shifted in bit by bit through bit 0 of the shift register. When bits R31 to P have been collected after input of the required number of cycles of RSPCK, the value in the shift register is copied to the receive buffer. On copying of data to the shift register, the data from R31 to P are checked by judging the parity.

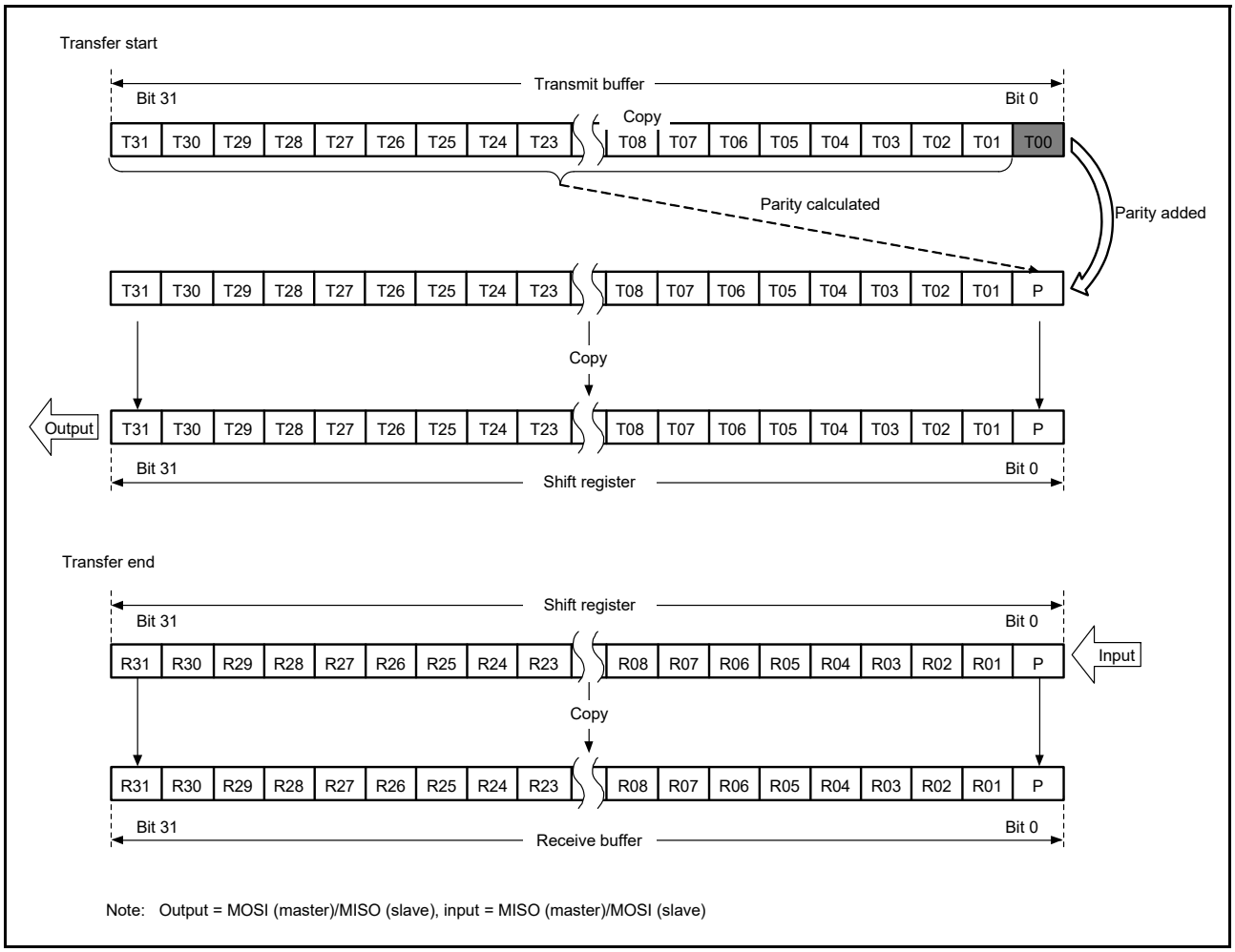


Figure 44.18 MSB First Transfer (32-Bit Data, Parity Enabled)

(2) MSB First Transfer (24-Bit Data)

Figure 44.19 shows details of operations by the RSPI data register (SPDR) and the shift register in transfer with parity enabled, 24 bits as the RSPI data length for an example that is not 32 bits, and MSB first selected.
In transmission, the value of the parity bit (P) is calculated from bits T23 to T01. This replaces the final bit, T00, and the whole is copied to the shift register. Data are transmitted in the order T23, T22, …, T01, and P.
In reception, received data are shifted in bit by bit through bit 0 of the shift register. When bits R23 to P have been collected after input of the required number of cycles of RSPCK, the value in the shift register is copied to the receive buffer. On copying of data to the shift register, the data from R23 to P are checked by judging the parity. At this time, the upper 8 bits of the transmit buffer are stored in the upper 8 bits of the receive buffer. Writing 0 to bits T31 to T24 at the time of transmission leads to 0 being set in the upper 8 bits of the receive buffer.

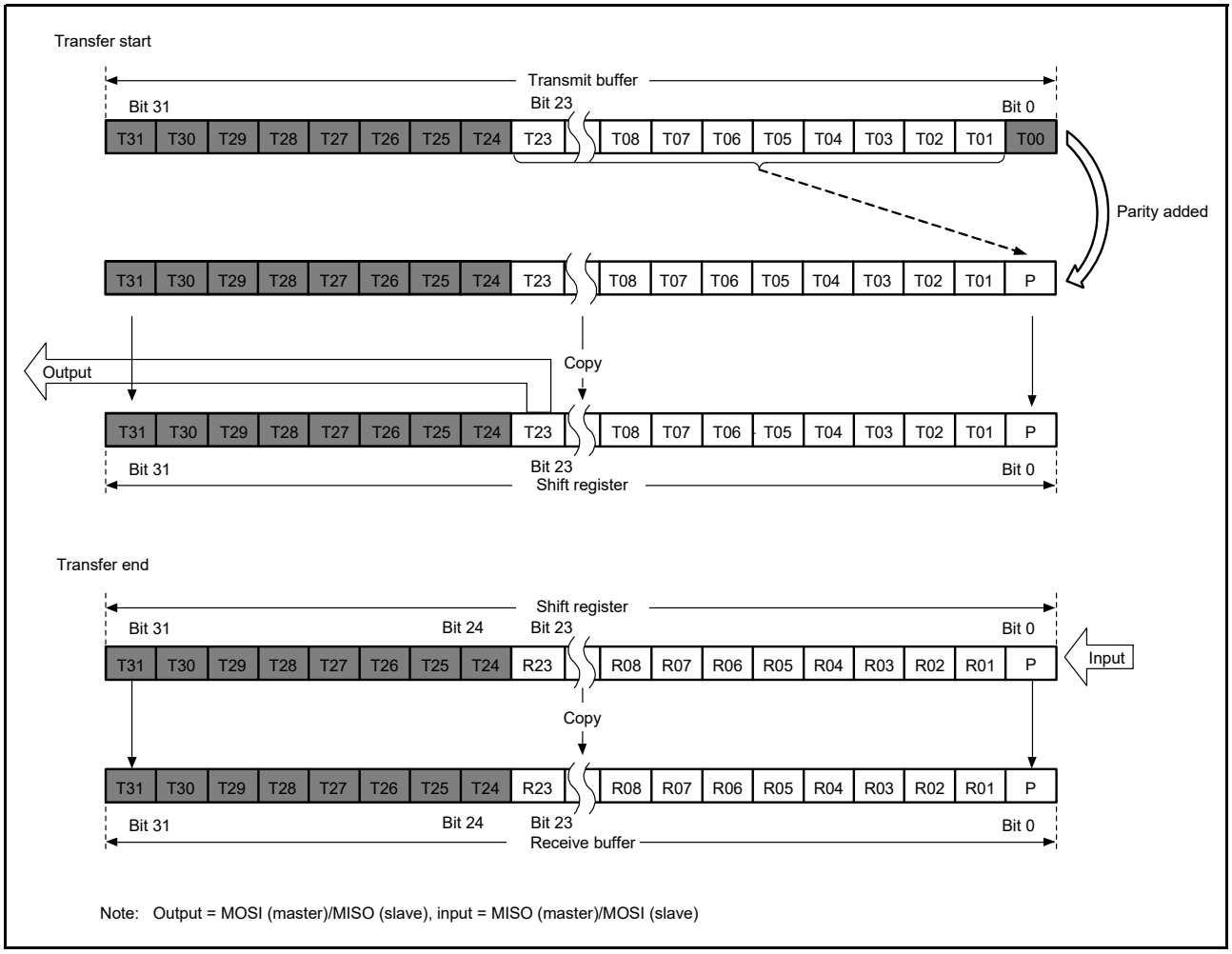


Figure 44.19 MSB First Transfer (24-Bit Data, Parity Enabled)

(3) LSB First Transfer (32-Bit Data)

Figure 44.20 shows details of operations by the RSPI data register (SPDR) and the shift register in transfer with parity enabled, an RSPI data length of 32 bits, and LSB first selected.

In transmission, the value of the parity bit (P) is calculated from bits T30 to T00. This replaces the final bit, T31, and the whole is copied to the shift register. Data are transmitted in the order T00, T01, …, T30, and P.

In reception, received data are shifted in bit by bit through bit 0 of the shift register. When bits R00 to P have been collected after input of the required number of cycles of RSPCK, the value in the shift register is copied to the receive buffer. On copying of data to the shift register, the data from R00 to P are checked by judging the parity.

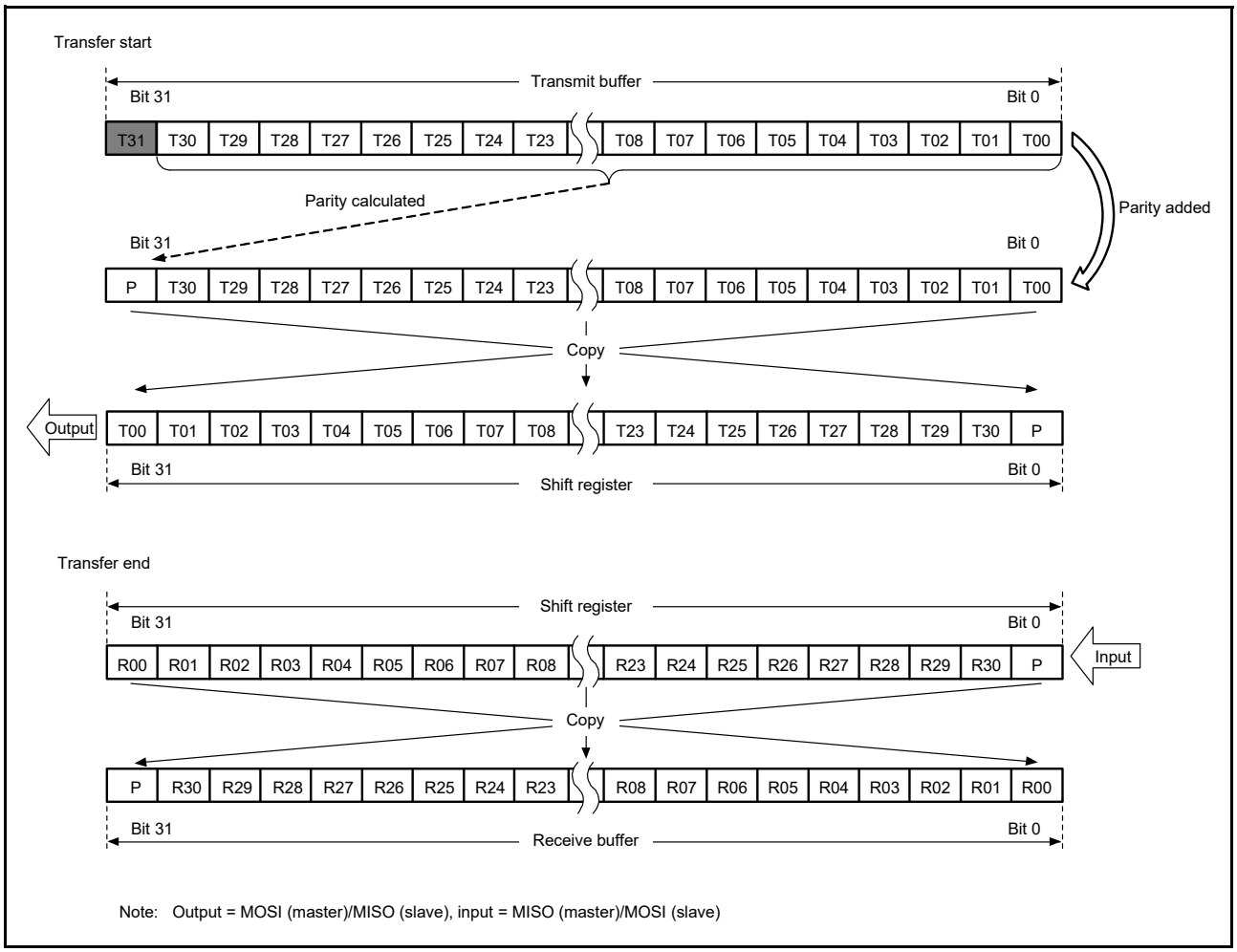


Figure 44.20 LSB First Transfer (32-Bit Data, Parity Enabled)

(4) LSB First Transfer (24-Bit Data)

Figure 44.21 shows details of operations by the RSPI data register (SPDR) and the shift register in transfer with parity enabled, 24 bits as the RSPI data length for an example that is not 32 bits, and LSB first selected.
In transmission, the value of the parity bit (P) is calculated from bits T22 to T00. This replaces the final bit, T23, and the whole is copied to the shift register. Data are transmitted in the order T00, T01, …, T22, and P.
In reception, received data are shifted in bit by bit through bit 8 of the shift register. When bits R00 to P have been collected after input of the required number of cycles of RSPCK, the value in the shift register is copied to the receive buffer. On copying of data to the shift register, the data from R00 to P are checked by judging the parity. At this time, the upper 8 bits of the transmit buffer are stored in the upper 8 bits of the receive buffer. Writing 0 to bits T31 to T24 at the time of transmission leads to 0 being set in the upper 8 bits of the receive buffer.

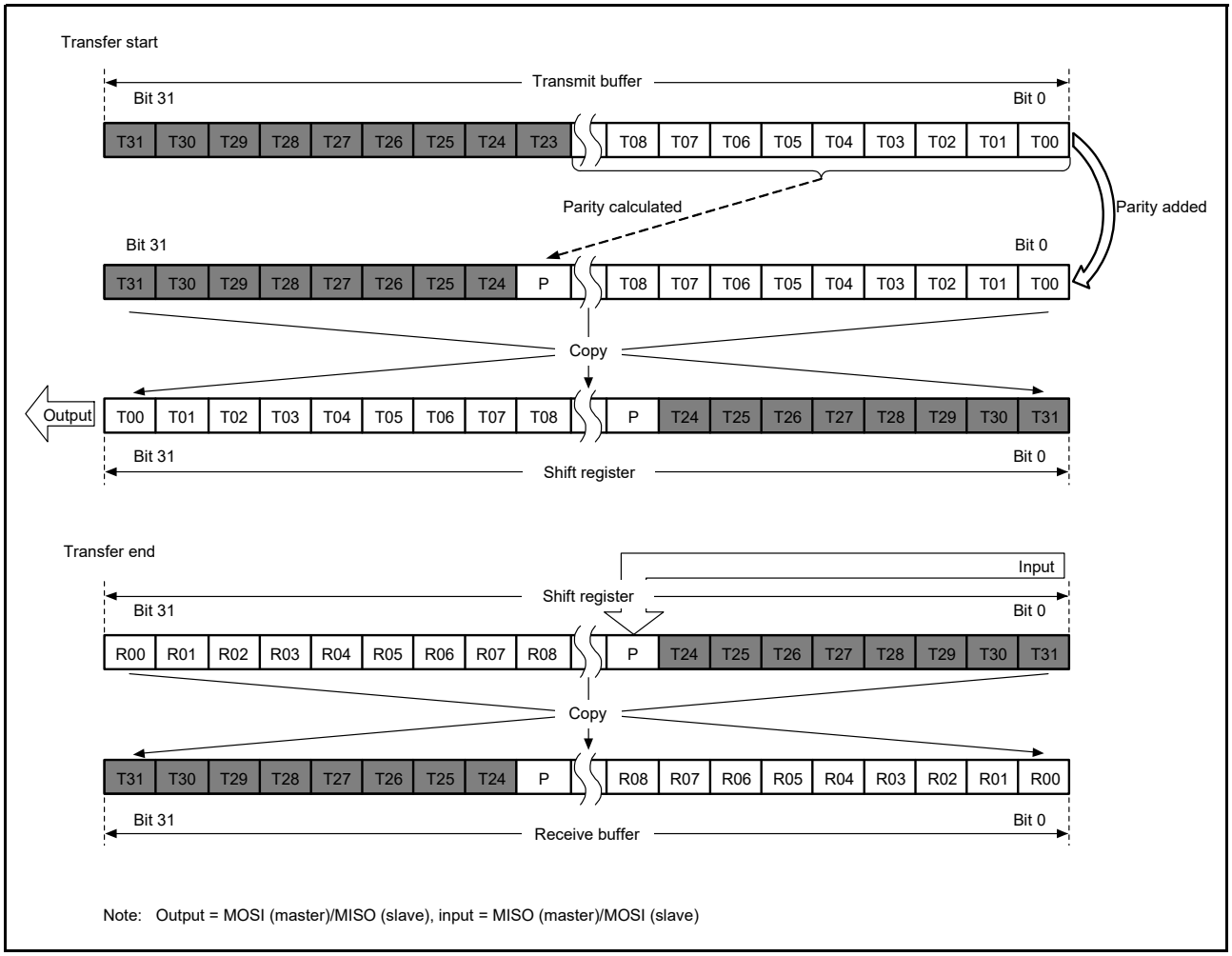


Figure 44.21 LSB First Transfer (24-Bit Data, Parity Enabled)

44.3.4.3 Byte Swap Transmission

When the SPDCR2.BYSW bit is 1 (byte swapping of SPDR data enabled), data bytes written in the transmit buffer (SPDR) will be swapped (in 8 bit units) when the data is transferred to the shift register. Figure 44.22 shows data transfer between the SPDR register and the shift register when data length is 32 bits.

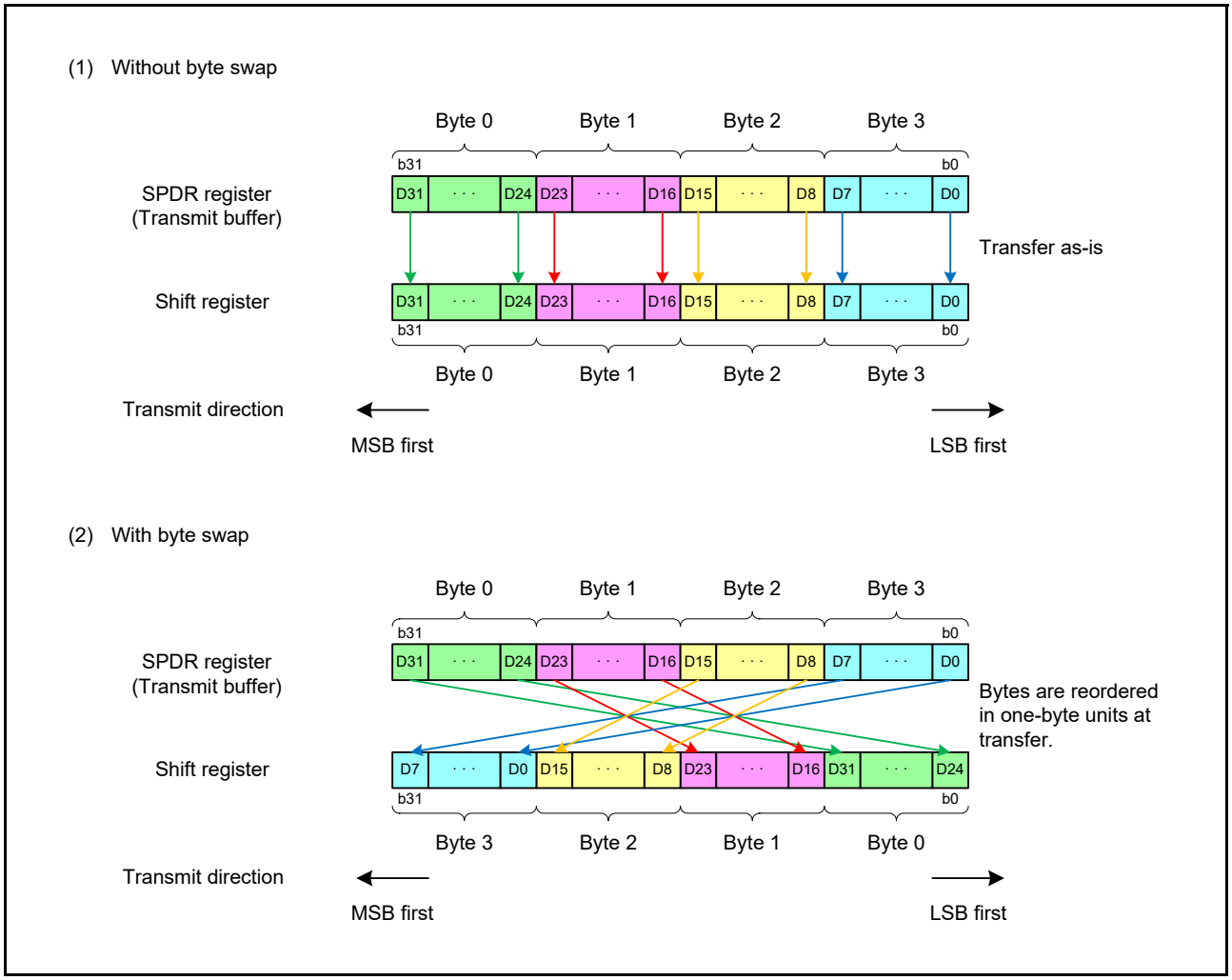


Figure 44.22 Transmit Data Handling in MSB/LSB First with Byte Swapping Enabled/Disabled

44.3.4.4 Byte Swap Reception

When the SPDCR2.BYSW bit is 1 (byte swapping of SPDR data enabled), data bytes in the shift register will be swapped (in 8 bit units) when the data is transferred to the receive buffer (SPDR). Figure 44.23 shows data transfer between the shift register and the SPDR register when data length is 32 bits.

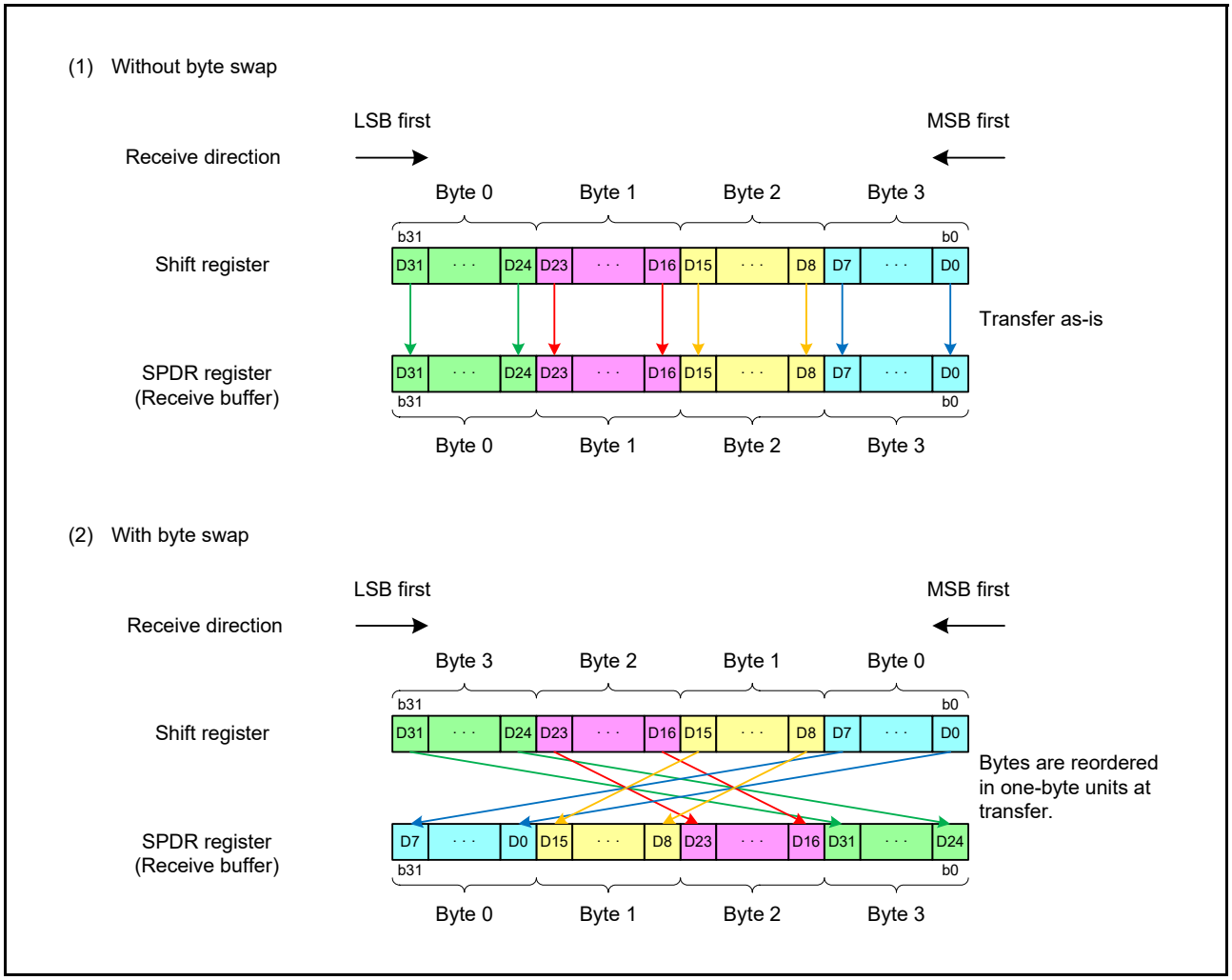


Figure 44.23 Receive Data Handling in MSB/LSB First with Byte Swapping Enabled/Disabled

44.3.5 Transfer Format

44.3.5.1 CPHA = 0

Figure 44.24 shows a sample transfer format for the serial transfer of 8-bit data when the SPCMDm.CPHA bit is 0. Note that clock synchronous operation (the SPCR.SPMS bit is 1) should not performed when the RSPI operates in slave mode (SPCR.MSTR = 0) and the CPHA bit is 0. In Figure 44.24, RSPCKx (CPOL = 0) indicates the RSPCKx signal waveform when the SPCMDm.CPOL bit is 0; RSPCKx (CPOL = 1) indicates the RSPCKx signal waveform when the CPOL bit is 1. The sampling timing represents the timing at which the RSPI fetches serial transfer data into the shift register. The I/O directions of the signals depend on the RSPI settings. For details, refer to section 44.3.2, Controlling RSPI Pins.

When the SPCMDm.CPHA bit is 0, the driving of valid data to the MOSIx and MISOx signals commences at an SSLxi signal assertion timing. The first RSPCKx signal change timing that occurs after the SSLxi signal assertion becomes the first transfer data fetch timing. After this timing, data is sampled at every 1 RSPCK cycle. The change timing for MOSIx and MISOx signals is 1/2 RSPCK cycles after the transfer data fetch timing. The CPOL bit setting does not affect the RSPCKx signal operation timing; it only affects the signal polarity.

t1 denotes a period from an SSLxi signal assertion to RSPCKx oscillation (RSPCK delay). t2 denotes a period from the termination of RSPCKx oscillation to an SSLxi signal negation (SSL negation delay). t3 denotes a period in which SSLxi signal assertion is suppressed for the next transfer after the end of serial transfer (next-access delay). t1, t2, and t3 are controlled by a master device running on the RSPI system. For a description of t1, t2, and t3 when the RSPI of this MCU is in master mode, refer to section 44.3.11.1, Master Mode Operation.

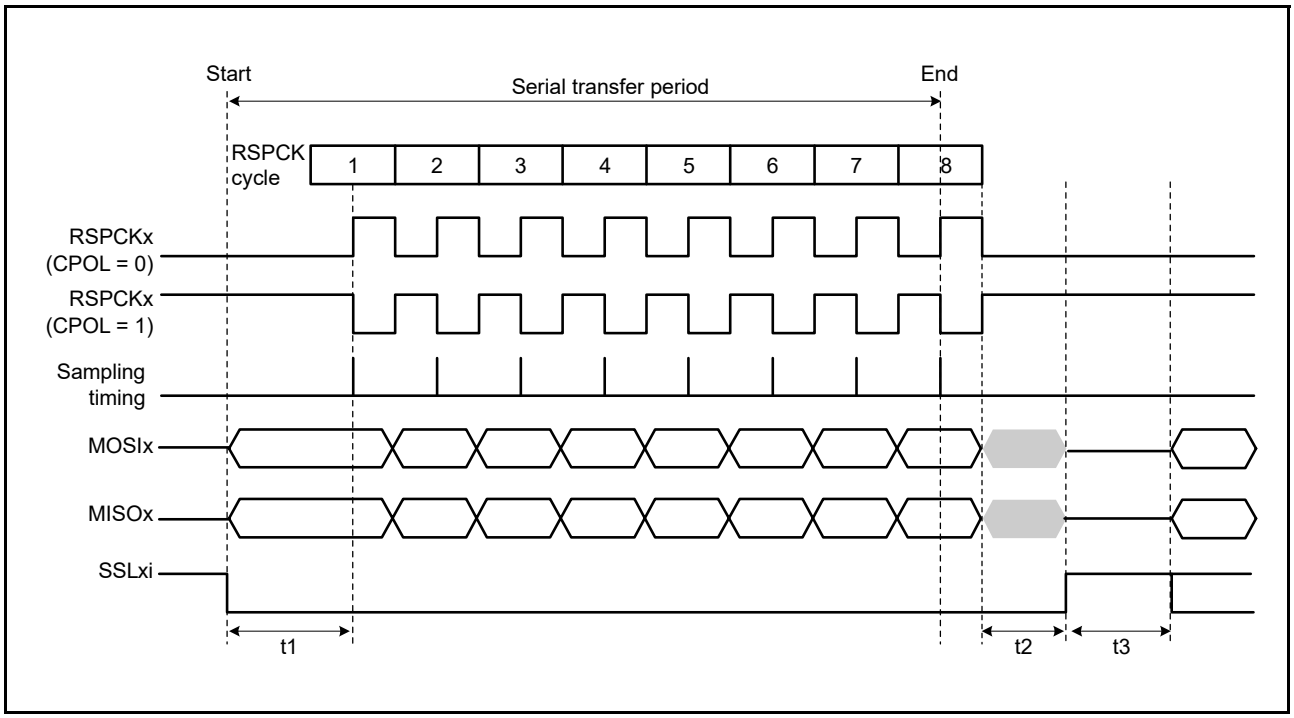


Figure 44.24 RSPI Transfer Format (CPHA = 0)

44.3.5.2 CPHA = 1

Figure 44.25 shows a sample transfer format for the serial transfer of 8-bit data when the SPCMDm.CPHA bit is 1. However, when the SPCR.SPMS bit is 1, the SSLxi signals are not used, and only the three signals RSPCKx, MOSIx, and MISOx handle communications. In Figure 44.25, RSPCK (CPOL = 0) indicates the RSPCKx signal waveform when the SPCMDm.CPOL bit is 0; RSPCKx (CPOL = 1) indicates the RSPCKx signal waveform when the CPOL bit is 1. The sampling timing represents the timing at which the RSPI fetches serial transfer data into the shift register. The I/O directions of the signals depend on the RSPI mode (master or slave). For details, refer to section 44.3.2, Controlling RSPI Pins.

When the SPCMDm.CPHA bit is 1, the driving of invalid data to the MISOx signal commences at an SSLxi signal assertion timing. The output of valid data to the MOSIx and MISOx signals commences at the first RSPCKx signal change timing that occurs after the SSLxi signal assertion. After this timing, data is updated at every 1 RSPCK cycle. The transfer data fetch timing is 1/2 RSPCK cycles after the data update timing. The SPCMDm.CPOL bit setting does not affect the RSPCKx signal operation timing; it only affects the signal polarity.

t1, t2, and t3 are the same as those in the case of CPHA = 0. For a description of t1, t2, and t3 when the RSPI of this MCU is in master mode, refer to section 44.3.11.1, Master Mode Operation.

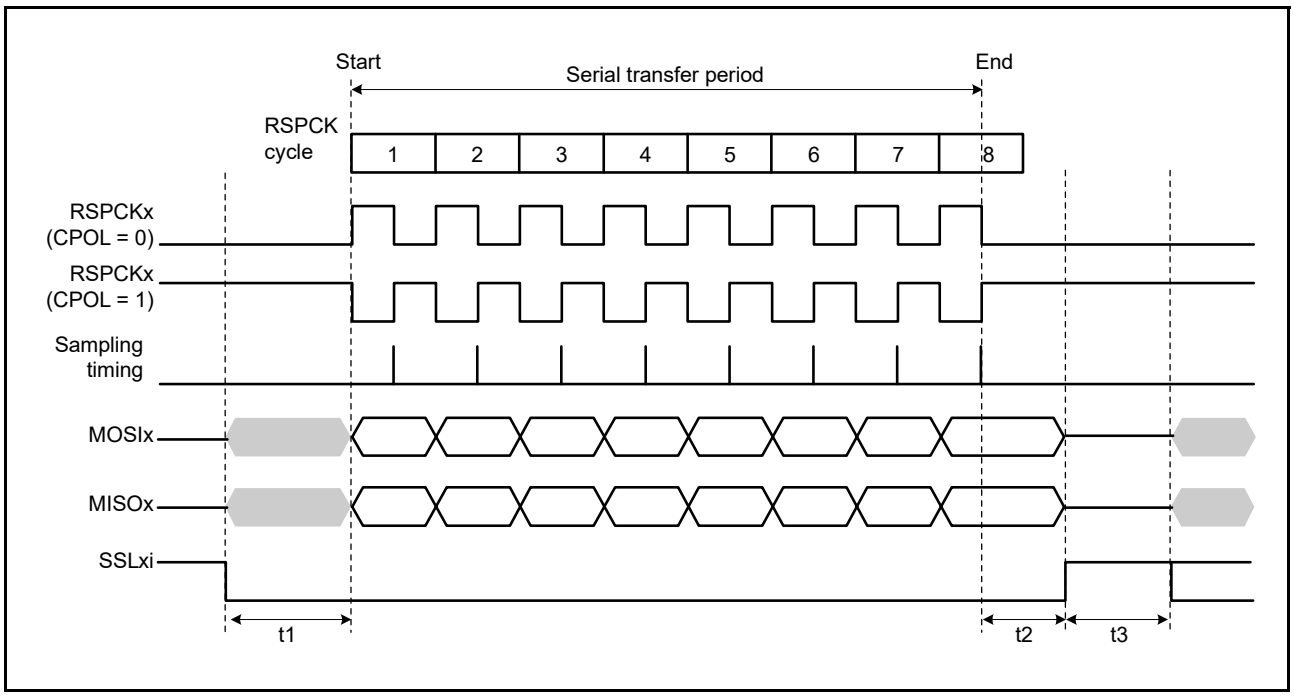


Figure 44.25 RSPI Transfer Format (CPHA = 1)

44.3.6 Communications Operating Mode

Full-duplex communications or transmit-only simplex communications can be selected by the SPCR.TXMD bit. ‘SPDR access’ shown in Figure 44.26 and Figure 44.27 indicate the condition of access to the SPDR register, where ‘W’ denotes a write cycle.

44.3.6.1 Full-Duplex Communications (SPCR.TXMD = 0)

Figure 44.26 shows an example of operation when the SPCR.TXMD bit is set to 0. In the example in Figure 44.26, the RSPI performs an 8-bit serial transfer in which the SPDCR.SPFC[1:0] bits are 00b, the SPCMDm.CPHA bit is 1, and the SPCMDm.CPOL bit is 0. The numbers given under the RSPCKx waveform represent the number of RSPCK cycles (i.e., the number of transferred bits).

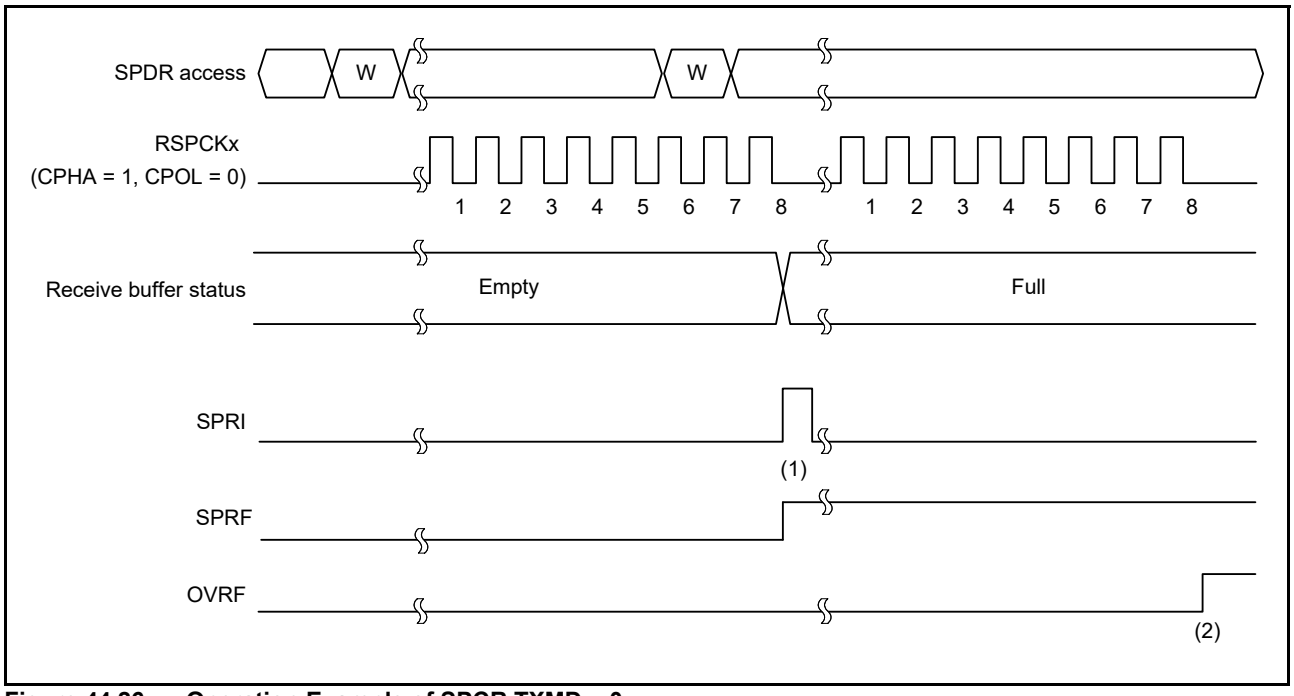


Figure 44.26 Operation Example of SPCR.TXMD = 0

The operation of the flags at timings shown in steps (1) and (2) in the figure is described below.

(1) When a serial transfer ends with the receive buffer of the SPDR register empty, the RSPI generates a receive buffer full interrupt request (SPRI) (sets the SPSR.SPRF flag to 1) and copies the received data in the shift register to the receive buffer.

(2) When a serial transfer ends with the receive buffer of the SPDR register holding data that was received in the previous serial transfer, the RSPI sets the SPSR.OVRF flag to 1 and discards the received data in the shift register.

When full-duplex communications (SPCR.TXMD = 0) is selected, reception occurs simultaneously with transmit operations. As such, the SPRF and OVRF flags in the SPSR register become 1 at the timing described in (1) and (2), respectively, according to the state of the receive buffer.

44.3.6.2 Transmit-only Simplex Communications (SPCR.TXMD = 1)

Figure 44.27 shows an example of operation when the SPCR.TXMD bit is set to 1. In the example in Figure 44.27, the RSPI performs an 8-bit serial transfer in which the SPDCR.SPFC[1:0] bits are 00b, the SPCMDm.CPHA bit is 1, and the SPCMDm.CPOL bit is 0. The numbers given under the RSPCKx waveform represent the number of RSPCK cycles (i.e., the number of transferred bits).

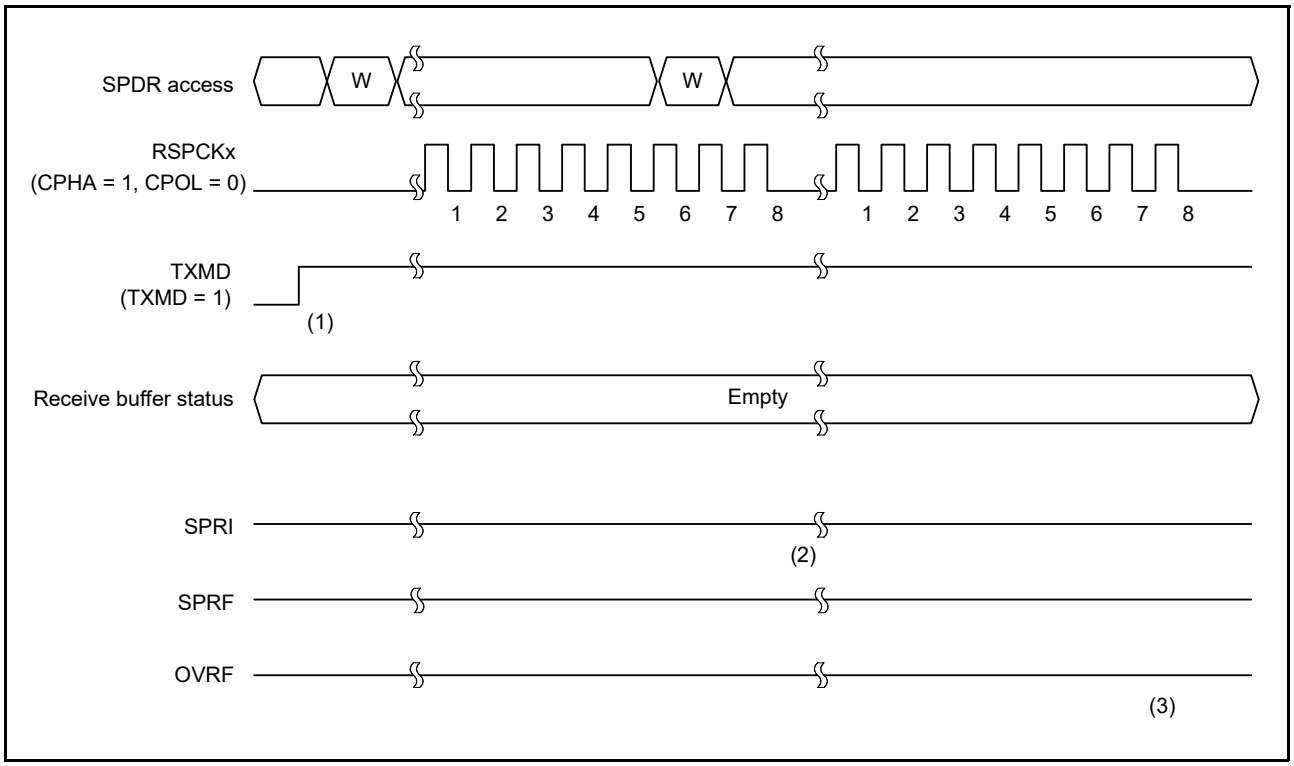


Figure 44.27 Operation Example of SPCR.TXMD = 1

The operation of the flags at timings shown in steps (1) to (3) in the figure is described below.

(1) Make sure there is no data left in the receive buffer and the SPSR.SPRF, OVRF flags are 0 before entering the mode of transmit-only simplex communications (SPCR.TXMD = 1).

(2) When a serial transfer ends with the receive buffer of the SPDR register empty, if the mode of transmit-only simplex communications is selected (SPCR.TXMD = 1), the SPRF flag remains 0 and the RSPI does not copy the data from the shift register to the receive buffer.

(3) Since the receive buffer of the SPDR register does not hold data that was received in the previous serial transfer, even when a serial transfer ends, the SPSR.OVRF flag retains the value of 0, and the data in the shift register is not copied to the receive buffer.

When performing transmit-only simplex communications (SPCR.TXMD = 1), the RSPI transmits data but does not receive data. Therefore, the SPSR.SPRF, OVRF flags remain 0 at the timings of (1) to (3).

44.3.7 Transmit Buffer Empty/Receive Buffer Full Interrupts

Figure 44.28 shows an example of operation of the transmit buffer empty interrupt (SPTI) and the receive buffer full interrupt (SPRI). 'SPDR access' shown in Figure 44.28 indicates the condition of access to the SPDR register, where 'W' denotes a write cycle, and 'R' a read cycle. In the example in Figure 44.28, the RSPI performs an 8-bit serial transfer in which the SPCR.TXMD bit is 0, the SPDCR.SPFC[1:0] bits are 00b, the SPCMDm.CPHA bit is 1, and the SPCMDm.CPOL bit is 0. The numbers given under the RSPCKx waveform represent the number of RSPCK cycles (i.e., the number of transferred bits).

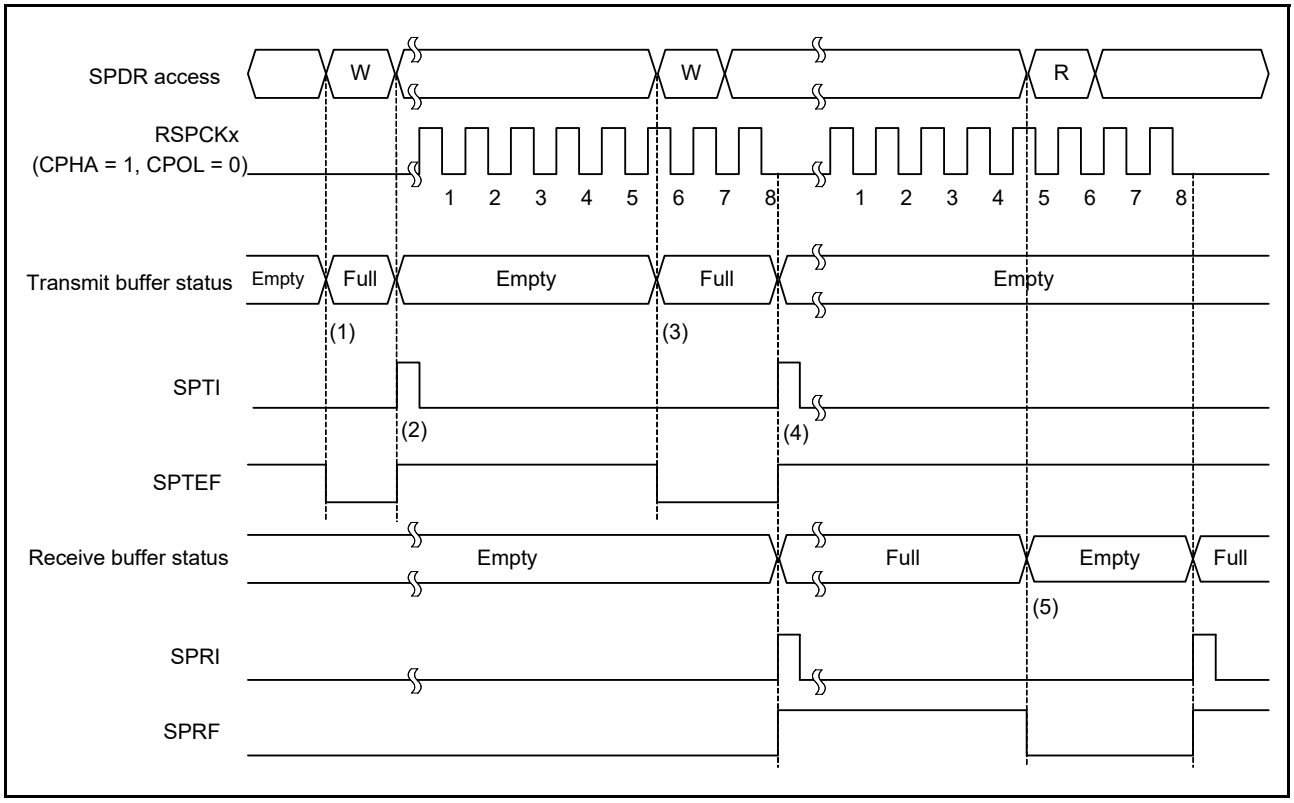


Figure 44.28 Operation Example of SPTI and SPRI Interrupts

The operation of the interrupts at timings shown in steps (1) to (5) in the figure is described below.

(1) When transmit data is written to the SPDR register when the transmit buffer of the SPDR register is empty (data for the next transfer is not set), the RSPI writes data to the transmit buffer and sets the SPSR.SPTEF flag to 0.

(2) If the shift register is empty, the RSPI copies the data from the transmit buffer to the shift register and generates a transmit buffer empty interrupt request (SPTI) and sets the SPSR.SPTEF flag to 1. How a serial transfer is started depends on the mode of the RSPI. For details, refer to section 44.3.11, SPI Operation, and section 44.3.12, Clock Synchronous Operation.

(3) When transmit data is written to the SPDR register in the transmit buffer empty interrupt routine or in the transmit buffer empty detecting process by polling the SPTEF flag, the data is transferred to the transmit buffer and the SPSR.SPTEF flag becomes 0. Because the data being transmitted is stored in the shift register, the RSPI does not copy the data from the transmit buffer to the shift register.

(4) When the serial transfer ends with the receive buffer of the SPDR register being empty, the RSPI copies the receive data from the shift register to the receive buffer, generates a receive buffer full interrupt request (SPRI), and sets the SPSR.SPRF flag to 1. Since the shift register becomes empty upon completion of serial transfer, when the transmit buffer had been full before the serial transfer ended, the RSPI sets the SPSR.SPTEF flag to 1 and copies the data from the transmit buffer to the shift register. Even when received data is not copied from the shift register to the receive buffer in an overrun error status, upon completion of the serial transfer, the RSPI determines that the shift register is empty, thus data transfer from the transmit buffer to the shift register is enabled.

(5) When the SPDR register is read in the receive buffer full interrupt routine or in the receive buffer full detecting process by polling the SPRF flag, the receive data can be read. When the receive data is read, the SPRF flag becomes 0.

If transmit data is written to the SPDR register while the transmit buffer holds data that has not yet been transmitted (the SPTEF flag is 0), the RSPI does not update the data in the transmit buffer. Transmit data should be written to the SPDR register in the transmit buffer empty interrupt request routine or in the transmit buffer empty detecting process by polling the SPTEF flag. To use a transmit buffer empty interrupt, set the SPTIE bit in SPCR to 1.
When setting the SPCR.SPE bit to 0 (RSPI disabled), the SPCR.SPTIE bit should also be set to 0. Otherwise (if the SPCR.SPE bit is 0 and the SPCR.SPTIE is 1), a transmit buffer empty interrupt request will occur.
When serial transfer ends with the receive buffer being full (the SPRF flag is 1), the RSPI does not copy data from the shift register to the receive buffer, and detects an overrun error (refer to section 44.3.9, Error Detection). To prevent a receive data overrun error, read the received data using a receive buffer full interrupt request before the next serial transfer ends. To use an receive buffer full interrupt, set the SPCR.SPRIE bit to 1.
Transmit and receive interrupts or the corresponding IRn.IR flags (where n is the interrupt vector number) in the ICU can be used to confirm the states of the transmit and receive buffers. Refer to section 15, Interrupt Controller (ICUD), for the interrupt vector numbers. The status of the transmit and receive buffers can be also confirmed by the SPTEF and SPRF flags.

44.3.8 Idle Interrupt

When the SPSR.IDLNF flag becomes 0 while the SPCR2.SPIIE bit is 1, an idle interrupt request (SPII) is generated.
In master mode, the IDLNF flag is 0 before transmission. Therefore, write data in the transmit buffer and set the SPIIE bit to 1 after the IDLNF flag becomes 1 so that an idle interrupt is not generated at this time. When the SSLx0 signal is negated after the transmission is completed and the next data is not supplied until next-access delay time (t3) elapses, the IDLNF flag becomes 0.

44.3.9 Error Detection

In the normal RSPI serial transfer, the data written to the transmit buffer of the SPDR register is transmitted, and the received data can be read from the receive buffer of the SPDR register. If access is made to the SPDR register, depending on the status of the transmit/receive buffer or the status of the RSPI at the beginning or end of serial transfer, in some cases non-normal transfers can be executed.

If a non-normal transfer operation occurs, the RSPI detects the event as an underrun error, overrun error, parity error, or mode fault error. Table 44.7 lists the relationship between non-normal transfer operations and the RSPI's error detection function.

Table 44.7 Relationship between Non-Normal Transfer Operations and RSPI Error Detection Function

	Occurrence Condition	RSPI Operation	Error Detection
1	The SPDR register is written when the transmit buffer is full.	• The contents of the transmit buffer are kept. • Missing write data.	None
2	The SPDR register is read when the receive buffer is empty.	If reception has completed, then the received data is output to the bus. Otherwise, data received previously is output instead.	None
3	Serial transfer is started when transmit data is still not loaded on the shift register while the RSPI is used in slave mode.	• Serial transfer is suspended • Transmit/receive data is missing • The MISO signal output is disabled • RSPI function is disabled	Underrun error
4	Serial transfer terminates when the receive buffer is full.	• The contents of the receive buffer are kept. • Missing receive data.	Overrun error
5	An incorrect parity bit is received when performing full-duplex communications with the parity function enabled.	The parity error flag is set.	Parity error
6	The SSLx0 input signal is asserted when the serial transfer is idle in multi-master mode.	• Driving of the RSPCKx, MOSIx, SSLx1 to SSLx3 output signals is stopped. • RSPI function is disabled.	Mode fault error
7	The SSLx0 input signal is asserted during serial transfer in multi-master mode.	• Serial transfer is suspended. • Missing transmit/receive data. • Driving of the RSPCKx, MOSIx, SSLx1 to SSLx3 output signals is stopped. • RSPI function is disabled.	Mode fault error
8	The SSLx0 input signal is negated during serial transfer in slave mode.	• Serial transfer is suspended. • Missing transmit/receive data. • Driving of the MISOx output signal is stopped. • RSPI function is disabled.	Mode fault error

On operation 1 described in Table 44.7, the RSPI does not detect an error. To prevent data omission during the writing to the SPDR register, the SPDR register should be written when a transmit buffer empty interrupt request occurs or while the SPSR.SPTEF flag is 1.

Likewise, the RSPI does not detect an error on operation 2. To prevent extraneous data from being read, the SPDR register should be read when a receive buffer full interrupt request occurs or while the SPSR.SPRF flag is 1.

An underrun error shown in 3 is described in section 44.3.9.4, Underrun Error. An overrun error shown in 4 is described in section 44.3.9.1, Overrun Error. A parity error shown in 5 is described in section 44.3.9.2, Parity Error. A mode fault error shown in 6 to 8 is described in section 44.3.9.3, Mode Fault Error.

For the transmit and receive interrupts, refer to section 44.3.7, Transmit Buffer Empty/Receive Buffer Full Interrupts.

44.3.9.1 Overrun Error

If a serial transfer ends when the receive buffer of the SPDR register is full, the RSPI detects an overrun error, and sets the SPSR.OVRF flag to 1. When the OVRF flag is 1, the RSPI does not copy data from the shift register to the receive buffer so that the data prior to the occurrence of the error is retained in the receive buffer. To set the OVRF flag to 0, write 0 to the OVRF flag after the CPU has read the SPSR register with the OVRF flag set to 1.
Figure 44.29 shows an example of operations of the SPRF and OVRF flags. 'SPSR access' and 'SPDR access' shown in Figure 44.29 indicate the condition of accesses to the SPSR and SPDR registers, respectively, where 'W' denotes a write cycle, and 'R' a read cycle. In the example in Figure 44.29, the RSPI performs an 8-bit serial transfer in which the SPCMDm.CPHA bit is 1 and the SPCMDm.CPOL bit is 0. The numbers given under the RSPCKx waveform represent the number of RSPCK cycles (i.e., the number of transferred bits).

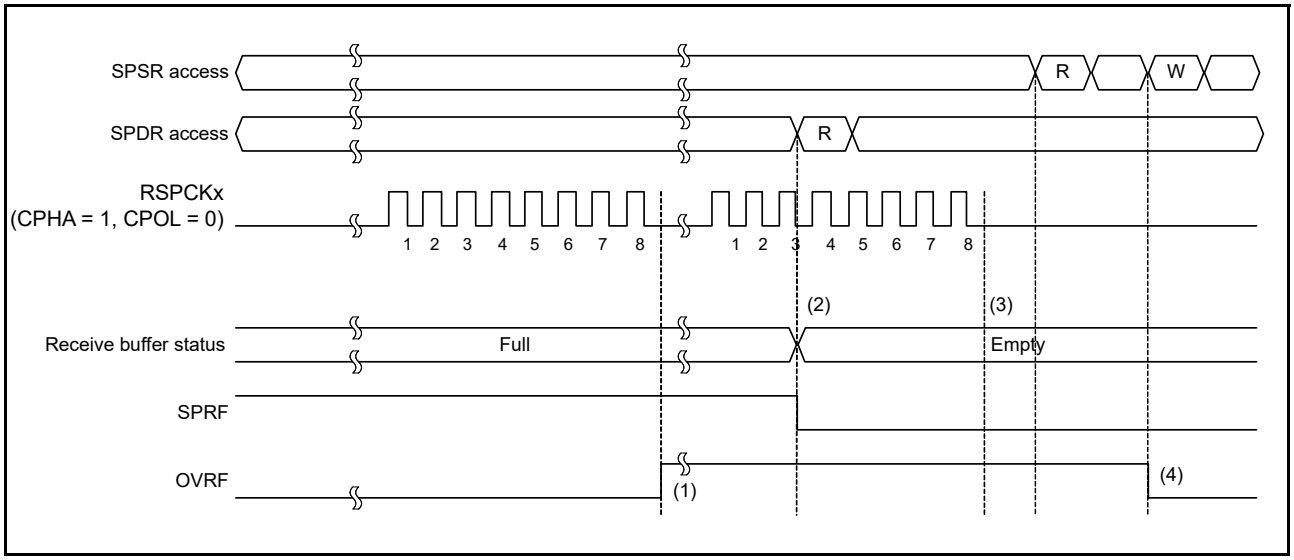


Figure 44.29 Operation Example of the SPRF and OVRF Flags

The operation of the flags at the timing shown in steps (1) to (4) in the figure is described below.

(1) If a serial transfer terminates with the receive buffer full (the SPRF flag is 1), the RSPI detects an overrun error, and sets the OVRF flag to 1. The RSPI does not copy the data in the shift register to the receive buffer. Even if the SPPE bit is 1, parity errors are not detected. In master mode, the RSPI copies the pointer value to the SPCMDm register to the SPSSR.SPECM[2:0] bits.

(2) When the SPDR register is read, the RSPI outputs the data in the receive buffer. At this time the SPRF flag becomes 0. Even if the receive buffer becomes empty, the OVRF flag does not become 0.

(3) If the serial transfer ends with the OVRF flag being 1 (an overrun error occurs), the RSPI does not copy the data in the shift register to the receive buffer (the SPRF flag remains 0). A receive buffer full interrupt is not generated. Even if the SPPE bit is 1, parity errors are not detected. When in master mode, the RSPI does not update the SPSSR.SPECM[2:0] bits. When in an overrun error state and the RSPI does not copy the received data from the shift register to the receive buffer, upon termination of the serial transfer, the RSPI determines that the shift register is empty; in this manner, data transfer from the transmit buffer to the shift register is enabled.

(4) If 0 is written to the OVRF flag after the SPSR register is read when the OVRF flag is 1, the OVRF flag is set to 0.

The occurrence of an overrun can be checked either by reading the SPSR register or by using an error interrupt and reading the SPSR register. When executing a serial transfer, measures should be taken to ensure the early detection of overrun errors, such as reading the SPSR register immediately after the SPDR register is read. When the RSPI is used in master mode, the pointer value to the SPCMDm register at the occurrence of the error can be checked by reading the SPSSR.SPECM[2:0] bits.
If an overrun error occurs and the OVRF flag is set to 1, normal reception operations cannot be performed until the OVRF flag is set to 0.

When the RSPCK auto-stop function is enabled in master mode, an overrun error does not occur. Figure 44.30 and Figure 44.31 show the clock stop waveform when a serial transfer continues while the receive buffer is full in master mode.

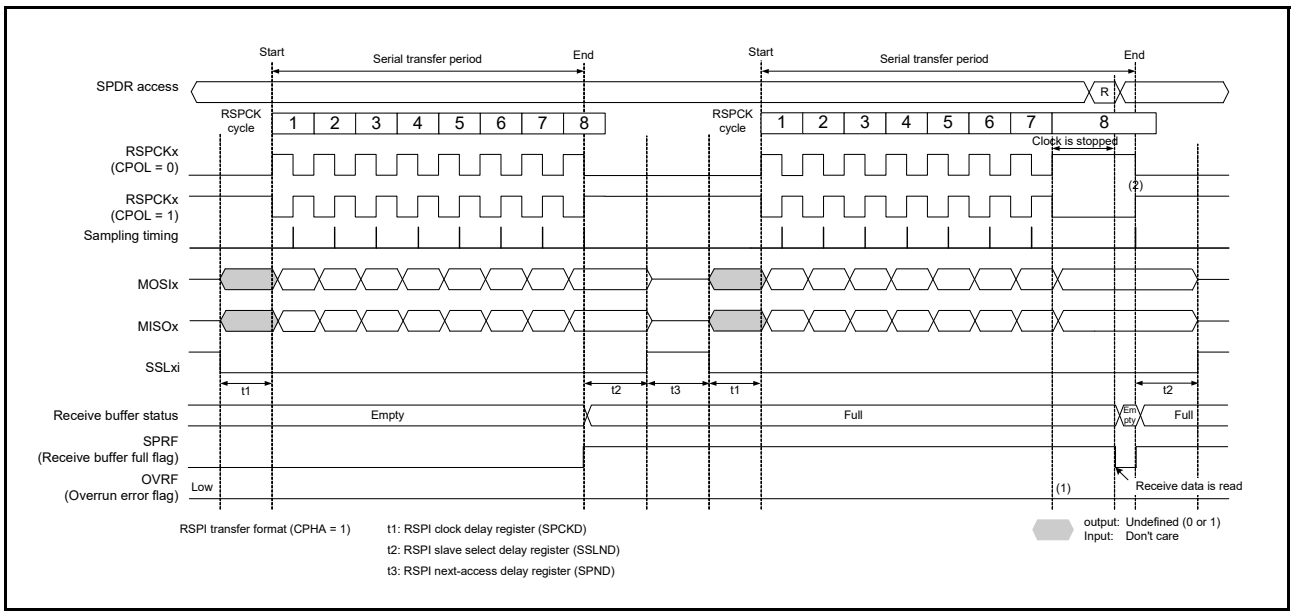


Figure 44.30 Clock Stop Waveform When a Serial Transfer Continues While the Receive Buffer is Full in Master Mode (CPHA = 1)

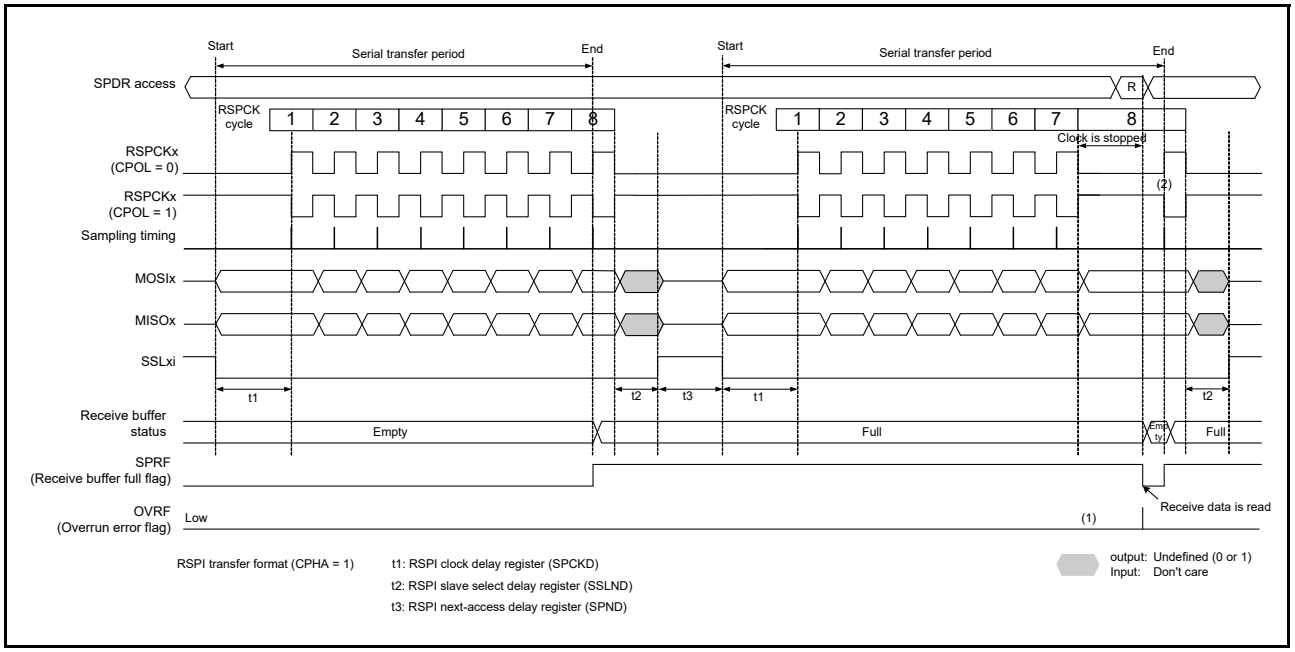


Figure 44.31 Clock Stop Waveform When a Serial Transfer Continues While the Receive Buffer is Full in Master Mode (CPHA = 0)

The operation of the flags at the timings shown in steps (1) and (2) in the figure is described below.

(1) When the receive buffer is full, an overrun error does not occur because the RSPCK clock is stopped.

(2) If the SPDR register is read while the clock is stopped, data in the receive buffer can be read. The RSPCK clock restarts after reading the receive buffer (after the SPRF flag becomes 0).

44.3.9.2 Parity Error

If full-duplex communication is performed with the SPCR.TXMD bit set to 0 and the SPCR2.SPPE bit set to 1, when serial transfer ends, the RSPI checks whether there are parity errors. Upon detecting a parity error in the received data, the RSPI sets the SPSR.PERF flag to 1. Since the RSPI does not copy the data in the shift register to the receive buffer when the SPSR.OVRF flag is set to 1, parity error detection is not performed for the received data. To set the PERF flag to 0, write 0 to the PERF flag after the SPSR register is read with the PERF flag set to 1.
Figure 44.32 shows an example of operation of the OVRF and PERF flags. 'SPSR access' shown in Figure 44.32 indicates the condition of access to the SPSR register, where 'W' denotes a write cycle, and 'R' a read cycle. In the example of Figure 44.32, full-duplex communication is performed while the SPCR.TXMD bit is 0 and the SPCR2.SPPE bit is 1. The RSPI performs an 8-bit serial transfer in which the SPCMDm.CPHA bit is 1 and the SPCMDm.CPOL bit is 0. The numbers given under the RSPCKx waveform represent the number of RSPCK cycles (i.e., the number of transferred bits).

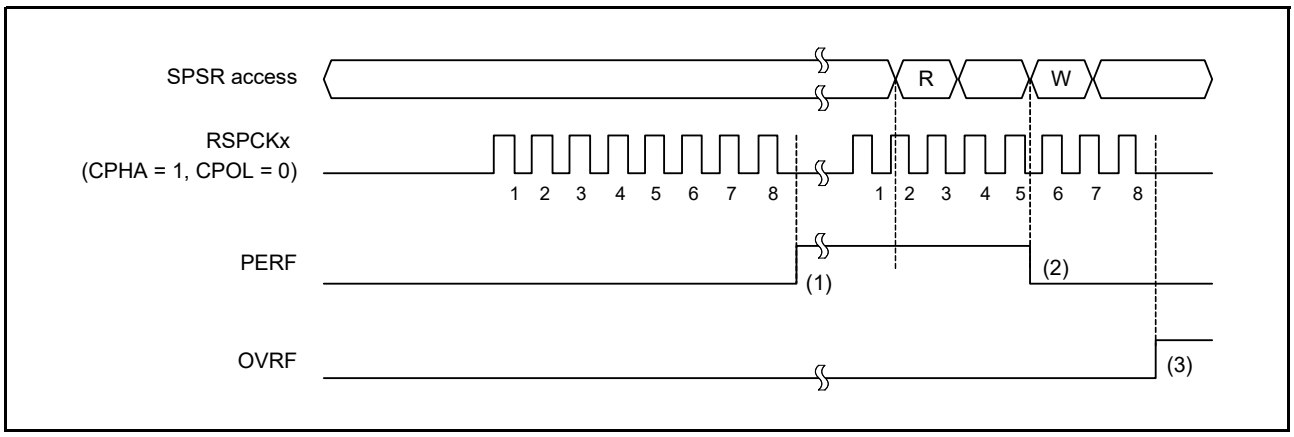


Figure 44.32 Operation Example of PERF Flag

The operation of the flags at the timing shown in steps (1) to (3) in the figure is described below.

(1) If a serial transfer terminates with the RSPI not detecting an overrun error, the RSPI copies the data in the shift register to the receive buffer. The RSPI judges the received data at this timing, and sets the PERF flag to 1 if a parity error is detected. In master mode, the RSPI copies the pointer value to the SPCMDm register to the SPSSR.SPECM[2:0] bits.

(2) If 0 is written to the PERF flag after the SPSR register is read when the PERF flag is 1, the PERF flag is set to 0.

(3) When the RSPI detects an overrun error and serial transfer is terminated, the data in the shift register is not copied to the receive buffer. The RSPI does not perform parity error detection at this timing.

The occurrence of a parity error can be checked either by reading the SPSR register or by using an error interrupt and reading the SPSR register. When executing a serial transfer, measures should be taken to ensure the early detection of parity errors, such as reading the SPSR register. When the RSPI is used in master mode, the pointer value to the SPCMDm register at the occurrence of the error can be checked by reading the SPSSR.SPECM[2:0] bits.

44.3.9.3 Mode Fault Error

The RSPI operates in multi-master mode when the SPCR.MSTR bit is 1, the SPCR.SPMS bit is 0, and the SPCR.MODFEN bit is 1. If the active level is input with respect to the SSLx0 input signal of the RSPI in multi-master mode, the RSPI detects a mode fault error irrespective of the status of the serial transfer, and sets the SPSR.MODF flag to 1. Upon detecting the mode fault error, the RSPI copies the value of the pointer to the SPCMDm register to the SPSSR.SPECM[2:0] bits. The active level of the SSLx0 signal is determined by the SSLP.SSL0P bit.

When the MSTR bit is 0, the RSPI operates in slave mode. The RSPI detects a mode fault error if the MODFEN bit of the RSPI in slave mode is 1, and the SPMS bit is 0, and if the SSLx0 input signal is negated during the serial transfer period (from the time the driving of valid data is started to the time the final valid data is fetched).

Upon detecting a mode fault error, the RSPI stops driving of the output signals and clears the SPCR.SPE bit to 0 (refer to section 44.3.10, Initializing RSPI). In the case of multi-master configuration, detection of a mode fault error is used to stop driving of the output signals and the RSPI function, which allows the master right to be released.

The occurrence of a mode fault error can be checked either by reading the SPSR register or by using an error interrupt and reading the SPSR register. Detecting mode fault errors without utilizing the error interrupt requires polling of the SPSR register. When using the RSPI in master mode, the pointer value to the SPCMDm register at the occurrence of the error can be checked by reading the SPSSR.SPECM[2:0] bits.

When the MODF flag is 1, writing of the value 1 to the SPE bit is ignored by the RSPI. To enable the RSPI function after the detection of a mode fault error, set the MODF flag to 0. When setting the MODF flag to 0, the SPE bit becomes 1.

44.3.9.4 Underrun Error

If a serial transfer is started when the SPCR.SPE bit is 1 (RSPI function is enabled) and transmit data is still not loaded on the shift register while RSPI operates in slave mode (the SPCR.MSTR bit is 0), the RSPI detects an underrun error, and sets the SPSR.MODF and SPSR.UDRF flags to 1. Upon detecting an underrun error, the RSPI stops driving of the output signals and clears the SPCR.SPE bit to 0. Clearing of the SPE bit disables the RSPI function. (Refer to section 44.3.10, Initializing RSPI). The occurrence of an underrun error can be checked either by reading the SPSR register or by using an error interrupt and reading the SPSR register. Detecting underrun errors without utilizing the error interrupt requires polling of the SPSR register. When the MODF flag is 1, writing of the value 1 to the SPE bit is ignored by the RSPI. To enable the RSPI function after the detection of an underrun error, set the MODF flag to 0. When setting the MODF flag to 0, the SPE bit becomes 1.

44.3.10 Initializing RSPI

If 0 is written to the SPCR.SPE bit or the RSPI sets the SPE bit to 0 because of the detection of a mode fault error or an underrun error, the RSPI disables the RSPI function, and initializes some of the module functions. When a system reset is generated, the RSPI initializes all of the module functions. The following describes initialization by the clearing of the SPCR.SPE bit and initialization by a system reset.

44.3.10.1 Initialization by Clearing the SPE Bit

When the SPCR.SPE bit is set to 0, the RSPI performs the following initialization:

- Aborting the transmission and reception that is being executed
- Stopping the driving of output signals (Hi-Z) in slave mode
- Initializing the internal state of the RSPI
- Initializing the transmit buffer of the RSPI (Set the SPTEF flag to 1)

Initialization by the clearing of the SPE bit does not initialize the control bits of the RSPI. For this reason, the RSPI can be started in the same transfer mode as prior to the initialization if the SPE bit is set to 1 again.
The SPSR.SPRF, UDRF, PERF, MODF, and OVRF flags are not initialized, nor is the value of the RSPI sequence status register (SPSSR) initialized. For this reason, even after the RSPI is initialized, data from the receive buffer can be read and the status of error occurrence during the RSPI transfer can be checked.
The transmit buffer is initialized to an empty state (the SPTEF flag is 1). Therefore, if the SPCR.SPTIE bit is set to 1 after RSPI initialization, a transmit buffer empty interrupt is generated. When the RSPI is initialized, in order to disable any transmit buffer empty interrupt, 0 should be written to the SPTIE bit simultaneously with the writing of 0 to the SPE bit.

44.3.10.2 System Reset

The initialization by a system reset completely initializes the RSPI through the initialization of all bits for controlling the RSPI, initialization of the status bits, and initialization of data registers, in addition to the requirements described in section 44.3.10.1, Initialization by Clearing the SPE Bit.

44.3.11 SPI Operation

44.3.11.1 Master Mode Operation

The only difference between single-master mode operation and multi-master mode operation lies in mode fault error detection (refer to section 44.3.9, Error Detection). When operating in single-master mode, the RSPI does not detect mode fault errors whereas the RSPI running in multi-master mode does detect mode fault errors. This section explains operations that are common to single-master mode and multi-master mode.

(1) Starting a Serial Transfer

The RSPI updates the data in the transmit buffer (SPTX) when data is written to the RSPI data register (SPDR) with the RSPI transmit buffer being empty (the SPTEF flag is 1 and data for the next transfer is not set). When the shift register is empty after the number of frames set in the SPDCR.SPFC[1:0] bits are written to the SPDR register, the RSPI copies data from the transmit buffer to the shift register and starts serial transfer. Upon copying transmit data to the shift register, the RSPI changes the status of the shift register to “full”, and upon termination of serial transfer, it changes the status of the shift register to “empty”. The status of the shift register cannot be referenced.
For details on the RSPI transfer format, refer to section 44.3.5, Transfer Format. The polarity of the SSLxi output pins depends on the SSLP register settings.

(2) Terminating a Serial Transfer

Irrespective of the SPCMDm.CPHA bit, the RSPI terminates the serial transfer after transmitting an RSPCKx edge corresponding to the final sampling timing. If free space is available in the receive buffer (SPRX) (the SPRF flag is 0), upon termination of serial transfer, the RSPI copies data from the shift register to the receive buffer of the SPDR register. It should be noted that the final sampling timing varies depending on the bit length of transfer data. In master mode, the RSPI data length depends on the SPCMDm.SPB[3:0] bit setting. The polarity of the SSLxi output pin depends on the SSLP register settings.
For details on the RSPI transfer format, refer to section 44.3.5, Transfer Format.

(3) Sequence Control

The transfer format that is employed in master mode is determined by the SPSCR, SPCMDm, SPBR, SPCKD, SSLND, and SPND registers.

The SPSCR register is a register used to determine the sequence configuration for serial transfers that are executed by the RSPI in master mode. The following items are set in the SPCMDm register: SSLxi pin output signal value, MSB/LSB first, data length, some of the bit rate settings, RSPCK polarity/phase, whether the SPCKD register is to be referenced, whether the SSLND register is to be referenced, and whether the SPND register is to be referenced. The SPBR register holds some of the bit rate settings; the SPCKD register, an RSPI clock delay value; the SSLND register, an SSL negation delay; and the SPND register, a next-access delay value.

According to the sequence length that is assigned to the SPSCR register, the RSPI makes up a sequence comprised of a part or all of the SPCMDm register. The RSPI contains a pointer to the SPCMDm register that makes up the sequence. The value of this pointer can be checked by reading the SPSSR.SPCP[2:0] bits. When the SPCR.SPE bit is set to 1 and the RSPI function is enabled, the RSPI loads the pointer to the commands in the SPCMD0 register, and incorporates the SPCMD0 settings into the transfer format at the beginning of serial transfer. The RSPI increments the pointer each time the next-access delay period for a data transfer ends. Upon completion of the serial transfer that corresponds to the final command comprising the sequence, the RSPI sets the pointer in the SPCMD0 register, and in this manner the sequence is executed repeatedly.

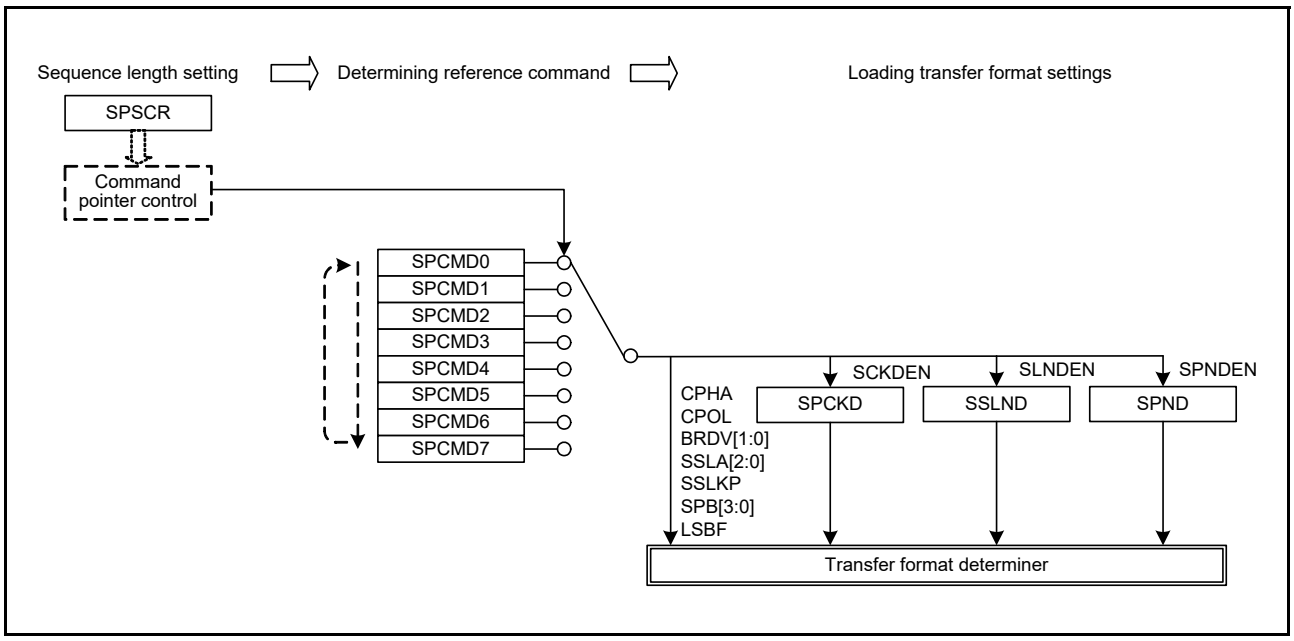


Figure 44.33 Procedure for Determining the Form of Serial Transfer in Master Mode

In this section, a frame is the combination of the data (SPDR) and the settings (SPCMDm).

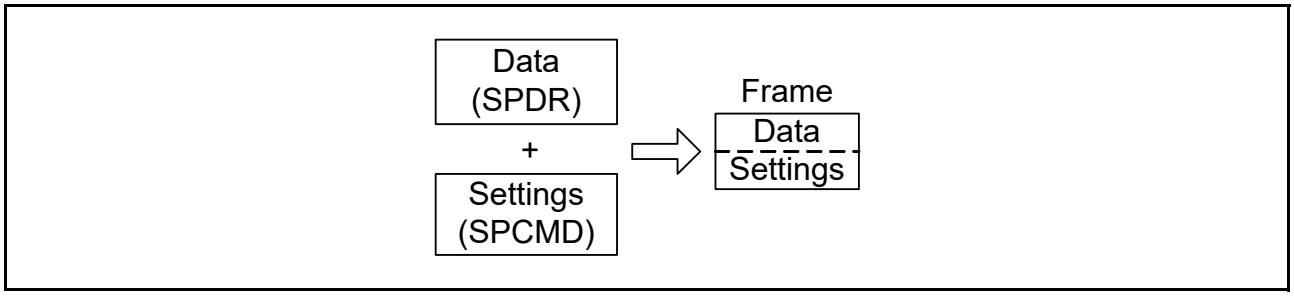


Figure 44.34 Concept of a Frame

Figure 44.35 shows the relationship between the command and the transmit and receive buffers in the sequence of operations specified by the settings in Table 44.4.

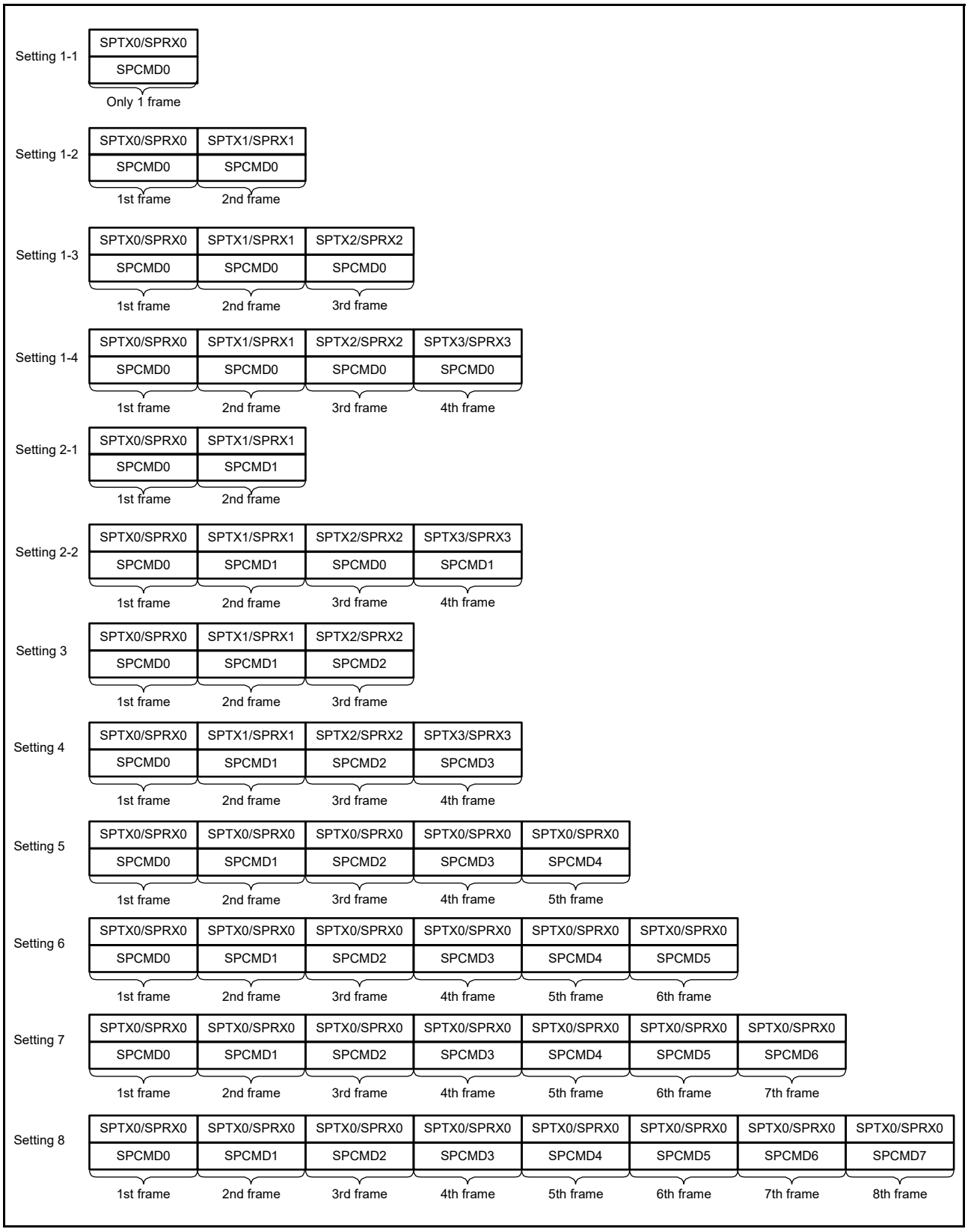


Figure 44.35 Correspondence between the RSPI Command Register and Transmit/Receive Buffers in Sequence Operations

(4) Burst Transfer

If the SPCMDm.SSLKP bit that the RSPI references during the current serial transfer is 1, the RSPI keeps the SSLxi signal level during the serial transfer until the beginning of the SSLxi signal assertion for the next serial transfer. If the SSLxi signal level for the next serial transfer is the same as the SSLxi signal level for the current serial transfer, the RSPI can execute continuous serial transfers while keeping the SSLxi signal assertion status (burst transfer).
Figure 44.36 shows an example of an SSLxi signal operation for the case where a burst transfer is implemented using SPCMD0 and SPCMD1 register settings. The text below explains the RSPI operations (1) to (7) as shown in Figure 44.36. It should be noted that the polarity of the SSLxi output signal depends on the SSLP register settings.

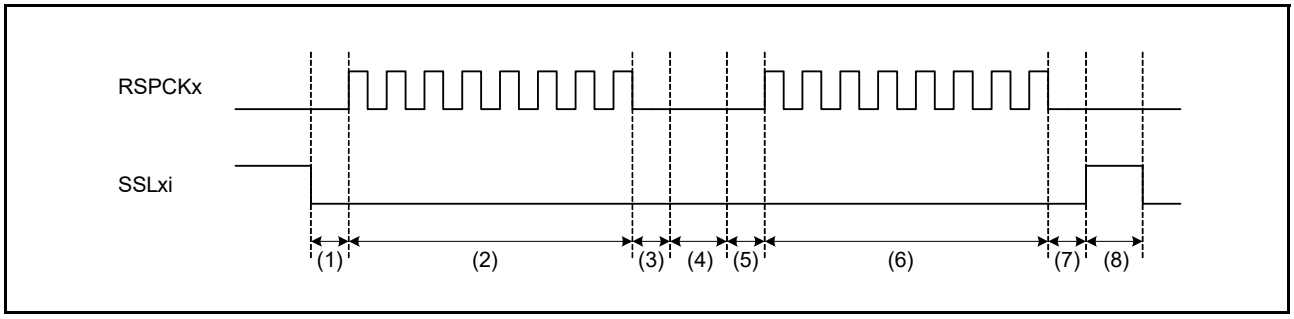


Figure 44.36 Example of Burst Transfer Operation Using SSLKP Bit (CPHA = 1, CPOL = 0)

(1) Based on the SPCMD0 register, the RSPI asserts the SSLxi signal and inserts RSPCK delays.
(2) The RSPI executes serial transfers according to the SPCMD0 register.
(3) The RSPI inserts SSL negation delays.
(4) Since the SPCMD0.SSLKP bit is 1, the RSPI keeps the SSLxi signal value on the SPCMD0 register. This period is sustained, at the shortest, for a period equal to the next-access delay of the SPCMD0 register. If the shift register is empty after the passage of a minimum period, this period is sustained until the transmit data is stored in the shift register for the next transfer.
(5) Based on the SPCMD1 register, the RSPI inserts RSPCK delays.
(6) The RSPI executes serial transfers according to the SPCMD1 register.
(7) The RSPI inserts SSL negation delays.
(8) Because the SPCMD1.SSLKP bit is 0, the RSPI negates the SSLxi signal. In addition, a next-access delay is inserted according to the SPCMD1 register.

If the SSLxi signal output settings in the SPCMDm register in which 1 is assigned to the SSLKP bit are different from the SSLxi signal output settings in the SPCMDm register to be used in the next transfer, the RSPI switches the SSLxi signal status to SSLxi signal assertion ((5) in Figure 44.36) corresponding to the command for the next transfer. Note that if such an SSLxi signal switching occurs, the slaves that drive the MISOx signal compete, and collision of signal levels may occur.
The RSPI in master mode references the SSLxi signal operation within the module for the case where the SSLKP bit is not used. Even when the SPCMDm.CPHA bit is 0, the RSPI can accurately start serial transfers by using the SSLxi signal assertion for the next transfer that is detected internally.

(5) RSPCK Delay (t1)

The RSPCK delay value in master mode depends on the SPCMDm.SCKDEN bit setting and the SPCKD register setting. The RSPI determines the SPCMDm register to be referenced during serial transfer by pointer control, and determines an RSPCK delay value during serial transfer by using the SPCMDm.SCKDEN bit and the SPCKD register, as listed in Table 44.8. For a definition of RSPCK delay, refer to section 44.3.5, Transfer Format.

Table 44.8 Relationship among SCKDEN Bit, SPCKD, and RSPCK Delay Value

SPCMDm.SCKDEN Bit	SPCKD.SCKDL[2:0] Bits	RSPCK Delay Value
0	000b to 111b	1 RSPCK
1	000b	1 RSPCK
	001b	2 RSPCK
	010b	3 RSPCK
	011b	4 RSPCK
	100b	5 RSPCK
	101b	6 RSPCK
	110b	7 RSPCK
	111b	8 RSPCK

(6) SSL Negation Delay (t2)

The SSL negation delay value in master mode depends on the SPCMDm.SLNDEN bit setting and the SSLND register setting. The RSPI determines the SPCMDm register to be referenced during serial transfer by pointer control, and determines an SSL negation delay value during serial transfer by using the SPCMDm.SLNDEN bit and the SSLND register, as listed in Table 44.9. For a definition of SSL negation delay, refer to section 44.3.5, Transfer Format.

Table 44.9 Relationship among SLNDEN Bit, SSLND, and SSL Negation Delay Value

SPCMDm.SLNDEN Bit	SSLND.SLNDL[2:0] Bits	SSL Negation Delay Value
0	000b to 111b	1 RSPCK
1	000b	1 RSPCK
	001b	2 RSPCK
	010b	3 RSPCK
	011b	4 RSPCK
	100b	5 RSPCK
	101b	6 RSPCK
	110b	7 RSPCK
	111b	8 RSPCK

(7) Next-Access Delay (t3)

The next-access delay value in master mode depends on the SPCMDm.SPNDEN bit setting and the SPND setting. The RSPI determines the SPCMDm register to be referenced during serial transfer by pointer control, and determines a next-access delay value during serial transfer by using the SPCMDm.SPNDEN bit and SPND, as listed in Table 44.10. For a definition of next-access delay, refer to section 44.3.5, Transfer Format.

Table 44.10 Relationship among SPNDEN Bit, SPND, and Next-Access Delay Value

SPCMDm.SPNDEN Bit	SPND.SPNDL[2:0] Bits	Next-Access Delay Value
0	000b to 111b	1 RSPCK + 2 PCLK
1	000b	1 RSPCK + 2 PCLK
	001b	2 RSPCK + 2 PCLK
	010b	3 RSPCK + 2 PCLK
	011b	4 RSPCK + 2 PCLK
	100b	5 RSPCK + 2 PCLK
	101b	6 RSPCK + 2 PCLK
	110b	7 RSPCK + 2 PCLK
	111b	8 RSPCK + 2 PCLK

(8) Initialization Flowchart

Figure 44.37 is a flowchart illustrating an example of initialization in SPI operation when the RSPI is used in master mode. For a description of how to set up the interrupt controller, DMAC, and I/O ports, refer to the descriptions given in the individual blocks.

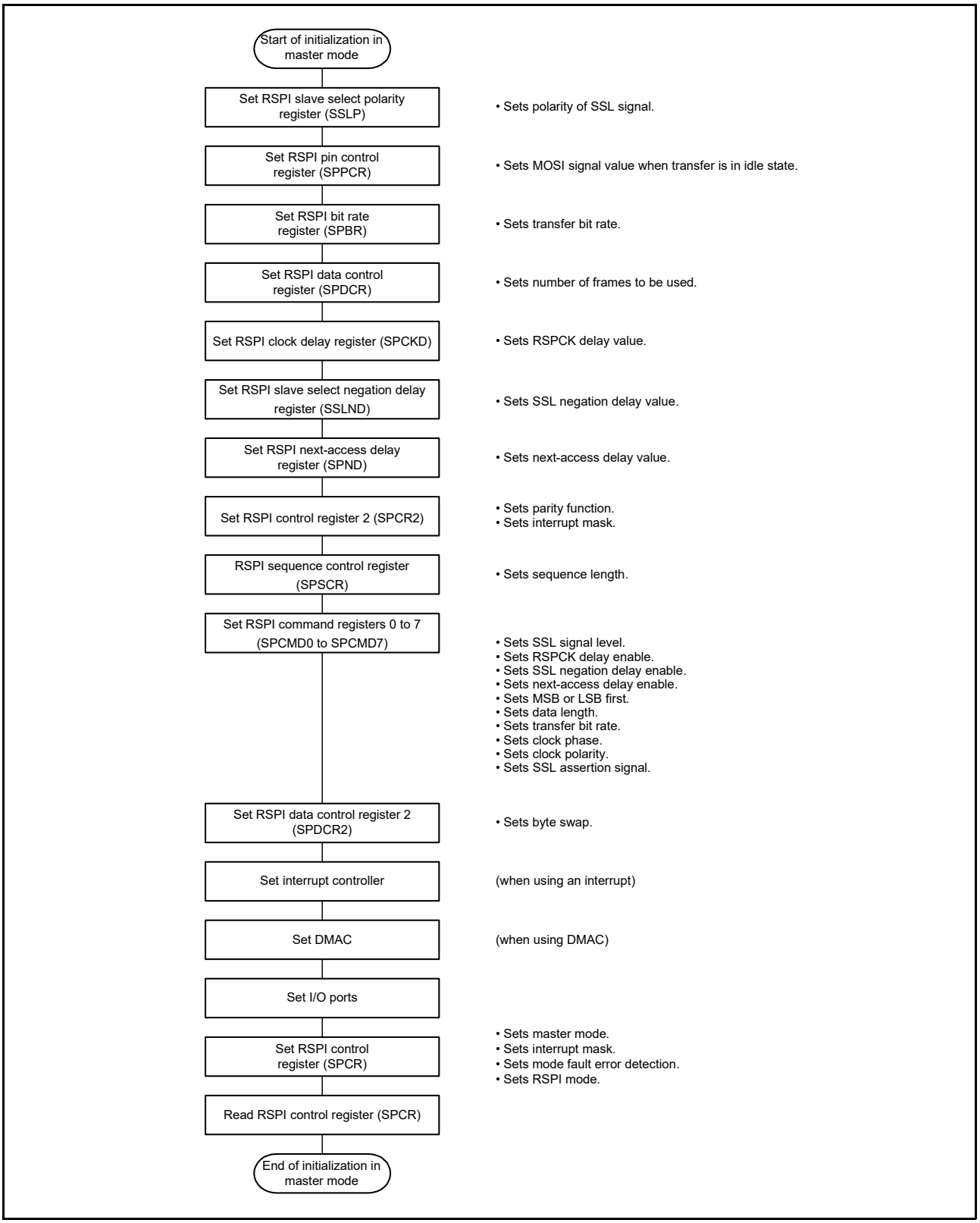


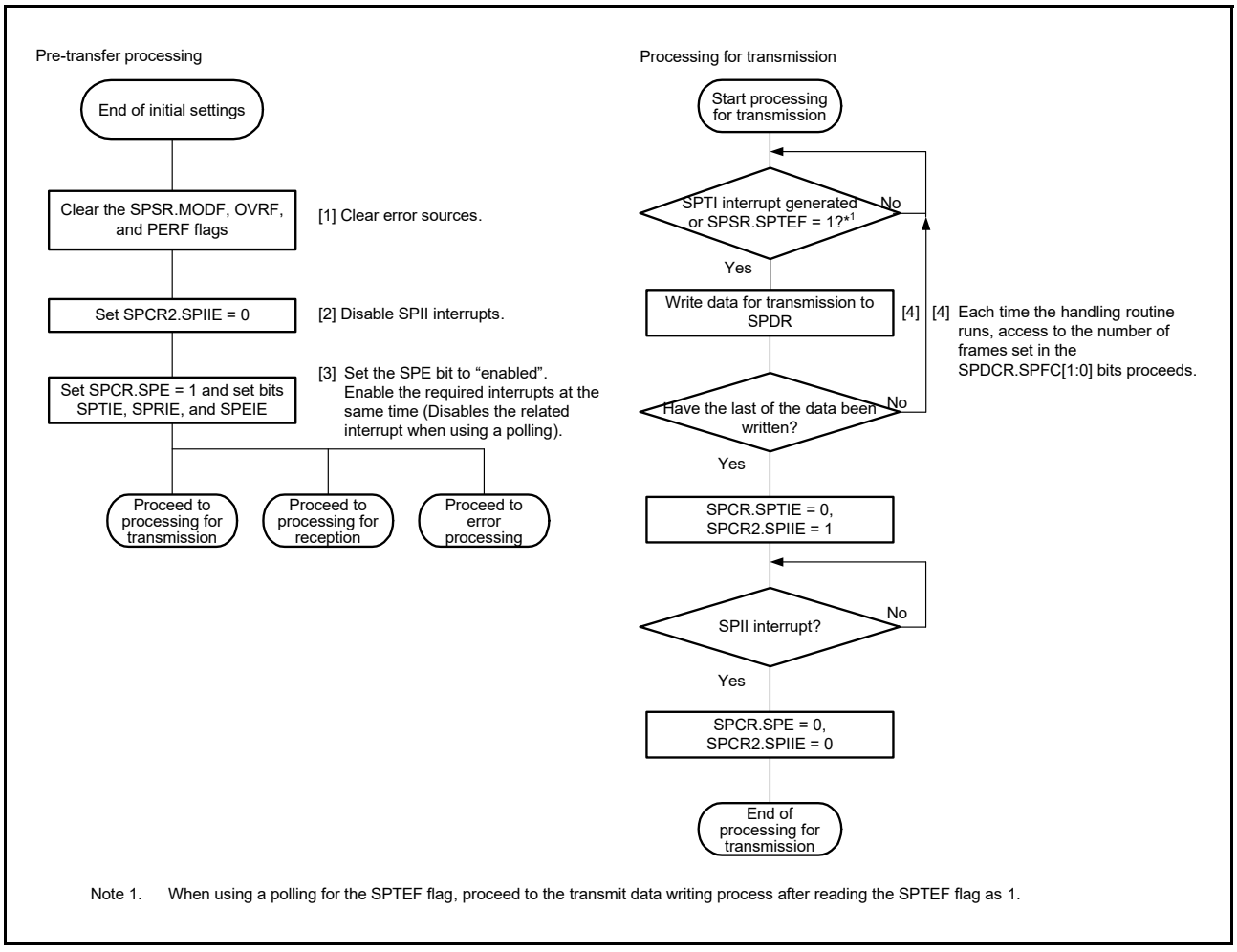
Figure 44.37 Example of Initialization Flowchart in Master Mode (SPI Operation)

(9) Software Processing Flow

Figure 44.38 to Figure 44.40 show examples of the flow of software processing.

(a) Transmit Processing Flow

When transmitting data, the CPU will be notified of the completion of data transmission by enabling the SPII interrupt after the last writing of data for transmission.
The completion of data transmission can also be checked by polling to see if the SPSR.IDLNF flag has become 0, instead of using the SPII interrupt. However, one cycle of PCLK is required for the time from when data for transmission is written in the SPDR register to when the IDLNF flag becomes 1. After the last data is written in the SPDR register, discard the value of the SPSR register once not to judge the condition with the IDLNF flag which has not yet become 1, and read and use the value of the IDLNF flag to confirm the completion of data transmission.



Note 1. When using a polling for the SPTEF flag, proceed to the transmit data writing process after reading the SPTEF flag as 1.

Figure 44.38 Flowchart in Master Mode (Transmission)

(b) Receive Processing Flow

The RSPI does not support receive-only simplex communications, so processing for transmission is required.

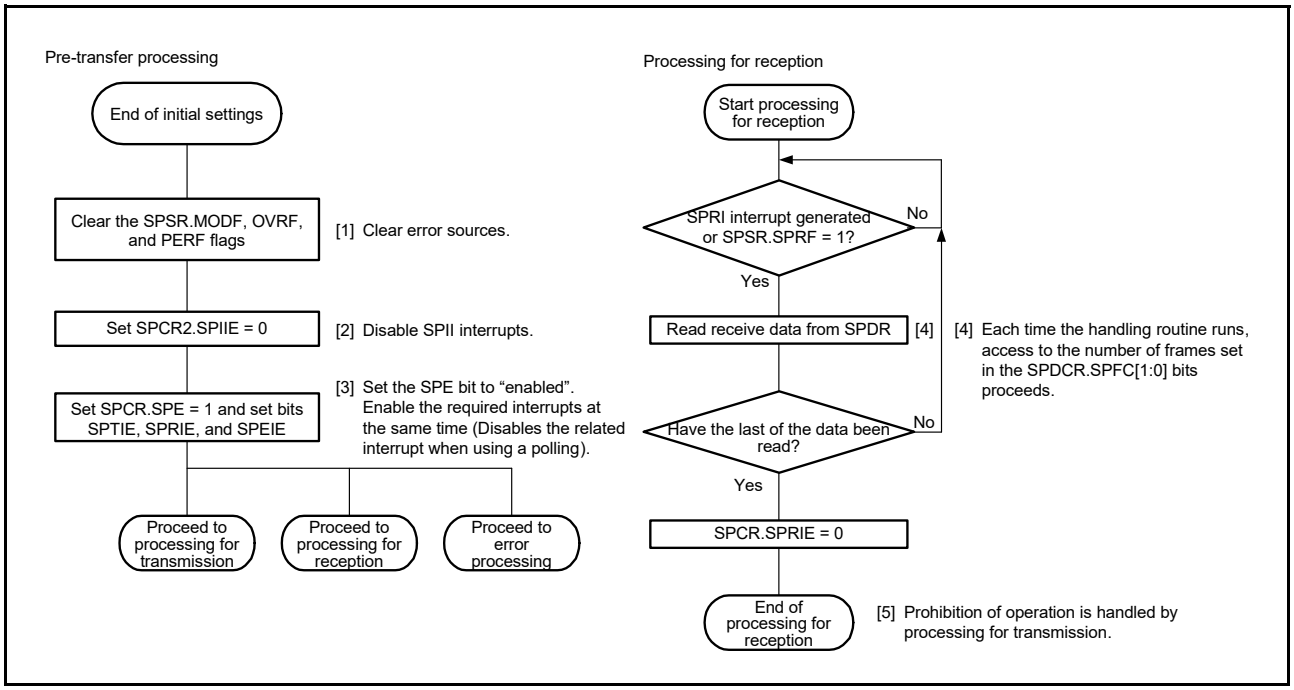


Figure 44.39 Flowchart in Master Mode (Reception)

(c) Flow of Error Processing

When a mode fault error is generated, the SPCR.SPE bit is automatically cleared, stopping operations for transmission and reception. For errors from other sources, however, the SPCR.SPE bit is not cleared and operations for transmission and reception continue; accordingly, we recommend clearing of the SPCR.SPE bit to stop operations in the case of errors other than mode fault errors. Not doing so will lead to updating of the SPSSR.SPECM[2:0] bits.

When interrupts are used and an error occurs, if the ICU.IRn.IR flag for the SPTI or SPRI interrupt request is set to 1, clear the ICU.IRn.IR flag in the error processing routine. If the SPRI interrupt request is indicated, read the receive buffer and initialize the sequencer in the RSPI.

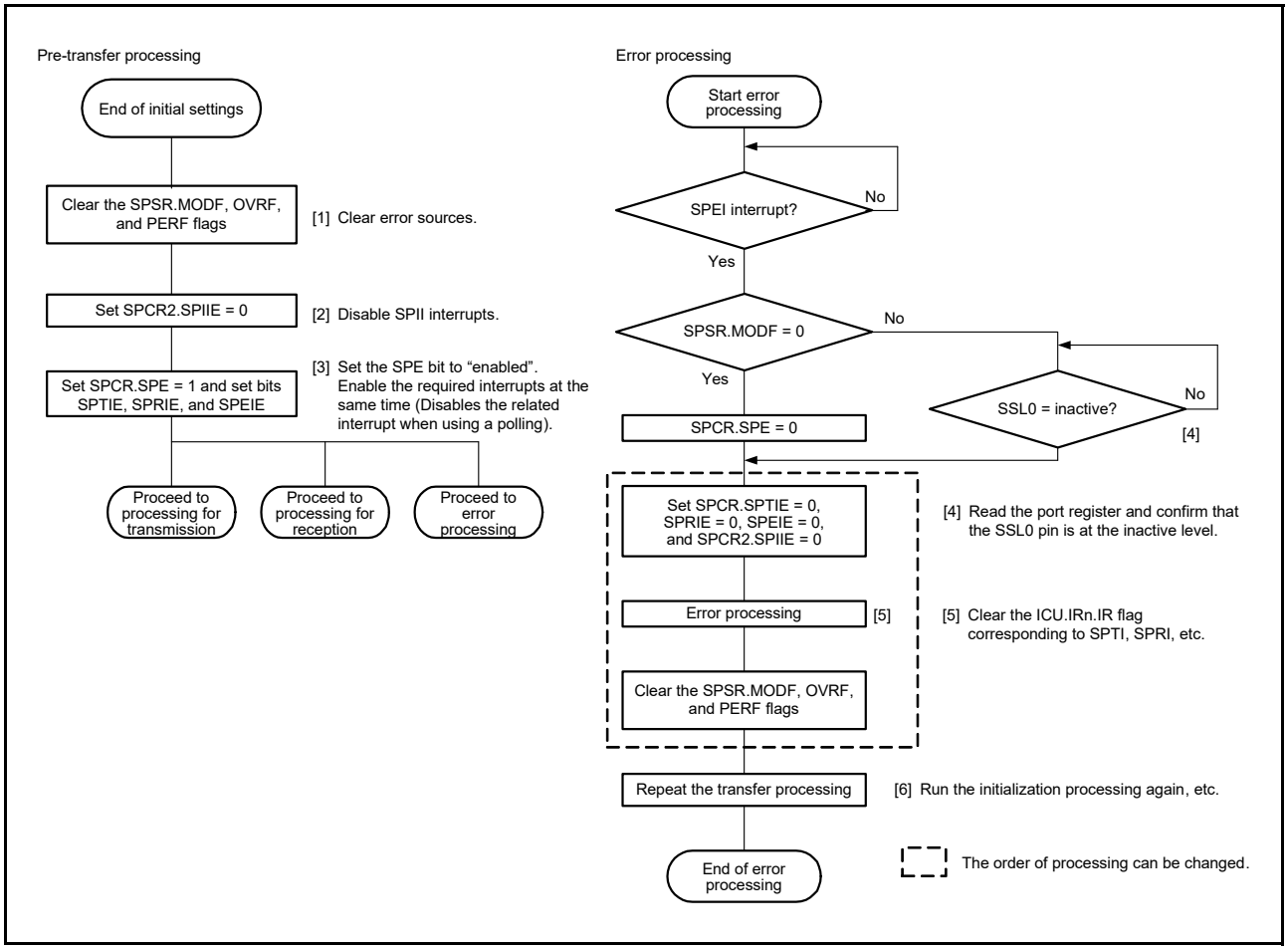


Figure 44.40 Flowchart for Master Mode (Error Processing)

44.3.11.2 Slave Mode Operation

(1) Starting a Serial Transfer

If the SPCMD0.CPHA bit is 0, when detecting an SSLx0 input signal assertion, the RSPI needs to start driving valid data to the MISOx output signal. For this reason, when the CPHA bit is 0, the assertion of the SSLx0 input signal triggers the start of a serial transfer.

If the CPHA bit is 1, when detecting the first RSPCKx edge in an SSLx0 signal asserted condition, the RSPI needs to start driving valid data to the MISOx output signal. For this reason, when the CPHA bit is 1, the first RSPCKx edge in an SSLx0 signal asserted condition triggers the start of a serial transfer.

Irrespective of the CPHA bit setting, the timing at which the RSPI starts driving of the MISOx output signal is the SSLx0 signal assertion timing. The data which is output by the RSPI is either valid or invalid, depending on the CPHA bit setting.

For details on the RSPI transfer format, refer to section 44.3.5, Transfer Format. The polarity of the SSLx0 input signal depends on the setting of the SSLP.SSL0P bit.

(2) Terminating a Serial Transfer

Irrespective of the SPCMD0.CPHA bit, the RSPI terminates the serial transfer after detecting an RSPCKx edge corresponding to the final sampling timing. When free space is available in the receive buffer (the SPRF flag is 0), upon termination of serial transfer the RSPI copies received data from the shift register to the receive buffer of the SPDR register. Upon termination of a serial transfer the RSPI changes the status of the shift register to “empty”, regardless of the receive buffer state. A mode fault error occurs if the RSPI detects an SSLx0 input signal negation from the beginning of serial transfer to the end of serial transfer (refer to section 44.3.9, Error Detection).

The final sampling timing changes depending on the bit length of transfer data. In slave mode, the RSPI data length depends on the SPCMD0.SPB[3:0] bit setting. The polarity of the SSLx0 input signal depends on the SSLP.SSL0P bit setting.

For details on the RSPI transfer format, refer to section 44.3.5, Transfer Format.

(3) Notes on Single-Slave Operations

If the SPCMD0.CPHA bit is 0, the RSPI starts serial transfers when it detects the assertion edge for an SSLx0 input signal. In the type of configuration shown in Figure 44.7 as an example, if the RSPI is used in single-slave mode, the SSLx0 signal is fixed at the active state. Therefore, when the CPHA bit is set to 0, the RSPI cannot correctly start a serial transfer. To correctly execute transmit/receive operations by the RSPI in slave mode in a configuration in which the SSLx0 input signal is fixed at the active state, the CPHA bit should be set to 1. If there is a need for setting the CPHA bit to 0, the SSLx0 input signal should not be fixed.

(4) Burst Transfer

If the SPCMD0.CPHA bit is 1, continuous serial transfer (burst transfer) can be executed while retaining the assertion state for the SSLx0 input signal. If the CPHA bit is 1, the period from the first RSPCKx edge to the sampling timing for the reception of the final bit in an SSLx0 signal active state corresponds to a serial transfer period. Even when the SSLx0 input signal remains at the active level, the RSPI can accommodate burst transfers because it can detect the start of an access.

If the CPHA bit is 0, the second and subsequent serial transfers during burst transfer cannot be executed correctly.

(5) Initialization Flowchart

Figure 44.41 is a flowchart illustrating an example of initialization in SPI operation when the RSPI is used in slave mode. For a description of how to set up the interrupt controller, DMAC, and I/O ports, refer to the descriptions given in the individual blocks.

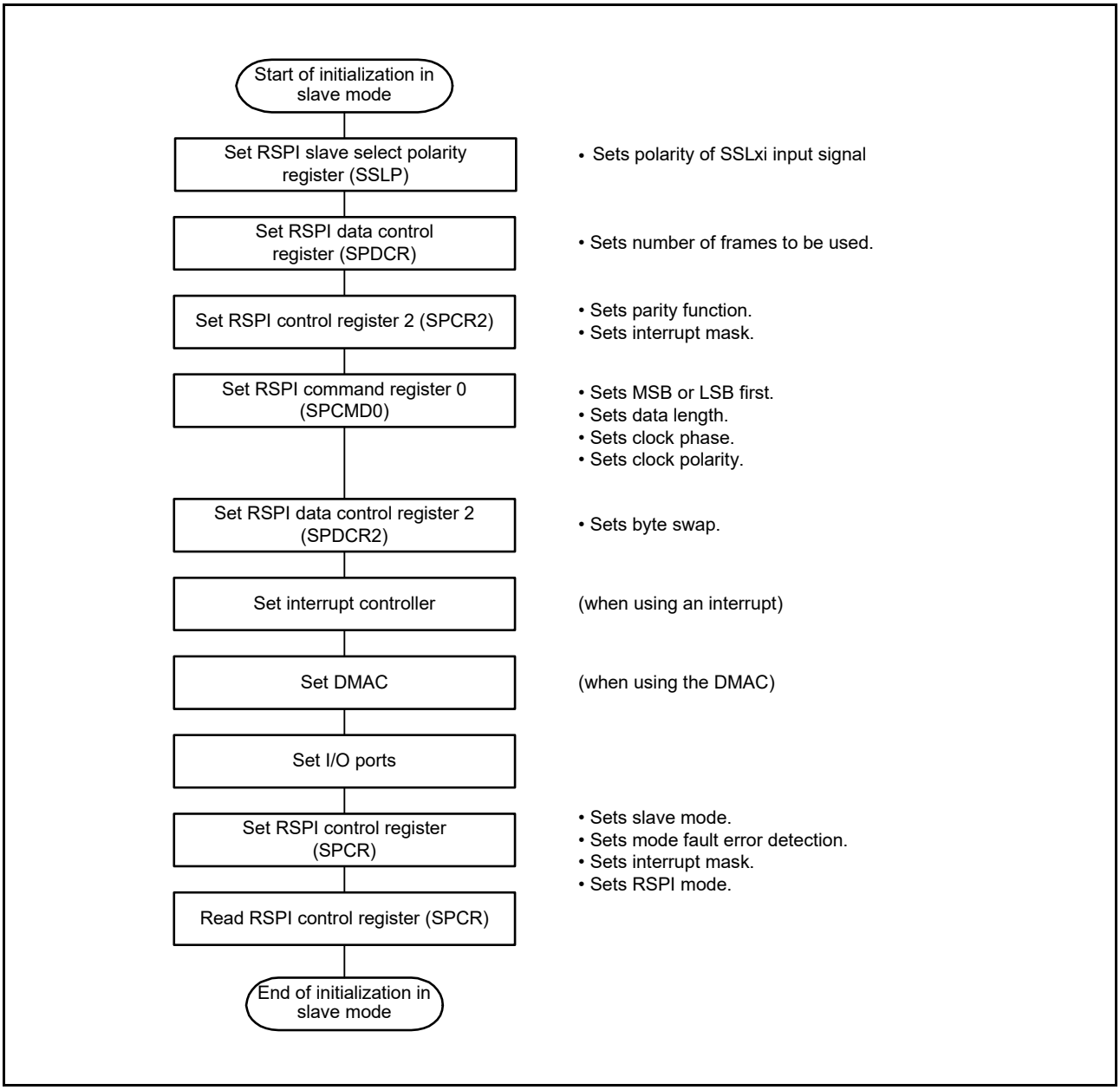


Figure 44.41 Example of Initialization Flowchart in Slave Mode (SPI Operation)

(6) Software Processing Flow

Figure 44.42 to Figure 44.44 show examples of the flow of software processing.

(a) Transmit Processing Flow

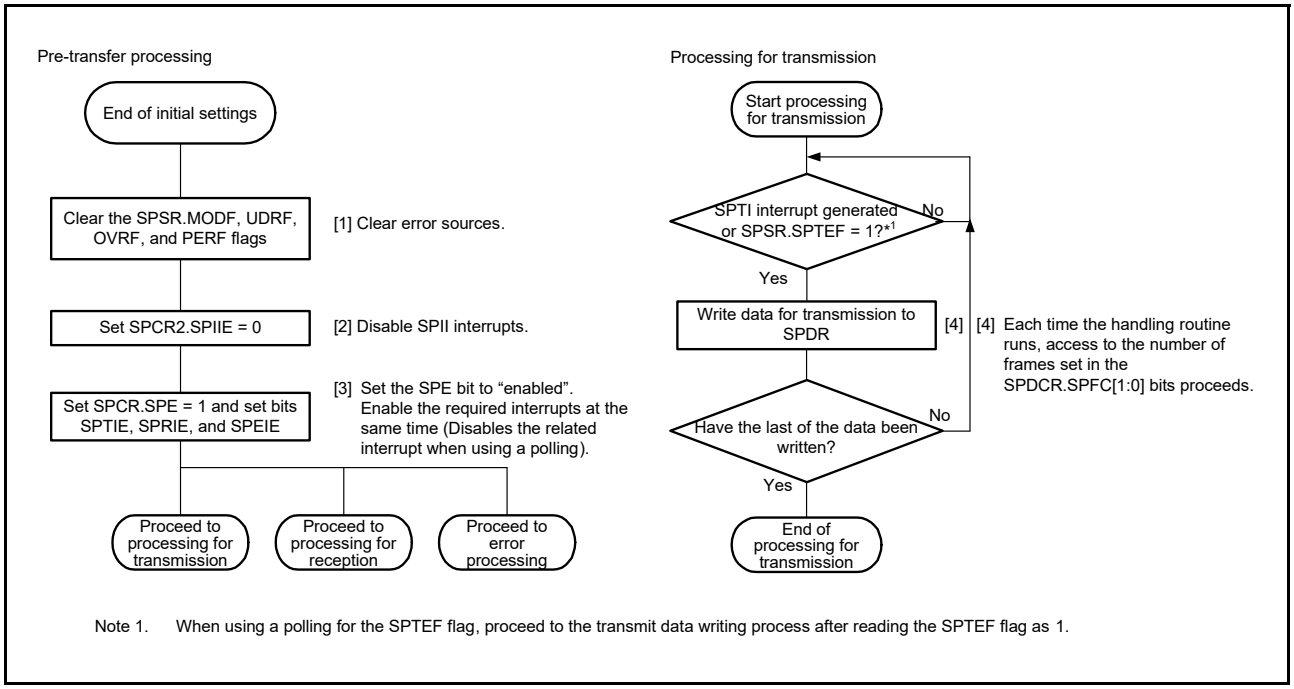


Figure 44.42 Flowchart in Slave Mode (Transmission)

(b) Receive Processing Flow

The RSPI does not support receive-only simplex communications, so processing for transmission is required.

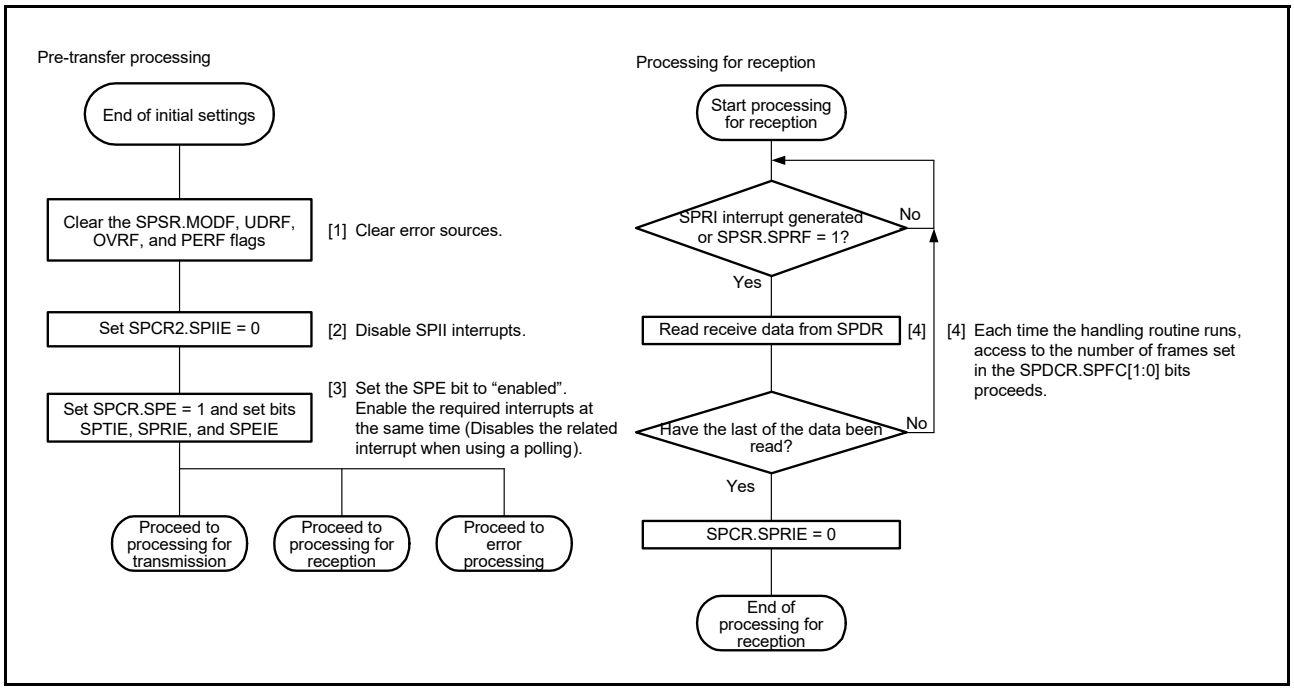


Figure 44.43 Flowchart in Slave Mode (Reception)

(c) Flow of Error Processing

In slave mode, even when a mode fault error is generated, the SPSR.MODF flag can be cleared regardless of the status of the SSLx0 pin.

When interrupts are used and an error occurs, if the ICU.IRn.IR flag for the SPTI or SPRI interrupt request is set to 1, clear the ICU.IRn.IR flag in the error processing routine. If the SPRI interrupt request is indicated, read the receive buffer and initialize the sequencer in the RSPI.

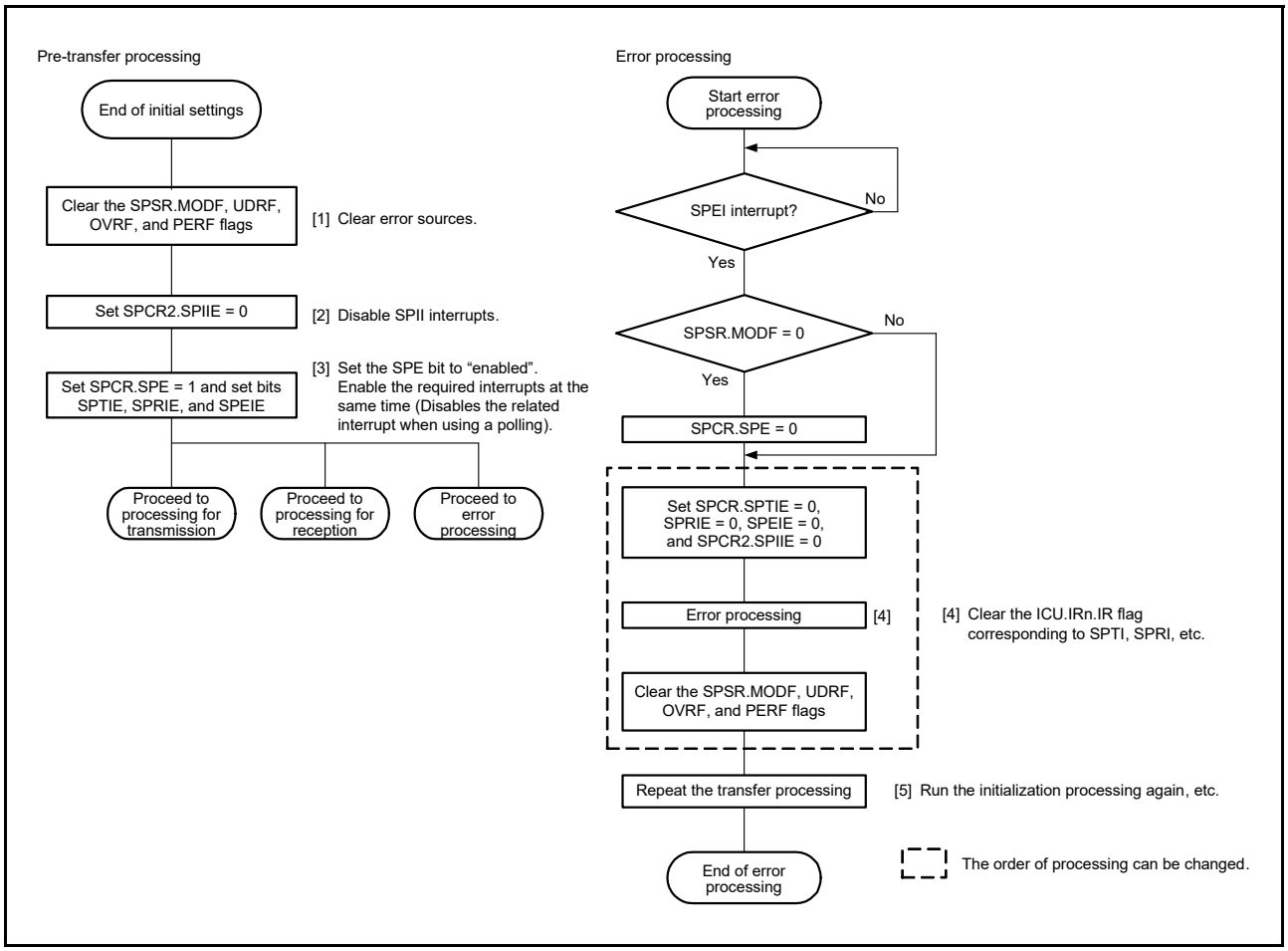


Figure 44.44 Flowchart for Slave Mode (Error Processing)

44.3.12 Clock Synchronous Operation

Setting the SPCR.SPMS bit to 1 selects clock synchronous operation of the RSPI. In clock synchronous operation, the SSLxi pin is not used, and the three pins of RSPCKx, MOSIx, and MISOx handle communications. The SSLxi pin is available as I/O port pins.
Although clock synchronous operation does not require use of the SSLxi pin, operation of the module is the same as in SPI operation. That is, in both master and slave mode operations, communications can be performed with the same flow as in SPI operation. However, mode fault errors are not detected because the SSLxi pin is not used.
Furthermore, do not set the SPCMDm.CPHA bit to 0 if clock synchronous operation is to proceed in slave mode (SPCR.MSTR = 0).

44.3.12.1 Master Mode Operation

(1) Starting a Serial Transfer

The RSPI updates the data in the transmit buffer (SPTX) of the SPDR register when data is written to the SPDR register with the transmit buffer being empty (the SPTEF flag is 1 and data for the next transfer is not set). When the shift register is empty after the number of frames set in the SPDCR.SPFC[1:0] bits are written to the SPDR register, the RSPI copies data from the transmit buffer to the shift register and starts serial transmission. Upon copying transmit data to the shift register, the RSPI changes the status of the shift register to “full”, and upon termination of serial transfer, it changes the status of the shift register to “empty”. The status of the shift register cannot be referenced.
For details on the RSPI transfer format, refer to section 44.3.5, Transfer Format.
However, transfer in clock synchronous operation is conducted without the SSLx0 output signal.

(2) Terminating a Serial Transfer

The RSPI terminates the serial transfer after transmitting an RSPCKx edge corresponding to the sampling timing. If free space is available in the receive buffer (SPRX) (the SPRF flag is 0), upon termination of serial transfer, the RSPI copies data from the shift register to the receive buffer of the RSPI data register (SPDR).
It should be noted that the final sampling timing varies depending on the bit length of transfer data. In master mode, the RSPI data length depends on the SPCMDm.SPB[3:0] bit setting.
For details on the RSPI transfer format, refer to section 44.3.5, Transfer Format.
However, transfer in clock synchronous operation is conducted without the SSLx0 output signal.

(3) Sequence Control

The transfer format employed in master mode is determined by SPSCR, SPCMDm, SPBR, SPCKD, SSLND, and SPND registers. Although the SSLxi signals are not output in clock synchronous operation, these settings are valid.
The SPSCR register is a register used to determine the sequence configuration for serial transfers that are executed by the RSPI in master mode. The following items are set in the SPCMDm register: SSLxi output signal value, MSB/LSB first, data length, some of the bit rate settings, RSPCKx polarity/phase, whether the SPCKD register is to be referenced, whether the SSLND register is to be referenced, and whether the SPND register is to be referenced. The SPBR register holds some of the bit rate settings; the SPCKD register, an RSPI clock delay value; the SSLND register, an SSL negation delay; and the SPND register, a next-access delay value.
According to the sequence length that is assigned to the SPSCR register, the RSPI makes up a sequence comprised of a part or all of the SPCMDm register. The RSPI contains a pointer to the SPCMDm register that makes up the sequence. The value of this pointer can be checked by reading the SPSSR.SPCP[2:0] bits. When the SPCR.SPE bit is set to 1 and the RSPI function is enabled, the RSPI loads the pointer to the commands in the SPCMD0 register, and incorporates the SPCMD0 register setting into the transfer format at the beginning of serial transfer. The RSPI increments the pointer each time the next-access delay period for a data transfer ends. Upon completion of the serial transfer that corresponds to the final command comprising the sequence, the RSPI sets the pointer in the SPCMD0 register, and in this manner the sequence is executed repeatedly.

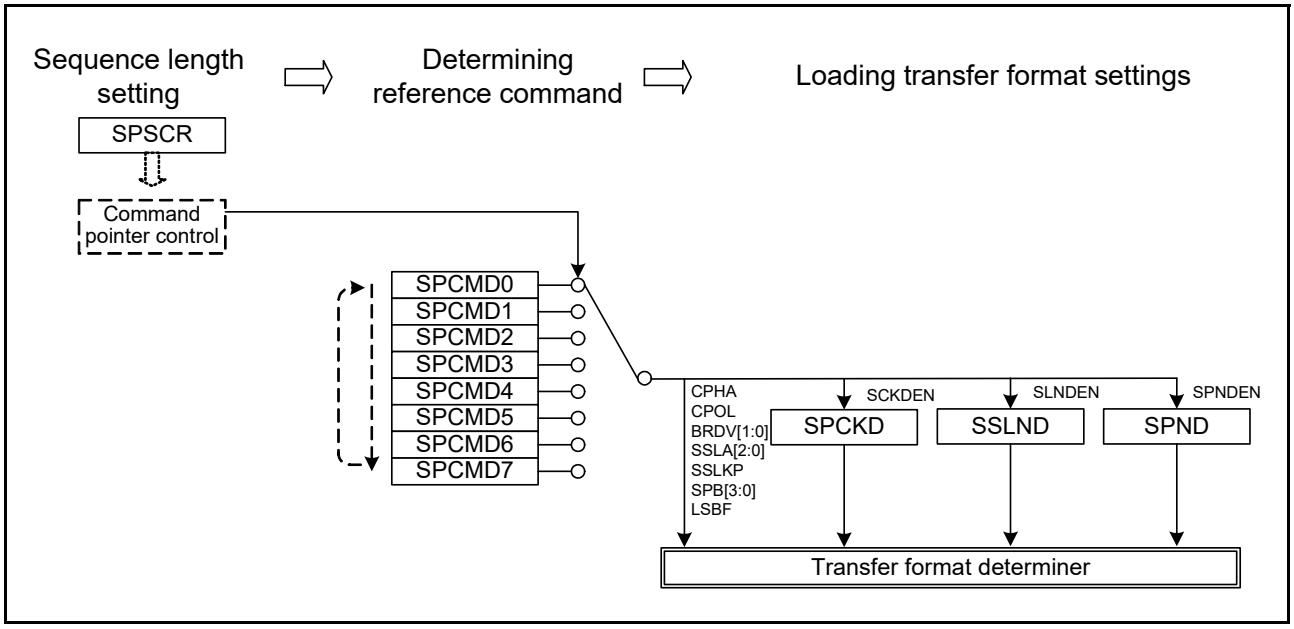


Figure 44.45 Procedure for Determining the Form of Serial Transmission in Master Mode

In this section, a frame is the combination of the data (SPDR) and the settings (SPCMDm).

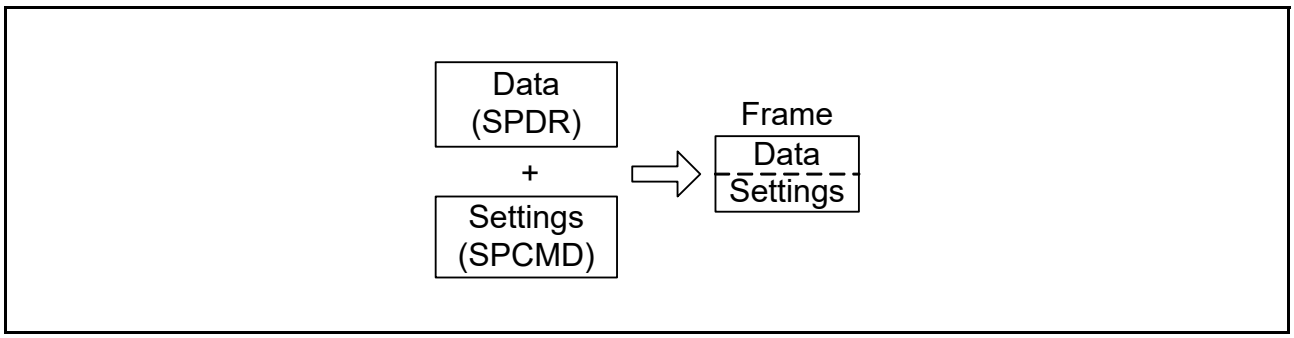


Figure 44.46 Concept of a Frame

Figure 44.47 shows the relationship between the command and the transmit and receive buffers in the sequence of operations specified by the settings in Table 44.4.

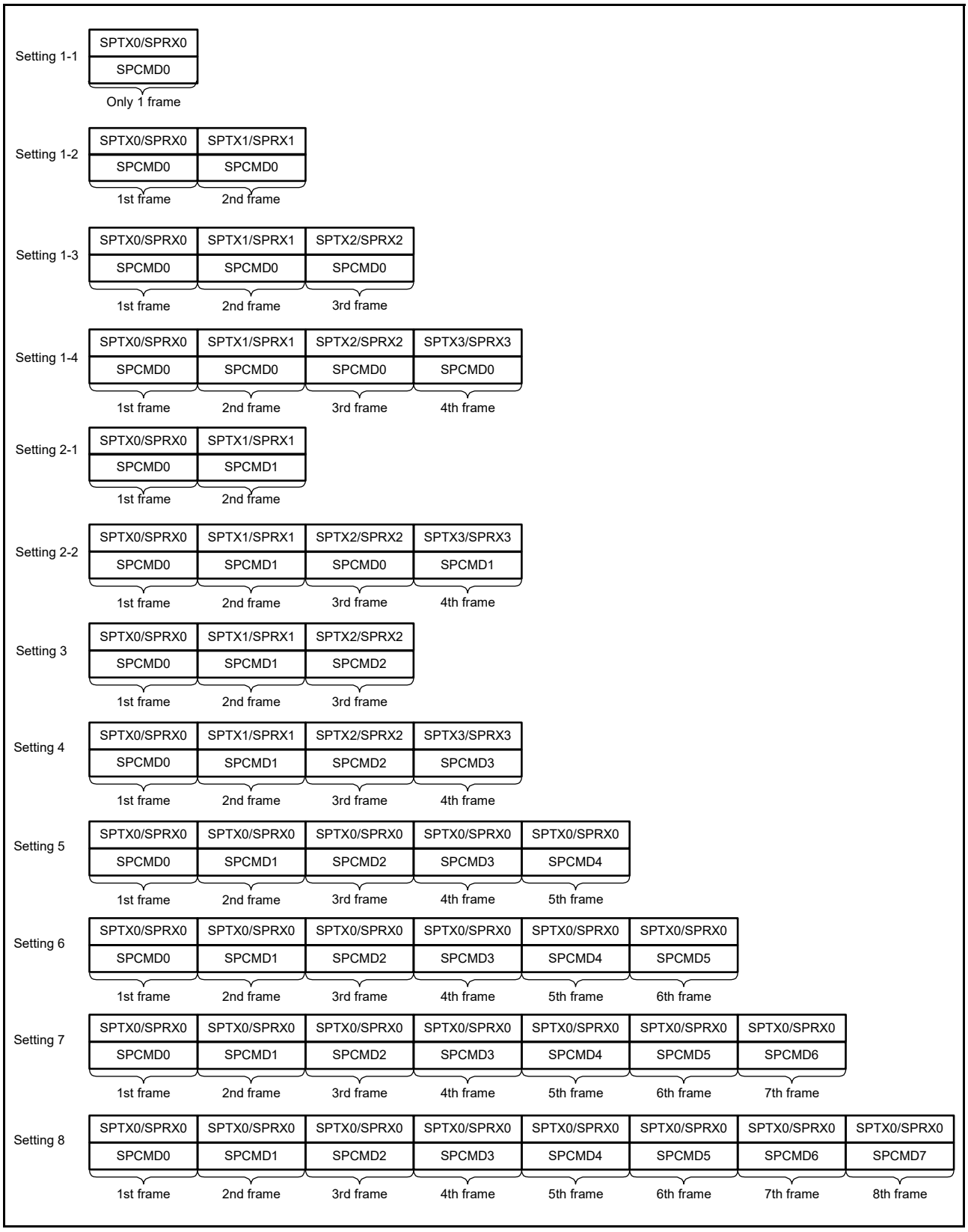


Figure 44.47 Correspondence between the RSPI Command Register and Transmit/Receive Buffers in Sequence Operations

(4) Initialization Flowchart

Figure 44.48 is a flowchart illustrating an example of initialization in clock synchronous operation when the RSPI is used in master mode. For a description of how to set up the interrupt controller, DMAC, and I/O ports, refer to the descriptions given in the individual blocks.

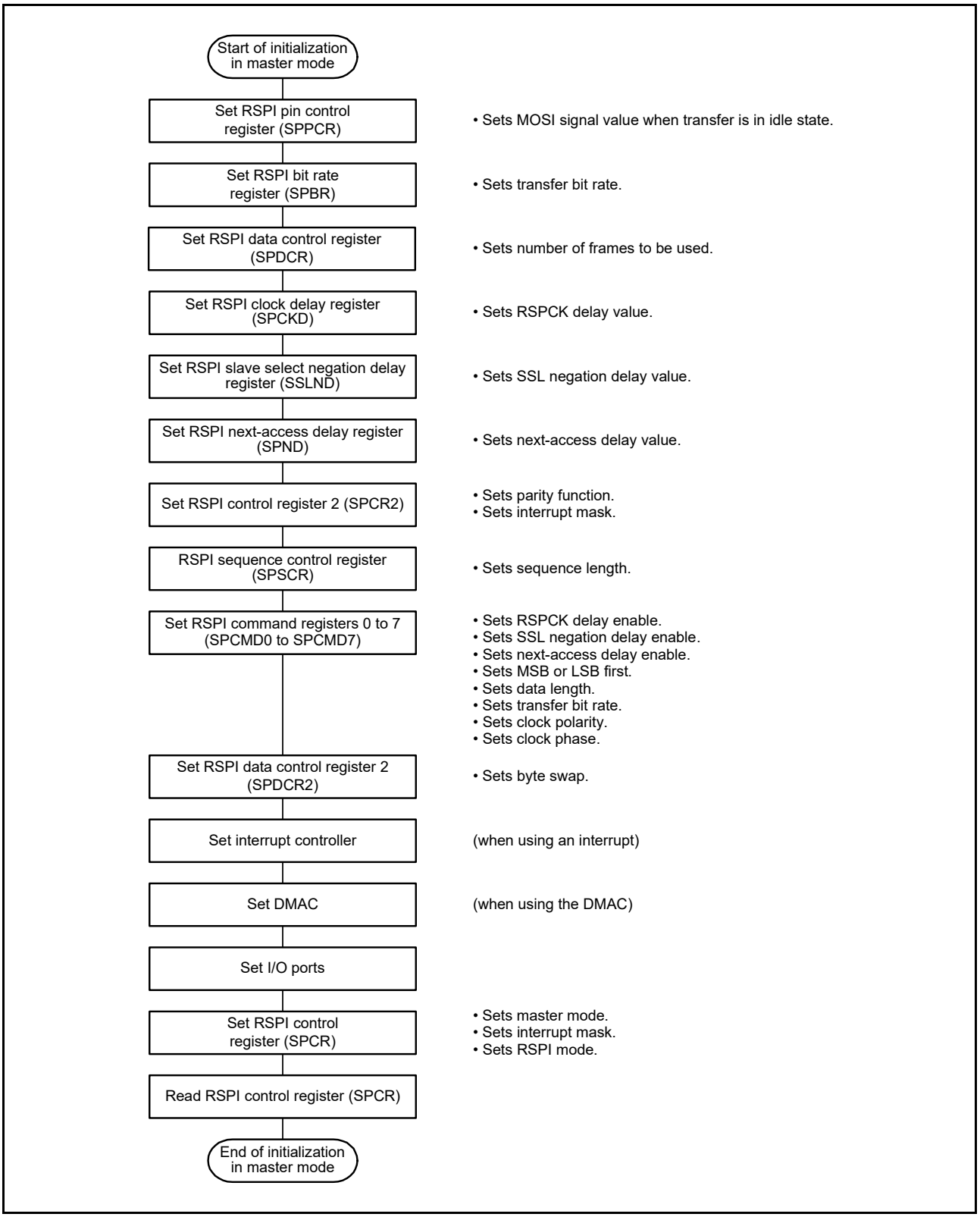


Figure 44.48 Example of Initialization Flowchart in Master Mode (Clock Synchronous Operation)

(5) Flow of Software Processing

Software processing during clock-synchronous operation when the RSPI is used in master mode is the same as that for SPI master mode operation. For details, refer to section 44.3.11.1, (9) Software Processing Flow. Note that mode fault errors will not occur.

44.3.12.2 Slave Mode Operation

(1) Starting a Serial Transfer

When the SPCR.SPMS bit is 1, the first RSPCKx edge triggers the start of a serial transfer in the RSPI.
When the SPMS bit is 1, the RSPI drives the MISOx output signal.
For details on the RSPI transfer format, refer to section 44.3.5, Transfer Format.
It should be noted that the SSLx0 input signal is not used in clock synchronous operation.

(2) Terminating a Serial Transfer

The RSPI terminates the serial transfer after detecting an RSPCKx edge corresponding to the final sampling timing. When free space is available in the receive buffer (the SPRF flag is 0), upon termination of serial transfer the RSPI copies received data from the shift register to the receive buffer of the SPDR register. Upon termination of a serial transfer the RSPI changes the status of the shift register to “empty” regardless of the receive buffer status. The final sampling timing changes depending on the bit length of transfer data. In slave mode, the RSPI data length depends on the SPCMD0.SPB[3:0] bit setting.
For details on the RSPI transfer format, refer to section 44.3.5, Transfer Format.

(3) Initialization Flowchart

Figure 44.49 is a flowchart illustrating an example of initialization in clock synchronous operation when the RSPI is used in slave mode. For a description of how to set up the interrupt controller, DMAC, and I/O ports, refer to the descriptions given in the individual blocks.

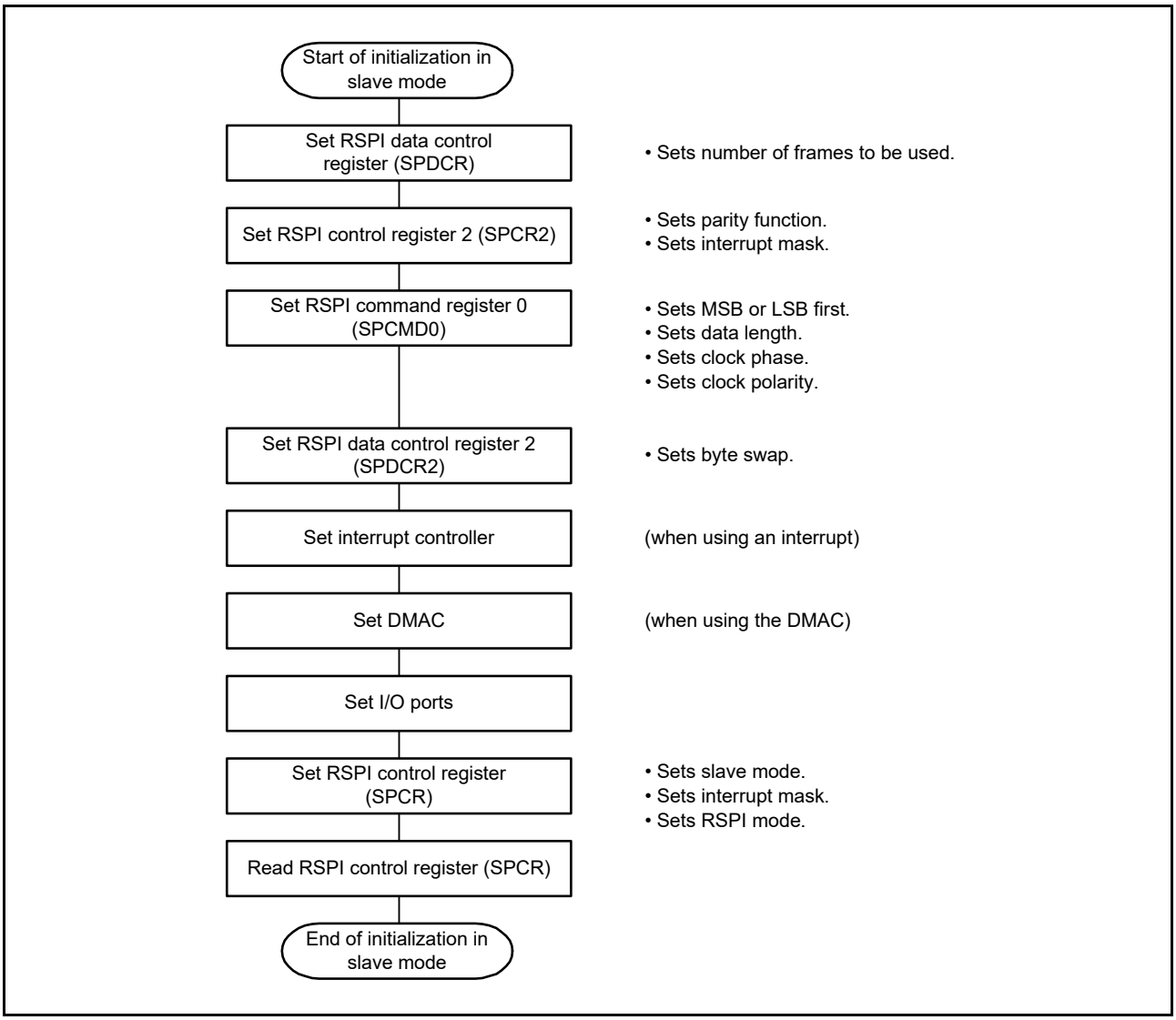


Figure 44.49 Example of Initialization Flowchart in Slave Mode (Clock Synchronous Operation)

(4) Flow of Software Processing

Software processing during clock-synchronous operation when the RSPI is used in slave mode is the same as that for SPI slave mode operation. For details, refer to section 44.3.11.2, (6) Software Processing Flow. Note that mode fault errors will not occur.

44.3.13 Loopback Mode

When 1 is written to the SPPCR.SPLP2 bit or SPPCR.SPLP bit, the RSPI shuts off the path between the MISOx pin and the shift register if the SPCR.MSTR bit is 1, and between the MOSIx pin and the shift register if the SPCR.MSTR bit is 0, and connects the input path and output path of the shift register. The RSPI does not shut off the path between the MOSIx pin and the shift register if the SPCR.MSTR bit is 1, and between the MISOx pin and the shift register if the SPCR.MSTR bit is 0. This is called loopback mode. When a serial transfer is executed in loopback mode, the transmit data for the RSPI or the reversed transmit data becomes the received data for the RSPI.
Table 44.11 lists the relationship among the SPLP2 and SPLP bits and the received data. Figure 44.50 shows the configuration of the shift register I/O paths for the case where the RSPI in master mode is set in loopback mode (SPPCR.SPLP2 = 0, SPPCR.SPLP = 1).

Table 44.11 SPLP2 and SPLP Bit Settings and Received Data

SPPCR.SPLP2 Bit	SPPCR.SPLP Bit	Received Data
0	0	Input data from the MOSIx pin or MISOx pin
0	1	Inverted transmit data
1	0	Transmit data
1	1	Transmit data

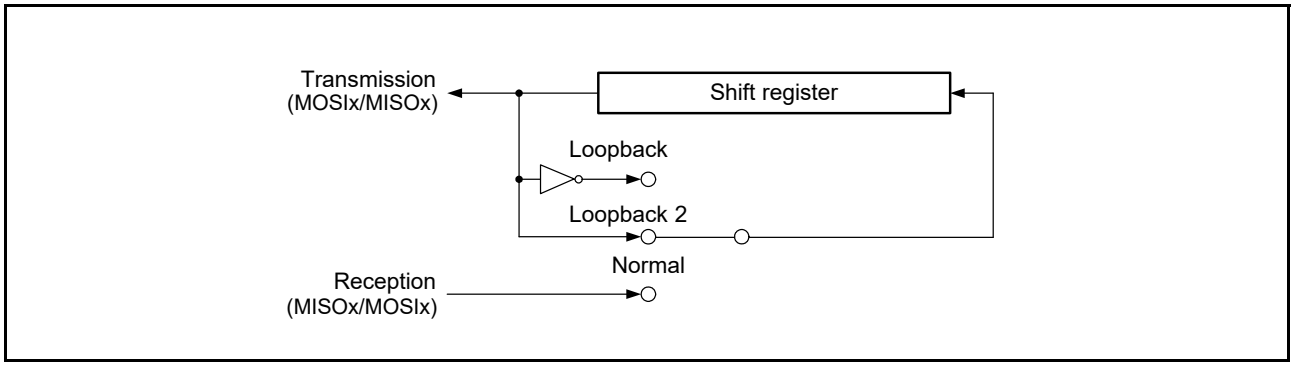


Figure 44.50 Configuration of Shift Register I/O Paths in Loopback Mode (Master Mode)

44.3.14 Self-Diagnosis of Parity Bit Function

The parity circuit consists of a parity bit adding unit used for transmit data and an error detecting unit used for received data. In order to detect defects in the parity bit adding unit and error detecting unit of the parity circuit, self-diagnosis is executed for the parity circuit following the flowchart shown in Figure 44.51.

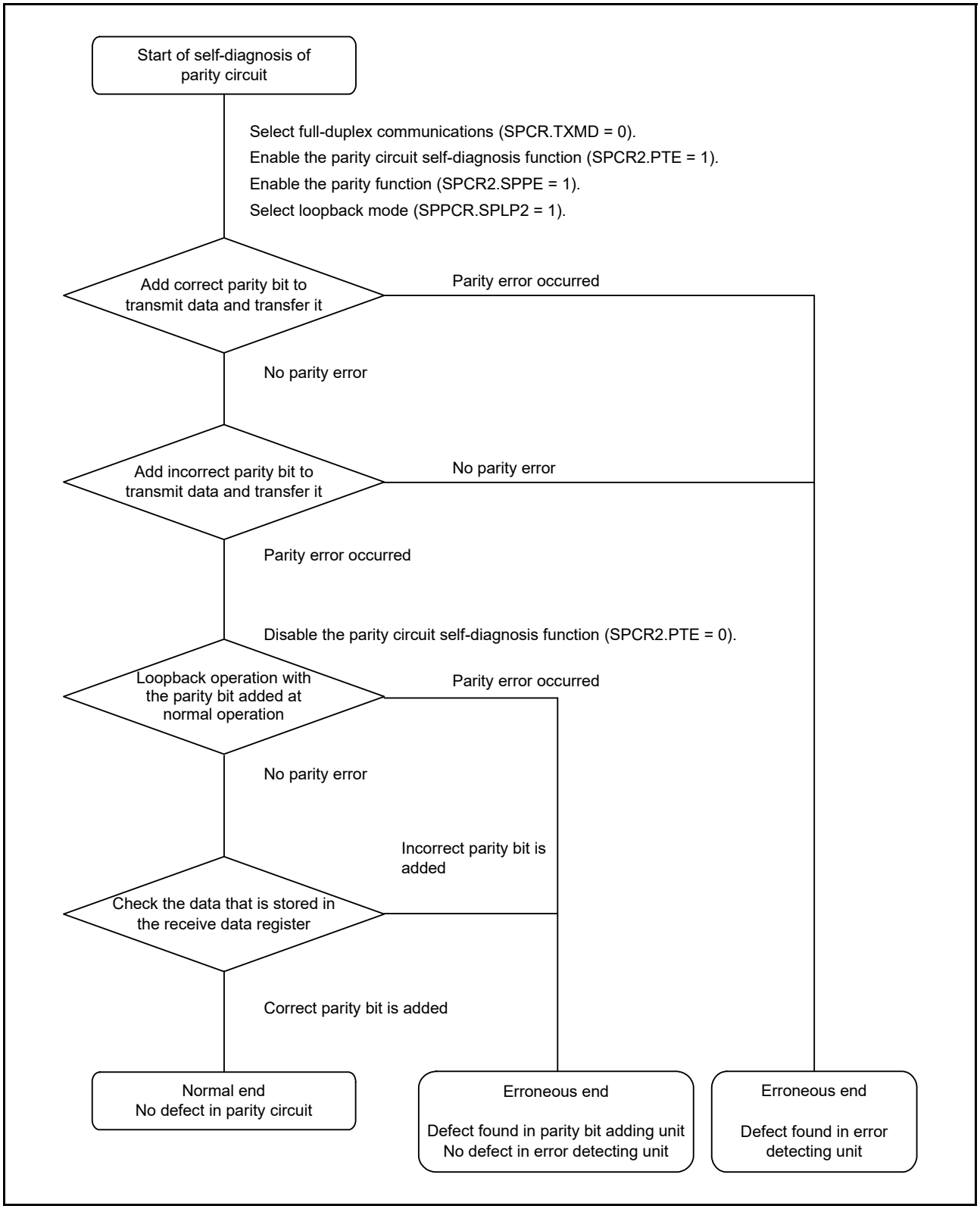


Figure 44.51 Flowchart for Self-Diagnosis of Parity Circuit

44.3.15 Interrupt Sources

The RSPI has interrupt sources of receive buffer full, transmit buffer empty, error (mode fault, underrun, overrun, and parity error), and idle. In addition, the DTC or DMAC can be activated by the receive buffer full or transmit buffer empty interrupt to perform data transfer.

Since the vector address for SPEI is allocated to interrupt requests due to mode fault, underrun, overrun, and parity errors, the actual interrupt source must be determined from the flags. Interrupt sources for the RSPI are listed in Table 44.12. An interrupt is generated on satisfaction of an interrupt condition in Table 44.12. Clear the receive buffer full and transmit buffer empty sources through data transfer.

When using the DTC or DMAC to perform data transmission/reception, the DTC or DMAC must be set up first to be in a status in which transfer is enabled before making the RSPI settings. For the method for setting the DTC or DMAC, refer to section 18, DMA Controller (DMACAa), or section 20, Data Transfer Controller (DTCb).

If the conditions for generating a transmit buffer empty or receive buffer full interrupt are generated while the ICU.IRn.IR flag is 1, the interrupt is not output as a request for ICU but is retained internally (the capacity for retention is one request per source). A retained interrupt request is output when the ICU.IRn.IR flag becomes 0. A retained interrupt request is automatically discarded once it is output as an actual interrupt request. The interrupt enable bit (the SPCR.SPTIE or SPCR.SPRIE bit) for an internally retained interrupt request can also be cleared to 0.

Table 44.12 Interrupt Sources of RSPI

Interrupt Source	Symbol	Interrupt Condition	DMAC/DTC Activation
Receive buffer full	SPRI	The receive buffer becomes full (the SPRF flag becomes 1) while the SPCR.SPRIE bit is 1.	Possible
Transmit buffer empty	SPTI	The transmit buffer becomes empty (the SPTEF flag becomes 1) while the SPCR.SPTIE bit is 1.	Possible
Errors (mode fault, underrun, overrun, and parity error)	SPEI	The SPSR.MODF, UDRF, OVRF, or PERF flag is set to 1 while the SPCR.SPEIE bit is 1.	Impossible
Idle	SPII	The SPSR.IDLNF flag is set to 0 while the SPCR2.SPIIE bit is 1.	Impossible

44.4 Link Operation by Event Linking

The RSPI0 supports the following event output for the event link controller (ELC). The event link output signal is output regardless of the interrupt enable bit setting.

44.4.1 Receive Buffer Full Event Output

This event signal is output when received data have been transferred from the shift register to the SPDR register on completion of serial transfer.

44.4.2 Transmit Buffer Empty Event Output

This event signal is output when data for transmission have been transferred from the transmit buffer to the shift register and when the value of the SPE bit has changed from 0 to 1.

44.4.3 Mode Fault, Underrun, Overrun, or Parity Error Event Output

(1) Mode Fault

Table 44.13 lists the occurrence conditions of a mode fault event.

Table 44.13 Occurrence Conditions of Mode Fault Event

	SPCR.MODFEN Bit	SSLx0 Pin	Remarks
Master (SPCR.MSTR bit = 1)	1	Active	Under the condition (the SPCR.MSTR bit is 1 and the MODFEN bit is 1), if the SPCR.SPMS bit is 0, mode fault error, overrun error, and parity error event output cannot be used. Do not set the ELSRn register to 52h.
Slave (SPCR.MSTR bit = 0)	1	Not active	Event is output only when the pin is deactivated during transmission.

(2) Underrun

The condition for this event signal being output in response to an underrun is the start of serial transfer with the transmit buffer containing no transmit data while the value of the SPCR.MSTR bit is 0 and the value of the SPCR.SPE bit is 1, in which case the UDRF and MODF flags are set to 1.

(3) Overrun

The condition for this event signal being output in response to an overrun is completion of serial transfer while the receive buffer contains data that have not been read and the value of the SPCR.TXMD bit is 0, in which case the OVRF flag is set to 1.

(4) Parity Error

The condition for this event signal being output in response to a parity error is detection of a parity error on completion of serial transfer while the value of the TXMD bit in SPCR is 0 and the value of the SPPE bit in SPCR2 is 1.

44.4.4 Idle Event Output

(1) In Master Mode

In master mode, an event is output when the condition for setting the IDLNF flag (idle flag) to 0 is satisfied.

(2) In Slave Mode

In slave mode, an event is output when the SPCR.SPE bit is set to 0 (RSPI is initialized).

44.4.5 Transmit End Event Output

During both SPI operation and clock synchronous operation in master mode, an event is output under the condition for setting the IDLNF flag (idle flag) from 1 to 0. In slave mode, an event is output under the conditions listed in Table 44.14.

Table 44.14 Generating Conditions of Transmit End Event (Slave mode)

RSPI Mode	Transmit Buffer State	Shift Register State	Others
SPI operation (SPMS = 0)	Empty	Empty	Negation of the SSLx0 input
Clock synchronous operation (SPMS = 1)	Empty	Empty	Even edge detection of the last RSPCKx for the last data

Whether the operation is in master mode or slave mode, an event is not output if 0 is written to the SPCR.SPE bit during transmission or the SPCR.SPE bit is cleared by the mode fault error.

44.5 Usage Notes

44.5.1 Setting Module Stop Function

Module stop control register B (MSTPCRB) and module stop control register C (MSTPCRC) can be used to enable or disable the RSPI. Immediately after a reset, operation of the RSPI is disabled. Register access is enabled by releasing the module stop state. For details, refer to section 11, Low Power Consumption.

44.5.2 Note on Low Power Consumption Functions

When using the module stop function and entering a low power consumption mode other than sleep mode, set the SPCR.SPE bit to 0 before completing communication.

44.5.3 Notes on Starting Transfer

If the ICU.IRn.IR flag is 1 at the time transfer is to be started, an interrupt request is internally retained after transfer starts, and this can lead to unanticipated behavior of the ICU.IRn.IR flag.
When the ICU.IRn.IR flag is 1 at the time transfer is to start, follow the procedure below to clear interrupt requests before enabling operations (by setting the SPCR.SPE bit to 1).

1. Confirm that transfer has stopped (i.e. that the SPCR.SPE bit is 0).
2. Set the relevant interrupt enable bit (the SPCR.SPTIE or SPCR.SPRIE bit) to 0.
3. Read the relevant interrupt enable bit (the SPCR.SPTIE or SPCR.SPRIE bit) and confirm that its value is 0.
4. Set the ICU.IRn.IR flag to 0.

44.5.4 Notes on the SPRF and SPTEF flags

When polling the SPSR.SPRF flag and/or SPSR.SPTEF flag, set the SPCR.SPRIE bit and/or SPCR.SPTIE bit to 0.